# Masked, Not Missing: Mid-Depth Retrieval Collapse in T5-Family Encoders

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## Abstract

Practitioners often read intermediate layers of off-the-shelf encoders, check them for anisotropy with mean pairwise cosine, and correct the embedding space post hoc. We test this practice on medieval Latin canon law, where a scholarly edition labels which manuscript copies (witnesses) share a source. In the raw T5 encoders LaTa, PhilTa and mT5-base, 26 of 36 layers rank same-source pairs near chance, although their first and last layers retrieve well. Mean pairwise cosine gives no reliable warning of this collapse, while the share of variance on the first principal component is high at every collapsed layer. The signal is masked, not missing: a projection fit on training embeddings alone brings all 100 layers of our six-encoder panel to AUROC 0.962–0.987. Of the six raw T5-family encoders we test, four collapse: these three and T5-v1.1-base. All four share the T5 v1.1 layout; encoder-only siblings pretrained on the same corpora and two checkpoints with the original T5 layout do not. Removing about three principal directions restores every collapsed layer, while zeroing the ten largest coordinates restores none. After the projection the six encoders route witnesses to their source or to "new" about equally well, but no frozen encoder at its train-selected layer beats a character n-gram reference.

## 1 Introduction

Off-the-shelf encoders are a common tool for finding reused text in digital-humanities collections (e.g., Manjavacas et al., 2019; Schelb et al., 2026). Practitioners often read an intermediate layer, because intermediate layers can outperform the final one (Skean et al., 2025). They check the space for anisotropy, usually with the mean cosine between embeddings (Ethayarajh, 2019). They then correct it, for example by removing the top principal components (Mu and Viswanath, 2018) or by whitening (Su et al., 2021). This practice assumes that the check warns when a layer retrieves poorly. Mean cosine is already known to be a poor isotropy measure (Rudman et al., 2022); we ask whether it, or any label-free statistic, warns where retrieval fails.

Medieval Latin suits this test. The Carolingian Canon Law project (CCL; Firey, 2009) keys each transcribed manuscript unit to the source it copies, with expert labels that no model shaped. Latin also offers T5 and encoder-only models from the same authors and pretraining corpora. Our testbed holds 1,705 *witnesses*, manuscript copies of 840 sources. We evaluate two tasks: *ranking* pairs of witnesses by cosine, scored by AUROC, and *routing* each witness to its source or to "new". Because witnesses copy their source's wording, a character n-gram reference ranks pairs at 0.987 AUROC, so a retrieval failure points to the representation.

Three raw T5 encoders, trained with no embedding objective, fail at mid-depth (Figure 1). In LaTa, PhilTa and mT5-base (Xue et al., 2021), 26 of 36 layers rank same-source pairs at AUROC 0.50 to 0.66, near chance. Their first and last layers still retrieve. Three embedding-trained encoders, which complete our six-model panel, never collapse. The usual check, mean pairwise cosine, gives no reliable warning in either direction. The share of variance on the first principal component (top-PC share) is high at every collapsed layer: necessary for collapse in our pretrained models, but not sufficient.

The signal is masked, not missing. All-but-the-Top (ABTT; Mu and Viswanath, 2018), a linear projection fit on training embeddings only, removes the mean and the top principal components from every embedding, with no parameter update. It brings all 100 panel layers to AUROC 0.962 to 0.987, the 26 collapsed layers included. Because the projection sees no test label, the ranking it recovers is already present in each collapsed layer.

Among the six raw T5-family encoders we test, the collapse tracks the T5 v1.1 layout: a gated-

GELU feed-forward block (Shazeer, 2020) and untied input and output embeddings (Google Research, 2020). LaBERTa and PhilBERTa, encoder-only siblings of LaTa and PhilTa, never fall below 0.826 and 0.883. T5-v1.1-base collapses on the same Latin text (lowest AUROC 0.489), while the original T5-base (Raffel et al., 2020) never falls below 0.816. T5-base also saw supervised tasks in pretraining, so we add T5-efficient-base (Tay et al., 2022), with T5-base's layout but pretrained on C4 alone, without dropout and on about 30 times fewer tokens; it does not collapse either. Of the six raw T5-family encoders we test, the four that collapse all have this layout. We do not isolate which change matters.

Inside the collapsed layers, about three directions mask the signal. Removing three principal components brings all 26 layers to AUROC 0.914 or above, and mT5-base keeps improving up to ten. Zeroing the ten largest coordinates restores none of them, although the first principal component (the dominant direction) sits mostly on such coordinates. In LaTa, pooling without the comma, the period and </s> restores every collapsed layer; in mT5-base, dropping up to its top 100 token types does not.

This paper makes four contributions. (1) A per-layer retrieval profile of T5 encoder states, the first we know of; it shows a mid-depth collapse that mean cosine misses, top-PC share flags and ABTT reverses (Section 4). (2) Controls showing that the collapse tracks the T5 v1.1 layout: T5-v1.1-base collapses where the original T5-base does not, encoder-only siblings on the same corpora do not, and the high-share middle layers recur on English text (Section 5). (3) A localization of the mask to about three directions, which zeroing the ten largest coordinates does not remove, with a token-level account in LaTa and none in mT5-base (Section 6). (4) A measure of what the projection gains for routing and where it stops, set against a character n-gram reference (Section 7). We will release the frozen testbed, split, evaluation code and character n-gram reference scores. We localize and repair the collapse but do not explain why it tracks the T5 v1.1 layout.

## 2 Related Work

**Anisotropy statistics.** Contextual embeddings occupy a narrow cone, usually checked by mean cosine over random pairs (Gao et al., 2019; Ethayarajh, 2019), which is not an appropriate measure of isotropy (Rudman et al., 2022). For pooled output embeddings, the variance share of the largest coordinate or top principal direction predicts how much rank-based similarity gains over cosine (Parupudi, 2026), and a topological signature led by effective rank moderately predicts retrieval (Rottach et al., 2025). Yet geometric statistics do not reliably predict downstream performance (Nastase and Merlo, 2025; Kulkarni et al., 2026). We relate mean cosine and top-PC share to retrieval at every layer (Section 4).

**Outlier dimensions and their carrier tokens.** One to three rogue dimensions dominate cosine similarity in pretrained transformers (Timkey and van Schijndel, 2021). In BERT they are tied to a few LayerNorm weights and to token frequency (Kovaleva et al., 2021; Puccetti et al., 2022). Clipping them improves mean-pooled sentence embeddings (Luo et al., 2021), though multilingual BERT is anisotropic without them (Rajaee and Pilehvar, 2022). Most BERT outliers sit on [SEP], the period and the comma (Bondarenko et al., 2023), and removing punctuation and special tokens before pooling improves sentence similarity (Fuster Baggetto and Fresno, 2022). Decoder language models carry massive activations on starting and delimiter tokens (Sun et al., 2024). The original T5's encoder carries outliers that grow with depth (Zhao et al., 2025), and outliers persist through fine-tuning (Rudman et al., 2023). On pooled T5 encoder states we compare coordinate zeroing, direction removal and token dropping (Section 6).

**Layerwise analysis and post-hoc correction.** Prior T5 embedding work reads the final layer (Ni et al., 2022) or the first and last layers averaged (Huang et al., 2021); Godey et al. (2024) trace anisotropy through T5-base's encoder and decoder layers but do not measure retrieval. Large decoder language models dip at mid-depth in retrieval and similarity and recover near the top, while two decoder embedding models stay roughly flat there before rising (Rizwan et al., 2026). Post-hoc corrections remove the mean and top principal components (Mu and Viswanath, 2018), whiten (Su et al., 2021), or down-weight frequent tokens (Arora et al., 2017). Timkey and van Schijndel (2021) apply them at every layer of four models and find that rogue dimensions obscure representational quality; our reading of the collapse as masked follows theirs, and we use ABTT as a probe. Whiten-

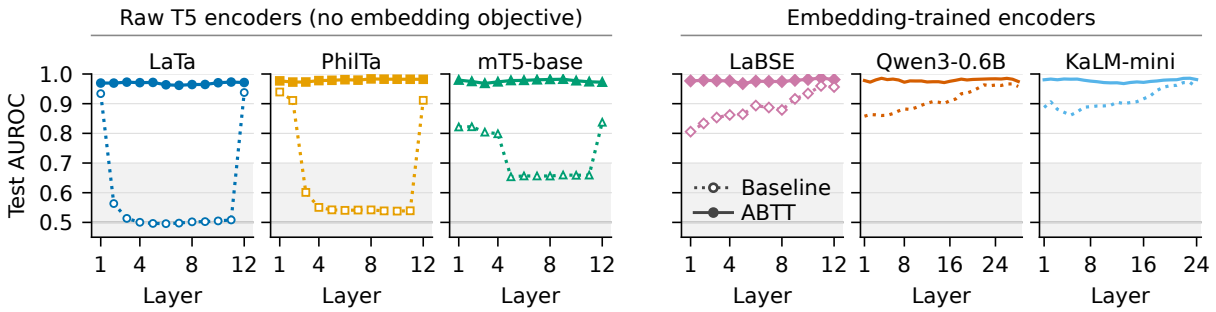



Figure 1: Test ranking AUROC at every layer of the six panel models, without correction (Baseline) and after ABTT, fit on training embeddings only. Gray band: collapsed (AUROC below 0.70); gray line: chance. The three raw T5-family panel models collapse at mid-depth; after ABTT every layer scores 0.962 to 0.987.

ing beats zeroing outlier dimensions in a multilingual encoder (Hämmerl et al., 2023). Centering alone gives embedding models gains under one point (Ren et al., 2026), and isotropy corrections stop helping after fine-tuning (Rajaee and Pilehvar, 2021). For routing, the nearest precedent is out-of-distribution detection in supervised classifiers, where low-variance components carry much of the signal (Kamoi and Kobayashi, 2020; Podolskiy et al., 2021), and nearest-neighbor scores depend on feature geometry (Sun et al., 2022). Lexical methods match or beat dense retrievers on Latin intertextuality (Schelb et al., 2026), while dense encoders reportedly help on paraphrastic reuse (de la Selle, 2026).

## 3 Experimental Setup

**Corpus and labels.** The testbed comes from CCL (Firey, 2009), an open-access edition of early medieval, mostly ninth-century, canon-law manuscripts. A *witness* is one manuscript's copy of a source text, kept as one file. Witnesses that copy the same source share a *directory*; the testbed holds 1,705 witnesses in 840 directories. The labels are expert-assigned by the project and checked by collation in its parallel display (Appendix A).

**Split and train-only rule.** We split the witnesses into 847 training and 858 test witnesses, with 565 and 596 same-source pairs; pairs form only within a split (Appendix B). The layer, routing threshold and ABTT are chosen or fit on the training split. The test split selects nothing.

**Tasks and metrics.** *Ranking* scores every pair of test witnesses by the cosine of their mean-pooled embeddings. We report AUROC with same-source pairs as positives. *Routing* is open-set: each test witness goes to an existing directory or to "new", and DirAcc@1 is the share of correct decisions (Section 7 gives the procedure).

| Model | Type | L | $\text{AUROC}_{\min}$ ($\ell$) | $\text{PC1}_{\max}$ | Coll. |
|---|---|---|---|---|---|
| *Raw T5 encoders, T5 v1.1 layout* | | | | | |
| LaTa | T5 | 12 | 0.496 (6) | 0.952 | 10 |
| PhilTa | T5 | 12 | 0.538 (10) | 0.858 | 9 |
| mT5-base | T5 | 12 | 0.654 (5) | 1.000 | 7 |
| T5-v1.1-base | T5 | 12 | 0.489 (2) | 0.975 | 10 |
| *Raw T5 encoders, original layout* | | | | | |
| T5-base | T5 | 12 | 0.816 (11) | 0.545 | 0 |
| T5-efficient-base | T5 | 12 | 0.736 (11) | 0.627 | 0 |
| *Raw encoder-only siblings* | | | | | |
| LaBERTa | Enc. | 12 | 0.826 (1) | 0.254 | 0 |
| PhilBERTa | Enc. | 12 | 0.883 (6) | 0.162 | 0 |
| *Embedding-trained* | | | | | |
| LaBSE | Enc. | 12 | 0.806 (1) | 0.524 | 0 |
| Qwen3-0.6B | Dec. | 28 | 0.858 (1) | 0.183 | 0 |
| KaLM-mini | Dec. | 24 | 0.861 (5) | 0.419 | 0 |

Table 1: The eleven main-text models, read as mean-pooled hidden states with no correction. Type: T5 encoder (T5), encoder-only (Enc.) or decoder (Dec.). L: blocks read. $\text{AUROC}_{\min}$: lowest test ranking AUROC over layers, at layer $\ell$. $\text{PC1}_{\max}$: peak top-PC share of training embeddings. Coll.: number of collapsed layers. Only the four raw T5 encoders with the T5 v1.1 layout collapse.

A *collapsed layer* has *baseline* (uncorrected) test AUROC below 0.70. The cutoff only describes layers; it selects no reported number. Two label-free statistics, computed on the training embeddings, describe each layer. The *mean pairwise cosine* is the usual anisotropy check (Ethayarajh, 2019). The *top-PC share* is the share of variance on the first principal component of the centered embeddings.

**Models.** Of the eleven models in Table 1, six form the *panel*, analyzed in full over 100 layers. Three are raw T5 encoders with no embedding objective: the Latin LaTa (Riemenschneider and Frank, 2023d), its Ancient Greek, Latin and English sibling PhilTa (Riemenschneider and Frank, 2023a), and mT5-base (Xue et al., 2021). Three are embedding-trained: LaBSE (Feng et al.,

2022), Qwen3-Embedding-0.6B (Qwen3-0.6B; Zhang et al., 2025) and KaLM-mini (Hu et al., 2025). The other five are raw: T5-efficient-base and four controls, each matched to another model in Section 5: the encoder-only siblings LaBERTa (Riemenschneider and Frank, 2023c) and PhilBERTa (Riemenschneider and Frank, 2023a), and the English T5 encoders T5-v1.1-base (Google Research, 2020) and the original T5-base (Raffel et al., 2020). We mean-pool the hidden states of every transformer block, dropping only padding and tokens that decode to no text, such as SentencePiece's bare word-boundary piece; special tokens such as </s> and punctuation are kept. Appendix J compares last-token pooling. Table 6 lists all thirteen models, adding Sentence-T5 (Ni et al., 2022) and SPhilBERTa (Riemenschneider and Frank, 2023b).

**Post-processing.** ABTT (Mu and Viswanath, 2018) subtracts the training mean and removes the top $D$ principal components of the centered training embeddings from every vector. We choose $D \in \{1, 2, 3, 5, 7, 10\}$ per layer by training DirAcc@1. ABTT is fit on training embeddings only and updates no parameter; we also call it *the projection*. We compare SIF frequency weighting (Arora et al., 2017) and reduced-rank whitening (Su et al., 2021) in Appendix K.

## 4 Collapse by Depth

In LaTa, PhilTa and mT5-base most middle layers rank near chance. Mean cosine gives no reliable warning, but top-PC share is high at every collapsed layer. ABTT restores ranking at every panel layer.

**LaTa, PhilTa and mT5-base fall to near chance at mid-depth.** Figure 1 gives baseline ranking AUROC at every layer of the six panel models. In LaTa, PhilTa and mT5-base, 26 of the 36 layers collapse, ranking at AUROC 0.50 to 0.66. They are layers 2–11 of LaTa, 3–11 of PhilTa and 5–11 of mT5-base. The lowest remaining panel layer ranks at 0.799 (mT5-base, layer 4), so any cutoff from 0.66 to 0.79 gives the same 26 layers. Training AUROC below 0.70 picks the same 26 layers. The first and last layers still retrieve; LaTa ranks at 0.934 at layer 1 and 0.938 at layer 12. The embedding-trained panel models never rank below 0.806.

**A train-only projection restores ranking at every layer of the six panel models.** ABTT (Section 3) lifts all 100 layers of the panel, the 26 collapsed

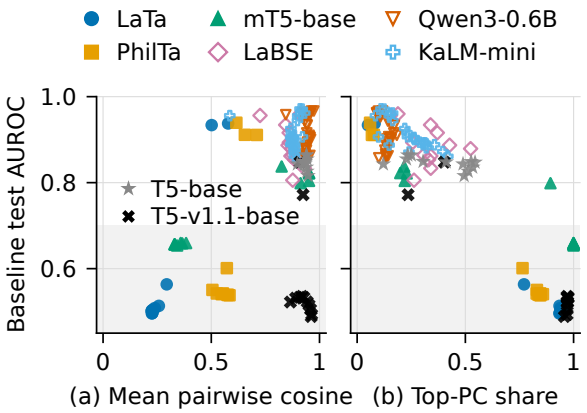



Figure 2: Baseline test AUROC against (a) mean pairwise cosine and (b) top-PC share of the training embeddings, one point per layer of the six panel models, T5-base and T5-v1.1-base. Filled: raw T5-family; open: embedding-trained. Gray band: collapsed (AUROC below 0.70). Collapsed layers span low and high mean cosine; all have a top-PC share of at least 0.76, as does one healthy layer.

layers included, to AUROC 0.962 to 0.987 (Figure 1, solid lines). Reduced-rank whitening (128 or 256 components), fit on the same training embeddings, ranks within 0.021 of ABTT at every panel layer (Appendix K). The narrow band holds for ranking only. Routing after ABTT still varies by layer, from 75.6 to 90.2 DirAcc@1 (Appendix I).

**Mean pairwise cosine gives no reliable warning.** Figure 2a plots training-set mean pairwise cosine against baseline test AUROC. Collapsed panel layers have a lower mean cosine (0.22 to 0.58) than any embedding-trained panel layer (0.59 to 0.97). But LaTa's healthy layers 1 and 12 sit in the collapsed range (0.50 and 0.58). T5-v1.1-base, which also collapses (Table 1; Section 5), sits at 0.87 to 0.96 at every layer, collapsed or not. Neither a high nor a low mean cosine marks a collapsed layer.

**Top-PC share is high at every collapsed layer of the pretrained models.** Top-PC share needs no labels either (Figure 2b). It is at least 0.76 at every collapsed layer of the panel and of T5-v1.1-base. In the three embedding-trained panel models it peaks at 0.524 (LaBSE). A high share is necessary for collapse in our pretrained models but not sufficient: of the 88 healthy layers in Figure 2, only mT5-base layer 4 reaches 0.76 (0.893, AUROC 0.799). Sentence-T5 (Ni et al., 2022), an embedding-trained T5 encoder, has a share of 0.76 or more at layers 2–9 (peak 0.94) yet ranks at 0.800 or above at every layer (Table 6). Among collapsed layers a higher share does not predict a lower AU-

ROC (Appendix C).

## 5 Collapse Across Models

Of the eleven models in Table 1, four collapse: LaTa, PhilTa, mT5-base and T5-v1.1-base. All four have the *T5 v1.1 layout*: a gated-GELU feed-forward (Shazeer, 2020) and untied input and output embeddings (Google Research, 2020) (Table 6). None of the other seven collapses at any layer. T5-base and T5-efficient-base keep a ReLU feed-forward and tied embeddings.

**Encoder-only siblings pretrained on the same corpora do not collapse.** LaBERTa and PhilBERTa share authors and pretraining corpora with LaTa and PhilTa (Riemenschneider and Frank, 2023d,c,a) and, like them, have no embedding objective. LaTa and PhilTa fall to 0.496 and 0.538 AUROC; the siblings never fall below 0.826 (LaBERTa) and 0.883 (PhilBERTa). Their top-PC share peaks at 0.254 and 0.162 (vs. 0.952 and 0.858 in LaTa and PhilTa). Shared pretraining text and a missing embedding objective do not produce the collapse on their own. Architecture, layout included, pretraining objective and tokenizer vary together within each pair, so the pairs cannot say which matters (Appendix D).

**T5-v1.1-base collapses; T5-base and T5-efficient-base, with the original layout, do not.** T5-v1.1-base and T5-base share their tokenizer, the span-corruption objective and C4 pretraining text (Raffel et al., 2020; Google Research, 2020). On Latin text, T5-v1.1-base falls to 0.489 at layer 2, and its layers 2 to 11 collapse. T5-base stays at 0.816 or above at every layer, with top-PC share peaking at 0.545. The two differ in layout and in pretraining mix: T5-base also saw supervised tasks (Raffel et al., 2020). To separate layout from that mix we add T5-efficient-base (Tay et al., 2022): T5-base's layout, but pretrained like T5-v1.1-base on C4 alone and without dropout, and on about 30 times fewer tokens than either. At that budget it does not collapse: its lowest AUROC (0.736) and its peak top-PC share (0.627) both fall at layer 11 (Table 8). On the four controls in Table 1, ABTT lifts every layer to AUROC 0.927 or above, the ten collapsed layers of T5-v1.1-base included, but eight of the 48 (seven in T5-v1.1-base) stay below the panel's band (0.962–0.987; Table 9). Among the six raw T5-family encoders we test, the collapse tracks the T5 v1.1 layout; we do not isolate which change is responsible.

**In three models with the T5 v1.1 layout, high top-PC share in the middle layers appears on English text too.** T5-v1.1-base was pretrained on English alone, so its high-share layers on Latin could reflect a language mismatch. We read mT5-base, PhilTa and T5-v1.1-base on 847 English passages of US court opinions (Harvard Law School Library Innovation Lab, 2024; Kandpal et al., 2025), paired with the Latin training witnesses and matched in mT5 token length. On English, in all three models, layers 4 to 11 have a top-PC share of at least 0.82 and the first and last layers stay at or below 0.34 (Table 10, Figure 7). At layers 2 to 11 of T5-v1.1-base, top-PC share is 0.931 to 0.950 on English and 0.958 to 0.975 on Latin. The check covers geometry only; without labels, it says nothing about English retrieval.

**In LaTa, contrastive fine-tuning leaves the middle layers collapsed.** In a small test, we fine-tune LaTa on 499 training pairs, adding the contrastive embedding objective it lacks (Appendix H). Fine-tuning lifts the last layer from 0.938 to 0.984 AUROC but leaves layers 2 to 11 at 0.50 to 0.57. Peak top-PC share barely moves (0.952 before, 0.945 after; Table 11). Hämmerl et al. (2023) report the same split: their sentence-tuned multilingual encoder keeps a large outlier dimension at a middle layer but none at its output layer.

## 6 Localizing the Collapse

In the collapsed layers about three directions mask the retrieval signal. Zeroing the ten largest coordinates does not remove this mask, and a few token types feed it in LaTa but not in mT5-base.

We run each test at the 26 collapsed layers, fit every intervention on training embeddings, and read test ranking AUROC. A layer counts as restored at AUROC 0.90 or above. Appendix E gives further tables and controls.

**Zeroing the ten largest coordinates restores no collapsed layer.** In pretrained transformers a few outlier coordinates dominate cosine similarity (Timkey and van Schijndel, 2021). Zeroing them improves pooled similarity in pretrained encoders (Luo et al., 2021), and whitening does better still (Hämmerl et al., 2023). In our encoders, zeroing up to ten coordinates, ranked on training embeddings by variance or by mean absolute value, restores none of the 26, and the best layer reaches 0.794

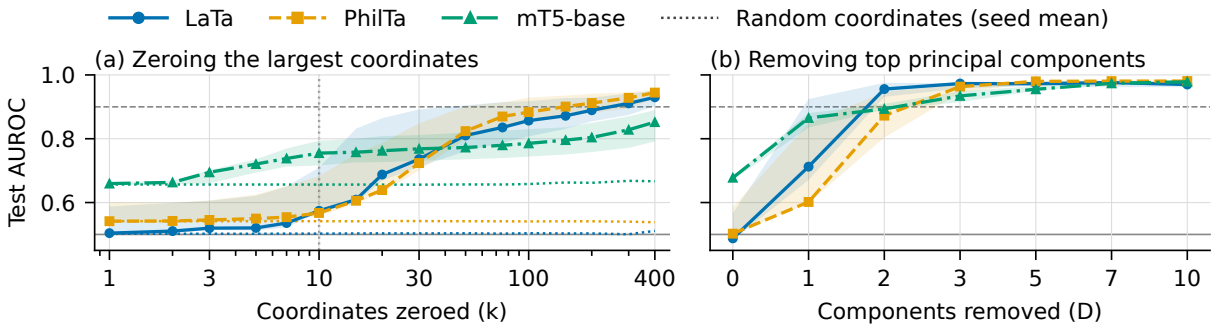



Figure 3: Test ranking AUROC at the 26 collapsed layers of LaTa (10 layers), PhilTa (9) and mT5-base (7). Lines and bands: median and range per model; dashed line: AUROC 0.90; gray line: chance. (a) Zeroing the $k$ coordinates of largest mean absolute value: no layer passes 0.90 by $k = 10$ (vertical line). (b) Removing the top $D$ principal components (ticks evenly spaced, $D$ values not): from $D = 3$ every layer passes 0.90.

(Figure 3a). The dominant direction loads mostly on these coordinates: the ten of highest variance carry a median 92% of its squared loading. Yet after zeroing them, its remainder still holds a median 72% of the remaining variance. Zeroing far more coordinates does restore LaTa and PhilTa. Ranked by variance, recovery starts at 50 coordinates, and their 19 collapsed layers all recover by 300 (by 400 under the mean-absolute-value ranking of Figure 3a). No mT5-base layer recovers even at 400. The dominant direction loads mostly on a few coordinates but acts beyond them, which extends the outlier-dimension account.

**Removing three directions restores every collapsed layer.** ABTT removes whole directions instead of coordinates, so we vary the number $D$ of principal components it removes (Figure 3b). Centering alone ($D = 0$) does not help: the median AUROC over the collapsed layers falls from 0.541 to 0.503. At these layers the first component holds a median 94% of the centered training variance. It is not mainly a passage-length direction: $|\rho| \leq 0.44$ with log token count at all 26 layers. Removing it alone recovers a median 45% of the AUROC gain from removing ten components, and 80% or more at only one of the 26 layers. Removing three components brings all 26 layers to 0.914 or above. mT5-base keeps improving up to ten components (median 0.934 at three, 0.978 at ten). The ten removed components rank at 0.489, the next ten at 0.919, and the retained subspace at 0.977 (Table 14). At these layers the signal lies outside the removed directions.

**In LaTa, three token types feed the removed directions.** A pooled vector is the mean of its token vectors, so a few token types with large values along the removed directions could dominate a witness's scores on those directions. We rank token types by their share of the variance of training-witness scores on the top three components, then pool without the top $m$ types, for $m$ up to 100. We ask whether some $m$ restores the collapsed layers. In LaTa, $m = 3$ does. Pooling without the comma, the period and `</s>` drops 11% of the test tokens and restores all 10 collapsed layers (median 0.919). Random token types of the same training share leave the median at 0.500. The tokens are carriers, not a cause, since LaBERTa also pools commas, periods and `</s>` and does not collapse (Section 5). PhilTa needs more types: dropping its top 30, about a third of its tokens, restores all 9 of its collapsed layers on test passages (settings chosen per layer on training AUROC restore every collapsed layer of both models, medians 0.926 and 0.928).

**In mT5-base, dropping up to its top 100 token types restores no collapsed layer.** Dropping up to 100 token types, ranked the same way, restores none of its 7 collapsed layers, the best reaching 0.748. mT5-base also resists coordinate zeroing, yet removing three to ten directions repairs it. Its mask sits neither in a few coordinates nor in a few token types, and we have no token-level account.

## 7 Routing and Limits

At their train-selected layers after ABTT, the six frozen encoders route about equally well, but none beats a character n-gram reference.

Of the 858 test witnesses, 535 have a same-source partner in test (*existing*) and 323 do not (*new*). Each test directory scores a witness by the highest cosine between the witness and any other member. A pseudo-directory `NEW` scores a fixed threshold $\tau$, chosen to maximize training F1 at separating same-source from different-source pairs.

| | Ranking AUROC | | Routing DirAcc@1 | |
|---|---|---|---|---|
| Model | Base | ABTT | Base | ABTT |
| LaTa | 0.938 [.910,.963] | 0.971$^{\downarrow}$ [.948,.991] | 72.1 [68.4,75.8] | 86.1$^{\downarrow}$ [83.2,88.9] |
| PhilTa | 0.939 [.918,.958] | 0.983 [.969,.993] | 69.3 [65.2,73.5] | 88.3 [85.6,90.9] |
| mT5-base | 0.838 [.810,.866] | 0.975 [.954,.992] | 46.6 [42.5,51.0] | 88.5 [85.7,91.2] |
| LaBSE | 0.956 [.932,.977] | 0.987 [.977,.995] | 81.5 [78.1,84.7] | 88.5 [85.8,90.9] |
| Qwen3-0.6B | 0.966 [.950,.981] | 0.973$^{\downarrow}$ [.948,.990] | 80.3 [76.9,83.6] | 89.4 [86.7,92.0] |
| KaLM-mini | 0.972 [.955,.986] | 0.981 [.963,.993] | 85.9 [82.9,88.7] | 89.4 [86.7,91.9] |
| *Spread* | 0.134 | 0.016 | 39.3 | 3.3 |
| *Fine-tuned; Base = fine-tuned, ABTT = fine-tuned + ABTT* | | | | |
| LaTa | 0.984 | 0.970 | 81.6 | 85.2 |
| Qwen3-0.6B | 0.996 | 0.994 | 90.8 | 91.4 |
| KaLM-mini | 0.997 | 0.994 | 91.7 | 92.5 |
| Char. n-grams | 0.987 [.976,.996] | | 89.9 [87.3,92.3] | |

Table 2: Test ranking (AUROC) and routing (DirAcc@1, %), uncorrected (Base) and after ABTT, each at its train-selected layer (Table 21). Fine-tuned: contrastively fine-tuned encoders, intervals in Table 16. Char. n-grams: character 3–5-gram TF-IDF. Brackets: 95% bootstrap intervals over the 514 test directories. Spread: range over the six encoders. ↓: below the n-gram reference, paired interval excluding zero (Table 17); overlapping brackets do not imply a tie. ABTT cuts the routing spread from 39.3 to 3.3 points; no frozen encoder beats the reference.

The witness goes to the highest-scoring directory. DirAcc@1 counts a decision as correct if it names the labeled directory, or `NEW` for a new witness. Assignment accuracy scores only the existing-versus-new decision; always answering existing scores 62.4%. Each headline number reads a model at its *train-selected* layer: the one of highest training AUROC for ranking, or of highest training DirAcc@1 for routing (Table 21). The task is lexically solvable by design, since the witnesses of one source copy its wording. Our *character n-gram reference*, TF-IDF over character 3–5-grams fit on training witnesses, reaches 0.987 AUROC and 89.9 DirAcc@1 (Table 20).

**At train-selected layers, routing after the projection converges across the six encoders.** At the train-selected layers of Table 2, the DirAcc@1 spread across the six encoders falls from 39.3 points without correction (95% CI 34.6 to 43.7) to 3.3 after ABTT (1.8 to 6.0). Uncorrected, they run from 46.6 (mT5-base) to 85.9 (KaLM-mini); after ABTT, from 86.1 to 89.4. Every encoder's routing gain has a bootstrap interval excluding zero, from 3.5 points for KaLM-mini (1.0 to 6.0) to 41.8 for mT5-base (Table 17).

**For the embedding-trained encoders, the routing gain does not depend on the threshold.** LaBSE, Qwen3-0.6B and KaLM-mini already rank well without correction (AUROC 0.956 to 0.972), so their routing gain cannot come from a ranking repair. ABTT still raises their DirAcc@1 by 7.0, 9.1 and 3.5 points. Because the gain could come from a better-placed threshold $\tau$, we also score the existing-versus-new decision without one, as the AUROC of each test witness's best-match cosine, existing against new. ABTT raises it by 0.037 for LaBSE, 0.055 for Qwen3-0.6B and 0.027 for KaLM-mini, and every paired interval excludes zero (Appendix G; per-cell values in Table 18). ABTT's gain in assignment accuracy also holds at each setting's best test threshold and under a finer threshold grid (Appendix F). For these three encoders, centering alone ($D = 0$) scores 12 to 26 points lower in assignment accuracy than ABTT at the same layer (Table 17). We do not test what in the removed components produces this gain.

**No frozen encoder at its train-selected layer beats a character n-gram reference.** The best ABTT results in Table 2 tie the reference: LaBSE ranks at 0.987, and Qwen3-0.6B and KaLM-mini route at 89.4, with every difference interval including zero. Two ranking cells fall below it with intervals excluding zero, LaTa and Qwen3-0.6B (by 0.016 and 0.014 AUROC); mT5-base falls 0.012 below, its interval touching zero (0.000 to 0.027). LaTa's routing cell falls 3.7 points below it (Table 17). Our claims thus concern where the signal sits in the encoders and what masks it.

**Fine-tuning Qwen3-0.6B and KaLM-mini adds about two routing points over the projection.** As a supervised reference, we fine-tune LaTa, Qwen3-0.6B and KaLM-mini contrastively on 499 training pairs (Appendix H). For routing, fine-tuning plus ABTT beats zero-shot ABTT by 1.8 points for Qwen3-0.6B (92.3 against 90.5) and by 2.1 for KaLM-mini (93.0 against 90.9); on LaTa the two tie at 87.7. These are means over five re-seedings, each a new random draw of which test witnesses are queries and which are the references they are routed against (Table 19). In the single-split cells of Table 2 the gains are 2.0 and 3.1 points (only KaLM-mini's interval excludes zero), and

LaTa falls 0.9. These references are flattered: 206 of the 535 existing test witnesses (38.5%) sit in a directory that supplied a training pair. Fine-tuned Qwen3-0.6B and KaLM-mini rank above the n-gram reference in point estimate (0.996 and 0.997 AUROC). After fine-tuning, ABTT no longer helps ranking: AUROC falls slightly in point estimate in all three models (by 0.002 to 0.014; every interval includes zero), in line with Rajaee and Pilehvar (2021).

## 8 Discussion

Before reading an intermediate layer of a frozen encoder, fit a projection on training embeddings. In the six panel models it scored above the baseline in point estimate at all 100 layers in ranking and routing (Appendix I), which makes it cheap insurance. Routing after it still varies by layer, so the layer still matters. To screen uncorrected states, check top-PC share, not mean pairwise cosine. In our models mean cosine gave no reliable warning in either direction, while every collapsed layer of the pretrained models had a top-PC share of at least 0.76 (Section 4). The 0.76 is the lowest share we observed at a collapsed layer of a pretrained checkpoint, not a cutoff validated on held-out models. A high share is necessary for collapse here but not sufficient, and does not order layers inside the collapsed regime. Choosing among layers still needs a train-only task signal, such as training AUROC.

Why models with the T5 v1.1 layout collapse remains open. Gated feed-forward blocks are reported to amplify outlier activations in decoder language models (Fishman et al., 2025; Yang et al., 2025), but the original T5 also carries large outliers (Zhao et al., 2025). Our localization names carriers in LaTa but no cause. Concurrently, Mikkelsen (2026) reports a mid-depth retrieval trough on clinical text; retrieval-trained encoders recover in the last layers, and encoders without a retrieval objective recover only partly. In our models, by contrast, neither the embedding-trained encoders nor the encoder-only siblings collapse, and it tests no T5 model.

For the editor of a text collection, ABTT makes the choice among these six encoders matter little (Section 7; maps in Appendix L). The existing-versus-new decision is still wrong for about one test witness in ten (assignment accuracy 88.5 to 91.7; Table 16). The final call stays with the scholar. Because the task is lexically solvable, a practical tool can start from character n-grams; dense encoders are reported to help where reuse is paraphrastic (de la Selle, 2026), a case we do not test.

## Limitations

Four factors are uncontrolled. (i) We use one labeled corpus in one language and genre, early medieval Latin canon law; its 1,705 witnesses are many for this field but few by current NLP standards. (ii) Within each sibling pair, architecture, pretraining objective and tokenizer vary together, so the pairs do not say which of them matters. That LaTa and LaBERTa share authors and pretraining corpora rests on their model cards (Riemenschneider and Frank, 2023d,c). (iii) The T5 v1.1 changes (gated-GELU feed-forward with a narrower inner width, untied embeddings) vary together, so we do not say which of them matters. T5-efficient-base has the original layout without T5-base's supervised pretraining, but it is a single checkpoint pretrained on about 30 times fewer tokens; it separates layout from pretraining mix only at that budget. (iv) The token ablation accounts for the masking directions in LaTa with three token types and in PhilTa only with 30, about a third of its tokens; in mT5-base dropping up to its top 100 token types restores no collapsed layer, so its mask has no token-level account.

The scope of three results is narrow. We claim the narrow band for the six-model panel only: on the four controls and T5-efficient-base ABTT lifts every layer to AUROC 0.927 or above, but nine of their 60 layers, seven of them in T5-v1.1-base, stay below the panel's band (0.962 to 0.987). The English check measures geometry only; we have no labels for English retrieval. The task is lexically solvable by design, and a character n-gram reference ties the best encoders after ABTT, so we do not claim that encoders beat surface overlap on this corpus.

The fine-tuning test is small (499 pairs, at most eight epochs), and its references are flattered by directory overlap with the training pairs (Section 7). Our intervals resample test directories with the layer, threshold and ABTT fit held fixed; they cover test sampling and not the choice of split. ABTT needs enough text to estimate principal components, so smaller archives need separate study.

The labels are expert-assigned, but by one project. Its director or supervised graduate students assigned the keys, and verification is by collation in the project's parallel display, not by independent double annotation. No inter-annotator agreement figure therefore exists for this testbed. Collation needs at least two witnesses of a key, and 545 of the 840 directories hold only one in this testbed.

## Ethical Considerations

The testbed holds early medieval legal texts, transcribed from manuscripts and early printings, and contains no personal data. The Carolingian Canon Law project publishes its material under a Creative Commons Attribution-NonCommercial-ShareAlike 4.0 license (CCL, 2026a). The project has confirmed in writing that the license covers the transcriptions, annotations and markup in the exports we use and that, in the view of the project's Advisory Board, attribution to the project suffices; the project's interface credits the transcriber and proofreader of each unit. The project's corpus keeps growing and being corrected, so the testbed is a dated snapshot (version 1), fixed by a sha256 manifest of its 1,705 labeled files, and every number in this paper comes from it. We will release it under the same license, in coordination with the project's periodic versioned exports, and later versions will follow those exports. The intended use is research on retrieval and on assisted source identification: the system ranks candidates for an expert to judge and does not identify sources on its own. The English control sample consists of public-domain US court opinions from the Caselaw Access Project (Harvard Law School Library Innovation Lab, 2024), which name litigants; we use it only for label-free geometry statistics and do not release it. We used AI assistants for writing and checking code; the authors verified all code and numbers.

## References


Sanjeev Arora, Yingyu Liang, and Tengyu Ma. 2017. A simple but tough-to-beat baseline for sentence embeddings. In *Proceedings of the 5th International Conference on Learning Representations (ICLR)*.

Yelysei Bondarenko, Markus Nagel, and Tijmen Blankevoort. 2023. Quantizable transformers: Removing outliers by helping attention heads do nothing. In *Advances in Neural Information Processing Systems 36 (NeurIPS)*, pages 75067–75096.

CCL. 2026a. About the Project. Carolingian Canon Law Project home page, University of Kentucky. Accessed 2026-09-14.

CCL. 2026b. Submission Desks. Carolingian Canon Law Project documentation page, University of Kentucky. Accessed 2026-09-14.

Théotime de la Selle. 2026. From transcription to semantic corpus analysis: Unsupervised learning of sentence representations for ancient languages. *arXiv preprint arXiv:2607.24542*.

Melodie H. Eichbauer. 2014. Review of the Carolingian Canon Law Project, directed by Abigail Firey. *Digital Philology: A Journal of Medieval Cultures*, 3(1):165–168.

Kawin Ethayarajh. 2019. How contextual are contextualized word representations? Comparing the geometry of BERT, ELMo, and GPT-2 embeddings. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 55–65. Association for Computational Linguistics.

Fangxiaoyu Feng, Yinfei Yang, Daniel Cer, Naveen Arivazhagan, and Wei Wang. 2022. Language-agnostic BERT sentence embedding. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 878–891. Association for Computational Linguistics.

Abigail Firey. 2009. The Carolingian Canon Law Project. Collaboratory for Research in Computing for the Humanities, University of Kentucky. Open-access digital edition, ongoing. Accessed 2026-09-14.

Maxim Fishman, Brian Chmiel, Ron Banner, and Daniel Soudry. 2025. Scaling FP8 training to trillion-token LLMs. In *Proceedings of the 13th International Conference on Learning Representations (ICLR)*.

Alejandro Fuster Baggetto and Victor Fresno. 2022. Is anisotropy really the cause of BERT embeddings not being semantic? In *Findings of the Association for Computational Linguistics: EMNLP 2022*, pages 4271–4281, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.

Jun Gao, Di He, Xu Tan, Tao Qin, Liwei Wang, and Tie-Yan Liu. 2019. Representation degeneration problem in training natural language generation models. In *Proceedings of the 7th International Conference on Learning Representations (ICLR)*.

Nathan Godey, Éric de la Clergerie, and Benoît Sagot. 2024. Anisotropy is inherent to self-attention in transformers. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 35–48. Association for Computational Linguistics.

Google Research. 2020. Experimental T5 pre-trained model checkpoints (T5.1.1). T5 code repository on GitHub. Accessed 2026-10-02.

Katharina Hämmerl, Alina Fastowski, Jindřich Libovický, and Alexander Fraser. 2023. Exploring anisotropy and outliers in multilingual language models for cross-lingual semantic sentence similarity. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 7023–7037. Association for Computational Linguistics.

Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J. Smith, Robert Kern, Matti Picus, Stephan Hoyer, Marten H. van Kerkwijk, Matthew Brett, Allan Haldane, Jaime Fernández del Río, Mark Wiebe, Pearu Peterson, and 7 others. 2020. Array programming with NumPy. *Nature*, 585(7825):357–362.

Harvard Law School Library Innovation Lab. 2024. Caselaw access project. `https://case.law/`. Full public release, March 2024.

Xinshuo Hu, Zifei Shan, Xinping Zhao, Zetian Sun, Zhenyu Liu, Dongfang Li, Shaolin Ye, Xinyuan Wei, Qian Chen, Baotian Hu, Haofen Wang, Jun Yu, and Min Zhang. 2025. KaLM-embedding: Superior training data brings a stronger embedding model. *arXiv preprint arXiv:2501.01028*.

Junjie Huang, Duyu Tang, Wanjun Zhong, Shuai Lu, Linjun Shou, Ming Gong, Daxin Jiang, and Nan Duan. 2021. WhiteningBERT: An easy unsupervised sentence embedding approach. In *Findings of the Association for Computational Linguistics: EMNLP 2021*, pages 238–244. Association for Computational Linguistics.

Ryo Kamoi and Kei Kobayashi. 2020. Why is the Mahalanobis distance effective for anomaly detection? *arXiv preprint arXiv:2003.00402*.

Nikhil Kandpal, Brian Lester, Colin Raffel, Sebastian Majstorovic, Stella Biderman, Baber Abbasi, Luca Soldaini, Enrico Shippole, A. Feder Cooper, Aviya Skowron, and 1 others. 2025. The common pile v0.1: An 8TB dataset of public domain and openly licensed text. *arXiv preprint arXiv:2506.05209*.

Olga Kovaleva, Saurabh Kulshreshtha, Anna Rogers, and Anna Rumshisky. 2021. BERT busters: Outlier dimensions that disrupt transformers. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, pages 3392–3405. Association for Computational Linguistics.

Atharva Kulkarni, Jacob Mitchell Springer, Arjun Subramonian, and Swabha Swayamdipta. 2026. Disentangling geometry, performance, and training in language models. *arXiv preprint arXiv:2602.20433*.

Ziyang Luo, Artur Kulmizev, and Xiaoxi Mao. 2021. Positional artefacts propagate through masked language model embeddings. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 5312–5327, Online. Association for Computational Linguistics.

Enrique Manjavacas, Brian Long, and Mike Kestemont. 2019. On the feasibility of automated detection of allusive text reuse. In *Proceedings of the 3rd Joint SIGHUM Workshop on Computational Linguistics for Cultural Heritage, Social Sciences, Humanities and Literature*, pages 104–114. Association for Computational Linguistics.

Leland McInnes, John Healy, and James Melville. 2018. UMAP: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426*.

Yngve Mikkelsen. 2026. Effects of model choice, corpus context, and post hoc correction on layer-level embedding degradation in clinical document retrieval: Experimental study. *JMIR Medical Informatics*, 14:e99639.

Jiaqi Mu and Pramod Viswanath. 2018. All-but-the-Top: Simple and effective postprocessing for word representations. In *Proceedings of the 6th International Conference on Learning Representations (ICLR)*.

Vivi Nastase and Paola Merlo. 2025. Testing the assumptions about the geometry of sentence embedding spaces: The cosine measure need not apply. *arXiv preprint arXiv:2509.01606*.

Jianmo Ni, Gustavo Hernandez Abrego, Noah Constant, Ji Ma, Keith Hall, Daniel Cer, and Yinfei Yang. 2022. Sentence-T5: Scalable sentence encoders from pretrained text-to-text models. In *Findings of the Association for Computational Linguistics: ACL 2022*, pages 1864–1874. Association for Computational Linguistics.

V. S. Raghu Parupudi. 2026. Anisotropy decides cosine vs. rank metrics for text embeddings. *arXiv preprint arXiv:2606.29571*.

Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, Alban Desmaison, Andreas Kopf, Edward Yang, Zachary DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, and 2 others. 2019. PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems*, volume 32, pages 8024–8035. Curran Associates, Inc.

Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. 2011. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12(85):2825–2830.

Alexander Podolskiy, Dmitry Lipin, Andrey Bout, Ekaterina Artemova, and Irina Piontkovskaya. 2021. Revisiting Mahalanobis distance for transformer-based out-of-domain detection. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 13675–13682.

Giovanni Puccetti, Anna Rogers, Aleksandr Drozd, and Felice Dell'Orletta. 2022. Outlier dimensions that disrupt transformers are driven by frequency. In *Findings of the Association for Computational Linguistics: EMNLP 2022*, pages 1286–1304. Association for Computational Linguistics.

Miloš Radovanović, Alexandros Nanopoulos, and Mirjana Ivanović. 2010. Hubs in space: Popular nearest neighbors in high-dimensional data. *Journal of Machine Learning Research*, 11:2487–2531.

Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. 2020. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67.

Sara Rajaee and Mohammad Taher Pilehvar. 2021. How does fine-tuning affect the geometry of embedding space: A case study on isotropy. In *Findings of the Association for Computational Linguistics: EMNLP 2021*, pages 3042–3049. Association for Computational Linguistics.

Sara Rajaee and Mohammad Taher Pilehvar. 2022. An isotropy analysis in the multilingual BERT embedding space. In *Findings of the Association for Computational Linguistics: ACL 2022*, pages 1309–1316. Association for Computational Linguistics.

Xingyu Ren, Youran Sun, and Haoyu Liang. 2026. Correcting mean bias in text embeddings: A refined renormalization with training-free improvements on MMTEB. *arXiv preprint arXiv:2511.11041*. Version 2.

Frederick Riemenschneider and Anette Frank. 2023a. Exploring large language models for classical philology. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15181–15199. Association for Computational Linguistics.

Frederick Riemenschneider and Anette Frank. 2023b. Graecia capta ferum victorem cepit. Detecting Latin allusions to Ancient Greek literature. In *Proceedings of the Ancient Language Processing Workshop*, pages 30–38, Varna, Bulgaria. INCOMA Ltd., Shoumen, Bulgaria.

Frederick Riemenschneider and Anette Frank. 2023c. LaBerta. Hugging Face model card, `bowphs/LaBerta`. Accessed 2026-10-02.

Frederick Riemenschneider and Anette Frank. 2023d. LaTa. Hugging Face model card, `bowphs/LaTa`. Accessed 2026-10-02.

Hammad Rizwan, Muhammad Umair Haider, Nishant Subramani, Mona T. Diab, A. B. Siddique, and Hassan Sajjad. 2026. On the persistent effects of lexicality in large language models. *arXiv preprint arXiv:2606.02750*.

Florian Rottach, William Rudman, Bastian Rieck, Harrisen Scells, and Carsten Eickhoff. 2025. From topology to retrieval: Decoding embedding spaces with unified signatures. *arXiv preprint arXiv:2511.22150*.

Olivier Roy and Martin Vetterli. 2007. The effective rank: A measure of effective dimensionality. In *Proceedings of the 15th European Signal Processing Conference (EUSIPCO)*, pages 606–610. IEEE.

William Rudman, Catherine Chen, and Carsten Eickhoff. 2023. Outlier dimensions encode task specific knowledge. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 14596–14605, Singapore. Association for Computational Linguistics.

William Rudman, Nate Gillman, Taylor Rayne, and Carsten Eickhoff. 2022. IsoScore: Measuring the uniformity of embedding space utilization. In *Findings of the Association for Computational Linguistics: ACL 2022*, pages 3325–3339. Association for Computational Linguistics.

Julian Schelb, Michael Wittweiler, Marie Revellio, Barbara Feichtinger, and Andreas Spitz. 2026. Loci similes: A benchmark for extracting intertextualities in Latin literature. *arXiv preprint arXiv:2601.07533*.

Noam Shazeer. 2020. GLU variants improve transformer. *arXiv preprint arXiv:2002.05202*.

Oscar Skean, Md Rifat Arefin, Dan Zhao, Niket Nikul Patel, Jalal Naghiyev, Yann LeCun, and Ravid Shwartz-Ziv. 2025. Layer by layer: Uncovering hidden representations in language models. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, pages 55854–55875. PMLR.

Jianlin Su, Jiarun Cao, Weijie Liu, and Yangyiwen Ou. 2021. Whitening sentence representations for better semantics and faster retrieval. *arXiv preprint arXiv:2103.15316*.

Mingjie Sun, Xinlei Chen, J. Zico Kolter, and Zhuang Liu. 2024. Massive activations in large language models. In *Proceedings of the First Conference on Language Modeling (COLM)*.

Yiyou Sun, Yifei Ming, Xiaojin Zhu, and Yixuan Li. 2022. Out-of-distribution detection with deep nearest neighbors. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, volume 162 of *Proceedings of Machine Learning Research*, pages 20827–20840. PMLR.

Yi Tay, Mostafa Dehghani, Jinfeng Rao, William Fedus, Samira Abnar, Hyung Won Chung, Sharan Narang, Dani Yogatama, Ashish Vaswani, and Donald Metzler. 2022. Scale efficiently: Insights from pre-training and fine-tuning transformers. In *International Conference on Learning Representations*.

William Timkey and Marten van Schijndel. 2021. All bark and no bite: Rogue dimensions in transformer language models obscure representational quality. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 4527–4546. Association for Computational Linguistics.

Laurens van der Maaten and Geoffrey Hinton. 2008. Visualizing data using t-SNE. *Journal of Machine Learning Research*, 9(86):2579–2605.

Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Remi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, and 3 others. 2020. Transformers: State-of-the-art natural language processing. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 38–45. Association for Computational Linguistics.

Linting Xue, Noah Constant, Adam Roberts, Mihir Kale, Rami Al-Rfou, Aditya Siddhant, Aditya Barua, and Colin Raffel. 2021. mT5: A massively multilingual pre-trained text-to-text transformer. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 483–498. Association for Computational Linguistics.

Jaewoo Yang, Hayun Kim, Junyung Ji, and Younghoon Kim. 2025. Mitigating quantization errors due to activation spikes in gated linear unit-based large language models. *Future Internet*, 17(4):185.

Yanzhao Zhang, Mingxin Li, Dingkun Long, Xin Zhang, Huan Lin, Baosong Yang, Pengjun Xie, An Yang, Dayiheng Liu, Junyang Lin, Fei Huang, and Jingren Zhou. 2025. Qwen3 embedding: Advancing text embedding and reranking through foundation models. *arXiv preprint arXiv:2506.05176*.

Tianyang Zhao, Kunwar Yashraj Singh, Srikar Appalaraju, Peng Tang, Ying Nian Wu, and Li Erran Li. 2025. On the analysis and distillation of emergent outlier properties in pre-trained language models. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 8475–8507. Association for Computational Linguistics.

## A Corpus Provenance and Derivation

This appendix records where the testbed text comes from and how our files were derived from it, in the

detail that Section 3 leaves out.

**The project.** CCL (Firey, 2009) is an open-access digital edition founded in 2009; Eichbauer (2014) reviews it.[1] Its texts are early medieval canon-law manuscripts, most of them ninth-century and never edited to modern standards. Each manuscript is identified by its siglum (Place-Library-Shelfmark), and a witness is one manuscript's copy of a source text, transcribed as one unit (one canon or excerpt).

**Transcription.** Transcription is manual, from manuscript images or, for some texts, eighteenth-century printings, with one exception: HTR has been attempted on one manuscript (Paris, BnF, lat. 1454; 142 of the 1,705 witnesses), and its transcription is still in proofreading. Before 2019 transcribers worked from purchased microfilm into word-processed files, which CCL contributors converted to TEI-P5 (the Text Encoding Initiative's XML standard for scholarly editions). Since 2019 they transcribe in the CCL Transcription Desk beside a IIIF image of the folio (a zoomable web-served scan of the manuscript leaf); an approved proofreader on the Proofreading Desk then checks the text against the image to the letter and adds TEI markup (CCL, 2026b). That proofreading stage belongs to the post-2019 workflow. The interface credits the transcriber and proofreader of each unit (CCL, 2026a).

**Identifiers.** The project has two identifier systems, both designed at its launch in 2009 by its director in consultation with a mathematician, a software developer and a second scholar of early medieval canon law. The first is a location identifier. Every manuscript is a unique object, so every unit has exactly one place in one manuscript, and the identifier records it as the manuscript siglum, the folium, and the unit's position on that folium. Our file names are these identifiers. The second is the source key. Many units in different manuscripts copy the same prescription issued by the same authority, and the key names that common source. It encodes the type of source (a church council, a papal decretal, a royal capitulary, a theologian, a penitential), the point of origin (a particular council, pope or author), the date of origin, and the item's reference in modern scholarship, such as a canon, letter or register number. The key is designed so that a researcher can find the modern authoritative edition of the source where one exists.

**Labels.** CCL keys a unit to its original source, such as a fourth-century council or a papal decretal, once that source is identified, and our directories reuse that key as supplied. In a label, a four-letter code carries the type of source and its point of origin, followed by the year and the canon or chapter number, as in `CTOU.567.16`; some labels carry a trailing annotation, as in `CTOU.567.16 (15)` or `CARL.501?.18`. A label counts as key-named when it begins with a four-letter code, a period, a year and a period, whatever follows; by this rule 689 of the 840 directory labels are source keys. The other 151 are modern edition citations, the Apostolic Canons (`Can.apost.`$N$), references to other collections such as the Capitula Martini, biblical references, and a handful of unidentified sources; each unidentified source is a singleton directory and contributes no positive pairs.

**Assignment and verification.** Keys were assigned by the project's director, a specialist in early medieval canon law, or by graduate students working under the director's supervision, who had enough Latin to check a manuscript version against the modern authoritative version of the source. Verification, and occasional correction, runs through the project's parallel display, which sets side by side all units that share a key and shows how far their wording agrees. A misidentified unit stands out there as a text that cannot be a version of the others. This is verification by collation (see Limitations). The project supplied two label corrections during this study, and both are logged with the frozen testbed. We assigned no label ourselves and changed only those two. What remains unidentified is large: across the several hundred manuscripts that contain the project's texts, thousands of units still carry no key, and they are the queries that routing is meant to serve.

**Derivation.** We hold the witnesses as plain text derived from CCL TEI-P5 exports. The rule is one file per unit: the directory is the source key from the unit's `scholSource` division, which is stripped from the text so that no file carries its own label; the text is the manuscript rubric followed by the canon text, with inline markup flattened

[1] Founding year, pre-2019 workflow, proofreading standard, the one manuscript transcribed by handwritten text recognition (HTR), the design of the identifiers, the assignment and verification of keys, and the extent of unidentified material: unpublished documentation and correspondence supplied by the project, 2026.

(editorial notes kept, deletions dropped) and layout whitespace collapsed; a unit without a source key receives no directory. The rule is scripted. On a current export of one manuscript (Paris, BnF, lat. 2123), it reproduces all 279 units present in both the export and the corpus character for character after whitespace normalization, and places every one in the same directory.

## B Experimental Setup Details

This appendix gives the parts of the setup that Sections 3 and 7 leave out: how the split treats directories of each size, how KaLM-mini attends, the further layer statistics, the SIF weighting, and how the lexical reference fares by degree of overlap.

**Split.** Singleton directories (545 of 840) are divided file by file between the two splits. Doubletons go whole to one side. Larger directories are divided internally, so that every size class yields training positives. Pairs form only within a split.

**KaLM-mini attention.** KaLM-mini, a decoder, attends bidirectionally, as its checkpoint's own modeling code specifies.

**Further layer statistics.** Let $p_i = \lambda_i / \sum_j \lambda_j$ for the covariance eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots$ of the centered training embeddings; the top-PC share is $p_1$. The *effective rank* is $\exp(-\sum_i p_i \log p_i)$, the entropy effective rank of Roy and Vetterli (2007) applied to the covariance eigenvalues. It is 1 when one direction holds all the variance. The *cosine gap* $g$ is the mean same-directory test cosine minus the mean different-directory test cosine. The *standardized gap* $g/s$ divides it by the standard deviation $s$ of all pairwise test cosines, which puts models with different cosine spreads on one scale. The two gaps need labels; the effective rank, like the top-PC share and the mean pairwise cosine, does not. Our mean pairwise cosine averages over all pairs of pooled passages, an adaptation of the word-level statistic of Ethayarajh (2019), which uses random pairs.

**SIF.** SIF (Arora et al., 2017) reweights each token by $a/(a + p(w))$ before pooling, with $a = 10^{-3}$ and $p(w)$ estimated on training texts. It gives special tokens, such as the end-of-sequence token </s>, zero weight. We use only this weighting step of SIF, not its common-component removal. SIF+ABTT applies both; ABTT-only marks ABTT alone where we contrast the two.

**Software.** We extract hidden states with PyTorch (Paszke et al., 2019) and Hugging Face Transformers (Wolf et al., 2020), fit whitening and the character TF-IDF reference with scikit-learn (Pedregosa et al., 2011), and compute principal components and the remaining statistics with NumPy (Harris et al., 2020). BM25 is our own implementation.

**Lexical reference by overlap.** We cut the positive test pairs into terciles of n-gram overlap, using the TF-IDF score itself. In every tercile, the character n-gram reference ranks a witness's partner first at least as often as every baseline and ABTT encoder configuration at its train-selected layer. Table 20 gives all three lexical scorers.

## C Collapse by Depth: Further Diagnostics

This appendix adds detail to the four findings of Section 4. It gives the geometry of each panel model over its layers, the correlations behind the statement that top-PC share marks the collapsed regime without ordering layers inside it, two label-based gaps, the spread of cosines at the collapsed layers, the effect of SIF there, and the score distributions before and after ABTT. Every statistic is computed on baseline mean-pooled hidden states unless a paragraph says otherwise.

**Geometry by model.** Table 3 summarizes each panel model over its layers. At the 26 collapsed layers, effective rank lies between 1.0 and 4.6; in the embedding-trained models it ranges from 16 to 150. At mT5-base layers 5–11 one direction holds all the variance to three decimals. The table reads mean cosine on test passages, while Figure 2 reads it on training passages. On test passages the collapsed layers span 0.23 to 0.62, so the PhilTa layers overlap the bottom of the embedding-trained range. ABTT with ten components, fit on training vectors, removes the concentration on held-out passages too: at the collapsed layers, test top-PC share falls to 0.03–0.05 and effective rank rises to 148–173. Table 5 shows this at the most anisotropic layer of each T5 encoder. That table finds the most anisotropic layer on test passages, where LaTa peaks at layer 8; Table 3 and the main text read top-PC share on training passages, where LaTa peaks at layer 4. LaTa's training curve is flat, with a top-PC share of 0.93 to 0.95 over layers 3 to 11.

**Correlations with ranking.** Table 4 correlates each statistic with baseline AUROC, one observa-

| | | | Geometry (baseline) | | | | Baseline test AUROC | | |
|---|---|---|---|---|---|---|---|---|---|
| **Model** | **Type** | $L$ | Mean cos. | Cos. SD | $\mathrm{PC1}_{\max}$ | $\mathrm{Rank}_{\min}$ | $\ell^*$ | at $\ell^*$ | Min (layer) |
| LaTa | T5 | 12 | 0.23–0.58 | 0.10–0.63 | 0.952 | 1.36 | 4 | 0.500 | 0.496 (6) |
| PhilTa | T5 | 12 | 0.53–0.71 | 0.09–0.46 | 0.858 | 1.81 | 6 | 0.541 | 0.538 (10) |
| mT5-base | T5 | 12 | 0.30–0.95 | 0.05–0.76 | 1.000 | 1.00 | 5 | 0.654 | 0.654 (5) |
| LaBSE | Enc. | 12 | 0.73–0.93 | 0.04–0.10 | 0.524 | 16.19 | 8 | 0.879 | 0.806 (1) |
| Qwen3-0.6B | Dec. | 28 | 0.85–0.97 | 0.01–0.04 | 0.183 | 71.60 | 16 | 0.910 | 0.858 (1) |
| KaLM-mini | Dec. | 24 | 0.58–0.93 | 0.02–0.07 | 0.419 | 23.83 | 5 | 0.861 | 0.861 (5) |

Table 3: Geometry and baseline test ranking AUROC of the six panel models, raw above the rule and embedding-trained below. Type, $\mathrm{PC1}_{\max}$ and $\mathrm{Rank}_{\min}$ as in Table 6. Mean cos. and Cos. SD: range over layers of the mean and SD of test pairwise cosines. $\ell^*$: layer of $\mathrm{PC1}_{\max}$; layer 1 is the first transformer block.

| | **All** (100) | | **T5** (36) | | **Non-T5** (64) | | **LOMO** $\rho$ | |
|---|---|---|---|---|---|---|---|---|
| **Statistic** | $r$ | $\rho$ | $r$ | $\rho$ | $r$ | $\rho$ | min | max |
| *Target: baseline AUROC* | | | | | | | | |
| Mean pairwise cosine | 0.83 | 0.55 | 0.61 | 0.67 | −0.06 | 0.10 | 0.42 | 0.67 |
| Top-PC share | −0.92 | −0.77 | −0.86 | −0.50 | −0.47 | −0.43 | −0.83 | −0.70 |
| Effective rank | 0.80 | 0.77 | 0.87 | 0.53 | 0.48 | 0.40 | 0.70 | 0.85 |
| Raw cosine gap | 0.24 | 0.22 | 0.63 | 0.72 | 0.15 | −0.03 | −0.10 | 0.48 |
| Standardized gap | 0.86 | 0.97 | 0.87 | 0.96 | 0.89 | 0.93 | 0.96 | 0.98 |
| *Target: ABTT AUROC gain* | | | | | | | | |
| Mean pairwise cosine | −0.83 | −0.54 | −0.60 | −0.66 | 0.07 | −0.07 | −0.66 | −0.41 |
| Top-PC share | 0.92 | 0.76 | 0.86 | 0.50 | 0.45 | 0.41 | 0.69 | 0.83 |
| Effective rank | −0.80 | −0.76 | −0.88 | −0.52 | −0.44 | −0.37 | −0.84 | −0.68 |
| Raw cosine gap | −0.23 | −0.23 | −0.61 | −0.71 | −0.16 | 0.00 | −0.48 | 0.09 |
| Standardized gap | −0.85 | −0.97 | −0.87 | −0.95 | −0.88 | −0.92 | −0.98 | −0.96 |

Table 4: Correlation of per-layer statistics with baseline test ranking AUROC and with ABTT gain (ABTT AUROC at the train-selected $D$ minus baseline), over panel layers. $r$: Pearson; $\rho$: Spearman. T5: three raw T5-family encoders; Non-T5: three embedding-trained models. LOMO: range of $\rho$ over leave-one-model-out subsets. Top-PC share and effective rank use training passages, the rest test passages.

tion per model-layer. Across all 100 panel layers, top-PC share and effective rank reach Spearman $-0.77$ and $+0.77$. Both values are inflated by the two clusters of collapsed and healthy layers, and layers within a model are not independent. Within the 26 collapsed layers, top-PC share carries no ordering (Spearman $+0.24$, $p = 0.23$). Within the 74 other layers it orders them only moderately ($-0.49$). Mean pairwise cosine correlates positively with AUROC within the three panel T5 encoders ($+0.67$ on test passages, $+0.72$ on training passages). This pattern does not carry over to T5-v1.1-base, whose collapsed layers sit at mean cosines of 0.87 to 0.96 on training passages, around the 0.91 and 0.92 of its two healthy layers; within that model the correlation is $-0.64$. The lower block of the table adds no separate evidence. ABTT lifts every panel layer into a narrow band (0.962 to 0.987), so its gain is nearly a constant minus the baseline, and the gain correlations mirror the baseline ones. The absence of a reliable ordering agrees with reports that geometric statistics do not reliably predict downstream performance (Kulkarni et al., 2026; Nastase and Merlo, 2025).

**Raw and standardized gap.** The raw cosine gap, the margin a routing threshold works with, tracks ranking poorly: across the panel it correlates with AUROC at Spearman $+0.22$ (Figure 4). The standardized gap reaches $+0.97$, but it is a label-based signal-to-noise ratio close to the $d'$ from which a binormal model computes AUROC. It tracks AUROC by construction, so we do not count it as a predictor. It explains one pattern in mT5-base (Figure 5). There the raw gap rises from about 0.03 to 0.21–0.25 exactly where ranking fails. Over the same layers the standard deviation of all pairwise cosines grows from 0.05–0.07 to about 0.75. Divided by that spread, the gap falls to 0.28–0.33, below the 0.41–0.49 of layers 1–3 and the 0.78 of layer 12. The collapsed layers thus spread all cosines widely, and same-source pairs gain little relative to that spread. mT5-base layer 4 is an exception: its standardized gap is the lowest of the model (0.19), yet it ranks at 0.799. In Figure 4, baseline gaps stay below 0.29 at every layer of every panel model, and in the middle layers of LaTa and PhilTa they stay within 0.09 of zero. ABTT raises every model's gap to between 0.44 and 0.61

| Model | Most anisotropic layer | PC1 before | PC1 after | Eff. rank before→after |
|---|---|---|---|---|
| LaTa | 8 | 0.956 | 0.036 | 1.34→168.40 |
| PhilTa | 6 | 0.862 | 0.039 | 1.83→155.02 |
| mT5-base | 5 | 1.000 | 0.044 | 1.00→151.74 |

Table 5: Most anisotropic layer of each raw T5-family panel encoder, by top-PC share (PC1) on test passages; Table 3 reads top-PC share on training passages. PC1 and effective rank before and after ABTT with $D = 10$, fit on training passages.

at every depth. SIF raises the LaTa and PhilTa gaps and leaves the other four models near or below baseline.

**Cosine spread at the collapsed layers.** At the collapsed layers, test cosines spread across both signs. At mT5-base's collapsed layers the 5th percentile lies between $-0.83$ and $-0.77$ and the median between 0.84 and 0.90, with a standard deviation of about 0.75. At LaTa's collapsed layers the 5th percentile lies between $-0.50$ and $-0.26$, and the standard deviation reaches 0.63. In the embedding-trained models the standard deviation stays between 0.007 and 0.10 at every layer. Pooled passages at a collapsed layer therefore lie on both sides of one dominant direction, which Section 6 examines.

**SIF at the collapsed layers.** SIF pooling (Arora et al., 2017) down-weights frequent tokens and drops special tokens (Appendix B). At the collapsed layers it recovers about 80 percent of the ABTT gain in LaTa and PhilTa, and under 10 percent in mT5-base. This split agrees with the token ablation of Section 6: in LaTa, mean pooling without the comma, the period and </s> restores every collapsed layer, while in mT5-base dropping up to its top 100 token types, ranked on training passages, does not. SIF stays below ABTT at every collapsed layer, and its full sweeps are in Appendix K.

**Score distributions.** Figure 6 restates the collapse and the repair as score distributions rather than as summary metrics. At PhilTa's collapsed layer 8 the cosines of same-source and different-source pairs overlap almost completely, and ABTT separates them. We show layer 8 (test AUROC 0.542) because it has the highest training AUROC of PhilTa's layers 4 to 8 (30 to 70% of depth), all of them collapsed; PhilTa's worst layer is layer 10, at baseline test AUROC 0.538.

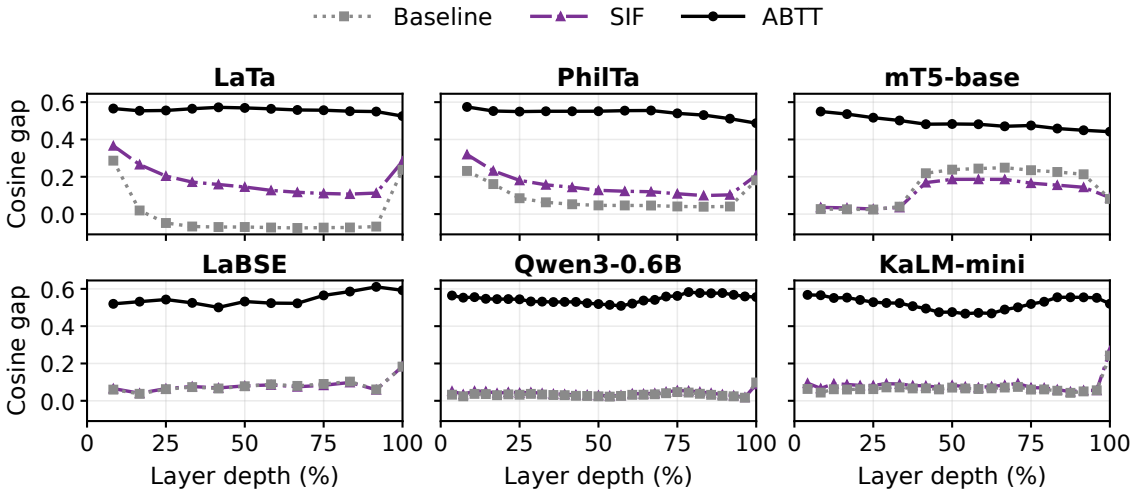



Figure 4: Raw cosine gap, the mean same-directory test cosine minus the mean different-directory one, against normalized layer depth for the six panel models under baseline pooling, SIF and ABTT (fit on training embeddings, $D$ chosen per layer). ABTT widens every model's gap at every depth.

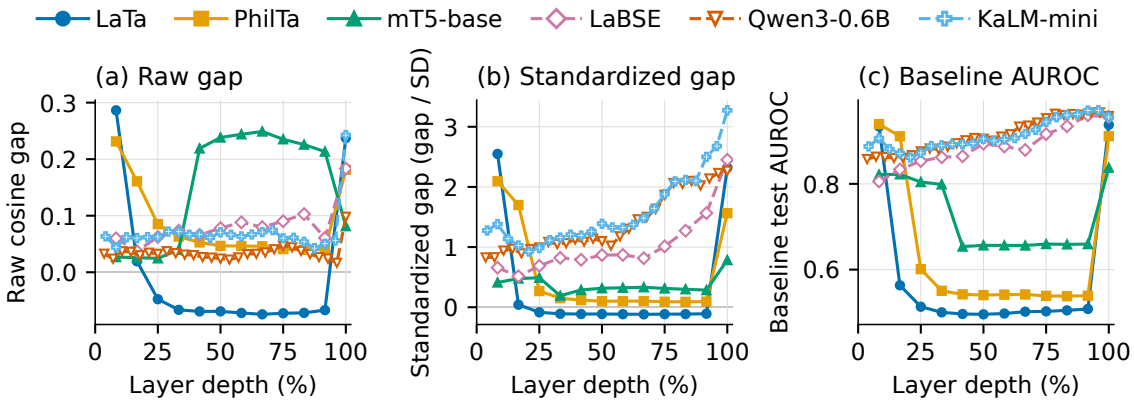



Figure 5: (a) Raw gap as in Figure 4, (b) standardized gap and (c) test ranking AUROC by depth for the six panel models, baseline mean pooling. Solid, filled: raw T5-family; dashed, hollow: embedding-trained. In mid-depth mT5-base the raw gap rises as AUROC falls; the standardized gap falls with it.

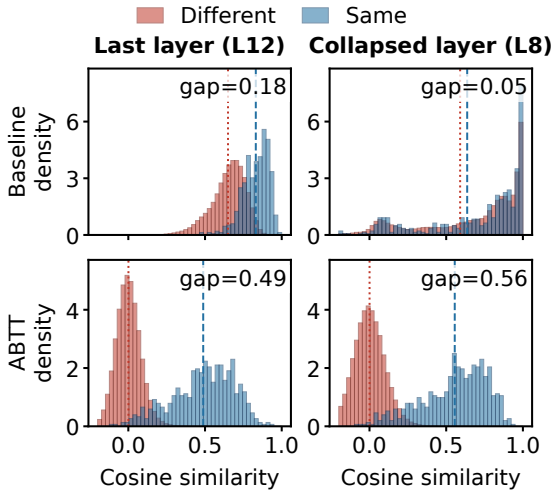



Figure 6: Test cosine distributions for PhilTa's last layer and collapsed layer 8, before and after ABTT. Same: pairs sharing a directory; Different: the rest; dashed, dotted lines: Same, Different means; gap: their difference. At layer 8 the two overlap almost completely until ABTT separates them.

## D All Thirteen Models

This appendix supports Section 5. Table 6 extends Table 1 in two ways. It adds two embedding-trained models that Table 1 leaves out, Sentence-T5 (Ni et al., 2022) and SPhilBERTa (Riemenschneider and Frank, 2023b). It also gives each checkpoint's feed-forward type and width, whether its embeddings are tied, its lowest effective rank and its range of mean pairwise cosine. None of the added models collapses. Every checkpoint has at most 0.6B parameters, and each is released under the Apache 2.0 license, as its Hugging Face model card states. The rest of the appendix gives the per-layer readouts behind the sibling pairs, T5-efficient-base, the projection on the four controls, the English-text check and the fine-tuned LaTa model.

**The sibling pairs.** Both encoder-only siblings stay above 0.80 AUROC at every layer (Table 7). LaBERTa clears that floor by 0.026, at layer 1, and rises to 0.910 at layer 12; PhilBERTa stays between 0.883 and 0.923. Effective rank falls to 1.36 in LaTa and 1.81 in PhilTa, and never below 57.87 in LaBERTa or 82.12 in PhilBERTa. Within each pair, several factors differ together. The T5 member is the encoder of an encoder-decoder, pretrained with span corruption; the sibling is an encoder-only RoBERTa-style model, pretrained with masked language modeling (Riemenschneider and Frank, 2023a). The T5 members use a SentencePiece tokenizer and the siblings a byte-level BPE tokenizer, so the tokens that enter the mean also differ. By their checkpoint configurations, the T5 members also use relative position biases, a gated-GELU feed-forward block and untied embeddings, and the siblings learned absolute positions, a GELU feed-forward block and tied embeddings. The pairs therefore rule out the shared pretraining text and the missing embedding objective as sufficient causes, and they cannot say which within-pair difference matters. Two input details of the siblings' tokenizers matter for Table 7. LaBERTa's tokenizer encodes every space as its own token, so the indentation in the corpus text becomes tokens, which the pooling filter drops from the mean. PhilBERTa's published tokenizer adds no sequence-boundary tokens, so the two whitespace-only witnesses pool to zero vectors; scoring without those two witnesses changes no AUROC by more than 0.0003.

**T5-base and T5-v1.1-base.** The two checkpoints share their tokenizer, which splits the Latin corpus identically, the span-corruption objective, relative position biases and pre-layer RMS normalization, at the same depth and width (Raffel et al., 2020; Google Research, 2020). They differ in feed-forward block (ReLU with inner width 3072, against gated GELU with 2048), in embedding tying, and in pretraining. T5-base was pretrained on C4 mixed with supervised tasks, with dropout (Raffel et al., 2020); T5-v1.1-base on C4 alone, without dropout by its release notes (Google Research, 2020), although its released pretraining configuration sets a dropout rate of 0.1. Dropout does not separate the collapsing models from the others, because PhilTa was also pretrained with dropout (Riemenschneider and Frank, 2023a). T5-base shows a weak form of the same depth profile: its top-PC share rises to 0.545 and its effective rank falls to 14.85 at layer 8, while its AUROC stays between 0.816 and 0.864. T5-v1.1-base recovers only partly at its last layer, to 0.772. T5-efficient-base (Tay et al., 2022) has T5-base's ReLU feed-forward with inner width 3072, tied embeddings, vocabulary, depth and width, and T5-v1.1-base's C4-only pretraining data; it was pretrained without dropout and on a much shorter run: 524,288 steps of 65,536 tokens, about 1/30 of the tokens of either, by the released training configurations. On the Latin corpus it never falls below 0.736 (layer 11), and its top-PC share stays at or below 0.627 (Table 8). It ranks below T5-base at every layer, by 0.016 to 0.080, and its layer 11 has effective rank 8.1, a mild form of the low-rank profile. Its tokenizer adds one whitespace token to 471 passages, which the pooling filter drops; reading it with T5-base's tokenizer changes no AUROC by more than 0.002 and no top-PC share by more than 0.07 (peak 0.615). Its mean pairwise cosine is 0.97 to 0.995 at layers 1 to 11, so a high mean cosine also flags a model that does not collapse. The T5-base column of Table 8 comes from the same CPU extraction run as T5-efficient-base.

**The projection on the controls.** Table 9 applies ABTT to the four controls with the panel's settings. LaBERTa (0.963 to 0.977) and PhilBERTa (0.974 to 0.980) stay inside the panel's band at every layer, T5-base misses it at layer 11 (0.959), and T5-v1.1-base reaches 0.927 to 0.970, below the band at layers 6 to 12. T5-efficient-base reaches 0.959 to 0.975 (Table 8), below the band at layer 11 only. $D = 10$, the top of the selection grid, is selected at 46 of the 48 control layers, so a larger $D$ might

| | | | Checkpoint | | | Over layers | | | | Layers | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Model | Type | Emb. obj. | Feed-forward | $d_{\text{ff}}$ | Tied | $\text{AUROC}_{\text{min}}$ | $\text{PC1}_{\text{max}}$ | $\text{Rank}_{\text{min}}$ | Mean cos. | Coll. | PC1≥0.76 |
| *Main text* | | | | | | | | | | | |
| LaTa | T5 | none | gated GELU | 2048 | no | 0.496 (6) | 0.952 (4) | 1.36 (8) | 0.22–0.58 | 2–11 | 2–11 |
| PhilTa | T5 | none | gated GELU | 2048 | no | 0.538 (10) | 0.858 (6) | 1.81 (6) | 0.51–0.71 | 3–11 | 3–11 |
| mT5-base | T5 | none | gated GELU | 2048 | no | 0.654 (5) | 1.000 (5) | 1.00 (5) | 0.33–0.95 | 5–11 | 4–11 |
| T5-v1.1-base | T5 | none | gated GELU | 2048 | no | 0.489 (2) | 0.975 (9) | 1.14 (9) | 0.87–0.96 | 2–11 | 2–11 |
| T5-base | T5 | none | ReLU | 3072 | yes | 0.816 (11) | 0.545 (8) | 14.85 (8) | 0.90–0.95 | none | none |
| T5-efficient-base | T5 | none | ReLU | 3072 | yes | 0.736 (11) | 0.627 (11) | 8.10 (11) | 0.89–0.99 | none | none |
| LaBERTa | Enc. | none | GELU | 3072 | yes | 0.826 (1) | 0.254 (9) | 57.87 (9) | 0.70–0.91 | none | none |
| PhilBERTa | Enc. | none | GELU | 3072 | yes | 0.883 (6) | 0.162 (6) | 82.12 (11) | 0.84–0.96 | none | none |
| LaBSE | Enc. | contrastive | GELU | 3072 | n/a | 0.806 (1) | 0.524 (8) | 16.19 (8) | 0.73–0.93 | none | none |
| Qwen3-0.6B | Dec. | contrastive | gated SiLU | 3072 | yes | 0.858 (1) | 0.183 (16) | 71.60 (16) | 0.84–0.97 | none | none |
| KaLM-mini | Dec. | contrastive | gated SiLU | 4864 | yes | 0.861 (5) | 0.419 (5) | 23.83 (5) | 0.59–0.93 | none | none |
| *Appendix only* | | | | | | | | | | | |
| Sentence-T5 | T5 | contrastive | ReLU | 3072 | n/a | 0.800 (11) | 0.943 (6) | 1.56 (6) | 0.72–0.98 | none | 2–9 |
| SPhilBERTa | Enc. | distillation | GELU | 3072 | n/a | 0.892 (1) | 0.127 (11) | 64.67 (11) | 0.61–0.95 | none | none |

Table 6: All thirteen models, read on the Latin corpus as mean-pooled hidden states with no correction; Type, $\text{AUROC}_{\text{min}}$ and $\text{PC1}_{\text{max}}$ as in Table 1, each extreme followed by its layer. Emb. obj.: embedding objective trained after pretraining, if any. Checkpoint columns follow each configuration file (T5-base, T5-efficient-base, LaBERTa and PhilBERTa leave tying at the library default, tied; n/a: no output head). $\text{Rank}_{\text{min}}$ and Mean cos.: lowest entropy effective rank and range of mean pairwise cosine, on training embeddings. Layers: the layers collapsed or with top-PC share of at least 0.76. Checkpoints: LaTa, `bowphs/LaTa`; PhilTa, `bowphs/PhilTa`; mT5-base, `google/mt5-base`; T5-v1.1-base, `google/t5-v1_1-base`; T5-base, `google-t5/t5-base`; T5-efficient-base, `google/t5-efficient-base`; LaBERTa, `bowphs/LaBerta`; PhilBERTa, `bowphs/PhilBerta`; LaBSE, `sentence-transformers/LaBSE`; Qwen3-0.6B, `Qwen/Qwen3-Embedding-0.6B`; KaLM-mini, `KaLM-Embedding/KaLM-embedding-multilingual-mini-instruct-v2.5`; Sentence-T5, `sentence-transformers/sentence-t5-base`; SPhilBERTa, `bowphs/SPhilBerta`.

| | LaBERTa | | | PhilBERTa | | |
|---|---|---|---|---|---|---|
| Layer | AUROC | PC1 | Rank | AUROC | PC1 | Rank |
| 1 | 0.826 | 0.170 | 87.15 | 0.893 | 0.092 | 134.77 |
| 2 | 0.832 | 0.193 | 75.32 | 0.905 | 0.096 | 109.71 |
| 3 | 0.839 | 0.184 | 78.12 | 0.913 | 0.112 | 97.87 |
| 4 | 0.830 | 0.191 | 74.89 | 0.903 | 0.124 | 98.85 |
| 5 | 0.828 | 0.191 | 72.73 | 0.885 | 0.151 | 95.27 |
| 6 | 0.834 | 0.198 | 68.46 | 0.883 | 0.162 | 97.28 |
| 7 | 0.836 | 0.228 | 61.98 | 0.896 | 0.149 | 97.34 |
| 8 | 0.844 | 0.248 | 58.96 | 0.905 | 0.157 | 94.21 |
| 9 | 0.857 | 0.254 | 57.87 | 0.911 | 0.140 | 92.12 |
| 10 | 0.873 | 0.245 | 61.55 | 0.919 | 0.120 | 89.92 |
| 11 | 0.892 | 0.233 | 65.75 | 0.913 | 0.143 | 82.12 |
| 12 | 0.910 | 0.154 | 85.69 | 0.923 | 0.083 | 105.56 |

Table 7: Per-layer readouts of LaBERTa and PhilBERTa, the encoder-only siblings of LaTa and PhilTa in Table 1, with no post-hoc correction. AUROC: test ranking AUROC. PC1 and Rank: top-PC share and effective rank of the 847 training passages. Both siblings stay above 0.80 AUROC at every layer.

| | T5-efficient-base | | | | | T5-base |
|---|---|---|---|---|---|---|
| Layer | AUROC | PC1 | Rank | Cos | ABTT | AUROC |
| 1 | 0.820 | 0.164 | 49.85 | 0.994 | 0.975 | 0.855 |
| 2 | 0.822 | 0.137 | 54.66 | 0.995 | 0.968 | 0.864 |
| 3 | 0.812 | 0.190 | 48.51 | 0.994 | 0.973 | 0.864 |
| 4 | 0.798 | 0.193 | 47.78 | 0.994 | 0.974 | 0.849 |
| 5 | 0.794 | 0.231 | 45.57 | 0.994 | 0.971 | 0.848 |
| 6 | 0.803 | 0.304 | 35.04 | 0.993 | 0.967 | 0.843 |
| 7 | 0.809 | 0.263 | 37.85 | 0.992 | 0.969 | 0.849 |
| 8 | 0.812 | 0.327 | 32.52 | 0.991 | 0.970 | 0.847 |
| 9 | 0.794 | 0.260 | 36.81 | 0.990 | 0.964 | 0.836 |
| 10 | 0.782 | 0.276 | 34.43 | 0.987 | 0.962 | 0.826 |
| 11 | 0.736 | 0.627 | 8.10 | 0.973 | 0.959 | 0.816 |
| 12 | 0.827 | 0.213 | 66.67 | 0.891 | 0.964 | 0.843 |

Table 8: Test ranking AUROC and geometry of T5-efficient-base (T5-base's layout, C4 alone, no dropout, 524,288 steps of 65,536 tokens) and AUROC of T5-base (pretrained for about 30 times as many tokens). PC1, Rank, Cos: top-PC share, effective rank, mean pairwise cosine of training passages. ABTT: AUROC after ABTT fit on them. T5-efficient-base does not collapse (lowest AUROC 0.736, layer 11).

narrow the gap; we did not test this.

**Sentence-T5.** Sentence-T5 has the original T5 layout and a contrastive embedding objective. At its layers 2 to 11, top-PC share is 0.659 to 0.943 while AUROC stays between 0.800 and 0.891. Sentence-T5 thus shows that a high top-PC share is not sufficient for collapse. It is necessary in our data: every one of the 36 collapsed layers of the pretrained checkpoints in Table 6 (26 in the panel, 10 in T5-v1.1-base) has a share of at least 0.76.

**English text.** Table 10 and Figure 7 give the label-free check of Section 5. The English sample comes from the public-domain US court opinions of the Caselaw Access Project (Harvard Law

| | LaBERTa | | PhilBERTa | | T5-base | | T5-v1.1-base | |
|---|---|---|---|---|---|---|---|---|
| Layer | Base | ABTT | Base | ABTT | Base | ABTT | Base | ABTT |
| 1 | 0.826 | 0.974 | 0.893 | 0.976 | 0.855 | 0.981 | 0.848 | 0.970 |
| 2 | 0.832 | 0.977 | 0.905 | 0.977 | 0.864 | 0.981 | 0.489 | 0.969 |
| 3 | 0.839 | 0.972 | 0.913 | 0.978 | 0.864 | 0.974 | 0.493 | 0.965 |
| 4 | 0.830 | 0.969 | 0.903 | 0.974 | 0.849 | 0.971 | 0.505 | 0.963 |
| 5 | 0.828 | 0.967 | 0.885 | 0.976 | 0.848 | 0.969 | 0.515 | 0.963 |
| 6 | 0.834 | 0.963 | 0.883 | 0.974 | 0.843 | 0.967 | 0.523 | 0.959 |
| 7 | 0.836 | 0.964 | 0.896 | 0.974 | 0.849 | 0.962 | 0.533 | 0.952 |
| 8 | 0.844 | 0.963 | 0.905 | 0.975 | 0.847 | 0.966 | 0.536 | 0.945 |
| 9 | 0.857 | 0.965 | 0.911 | 0.979 | 0.836 | 0.969 | 0.533 | 0.934 |
| 10 | 0.873 | 0.968 | 0.919 | 0.978 | 0.826 | 0.962 | 0.531 | 0.930 |
| 11 | 0.892 | 0.966 | 0.913 | 0.980 | 0.816 | 0.959 | 0.522 | 0.927 |
| 12 | 0.910 | 0.967 | 0.923 | 0.976 | 0.843 | 0.973 | 0.772 | 0.959 |

Table 9: Test ranking AUROC of the four controls before (Base) and after ABTT fit on training passages, with $D$ chosen per layer on training DirAcc@1 as for the panel. Seven layers of T5-v1.1-base and one of T5-base stay below the panel's band (0.962 to 0.987).

School Library Innovation Lab, 2024), in the redistribution of the Common Pile (Kandpal et al., 2025); every sampled document is marked public domain. Legal prose is the nearest openly licensed English register to canon law. Each non-empty Latin passage receives one English passage from a distinct opinion, and the two empty Latin files receive an empty one. The English passage is a run of whole words from a random sentence start in the body text, with an mT5 token count within $\max(2, 3\%)$ tokens of the Latin passage's, and it inherits its partner's split. The two texts then have the same mT5 length distribution (median 98 tokens). The match holds only under the mT5 tokenizer: under PhilTa's tokenizer the English passages are about 40 percent longer, and under T5-v1.1's English-only tokenizer the Latin passages are about 55 percent longer. Re-pairing the training passages on each model's own token lengths (653 to 847 pairs) moves top-PC share by at most 0.010 at the layers that Table 10 lists. At mid-depth the top-PC share is above 0.8 in every cell. The effective rank is near 1 for mT5-base and T5-v1.1-base but not for PhilTa, whose effective rank is 1.82 to 2.30 at English layers 4 to 11, and 1.81 to 4.34 at Latin layers 3 to 11. The two texts differ in language, era and genre together, and the English sample covers one register.

**Fine-tuned LaTa.** Table 11 gives fine-tuned LaTa at every layer, with no post-hoc correction, beside the pretrained model. Fine-tuning lifts the last layer from 0.938 to 0.984 AUROC and leaves layers 2 to 11 at 0.499 to 0.569. Their top-PC share stays at 0.759 to 0.945, against 0.771 to 0.952 before fine-tuning, so the collapsed geometry stays in place (fine-tuned layer 2, at 0.759, sits just below the 0.76 screen). The fine-tuning runs at most eight epochs on 499 pairs (Appendix H). It is therefore a weak test of whether an embedding objective repairs the middle layers: small-scale supervision at the output may leave them untouched whatever produces the collapse.

| | | | Layers with PC1 $\geq 0.76$ | | |
|---|---|---|---|---|---|
| Model | Text | $\mathrm{PC1}_{\max}$ ($\ell$) | Layers | PC1 | Eff. rank |
| mT5-base | Latin | 1.000 (5) | 4–11 | 0.893–1.000 | 1.00–2.15 |
| | English | 1.000 (5) | 4–11 | 0.822–1.000 | 1.00–3.10 |
| PhilTa | Latin | 0.858 (6) | 3–11 | 0.764–0.858 | 1.81–4.34 |
| | English | 0.894 (6) | 4–11 | 0.846–0.894 | 1.82–2.30 |
| T5-v1.1-base | Latin | 0.975 (9) | 2–11 | 0.958–0.975 | 1.14–1.20 |
| | English | 0.950 (9) | 2–11 | 0.931–0.950 | 1.28–1.36 |

Table 10: Label-free geometry of three raw T5-family encoders on the 847 Latin training passages and on as many English passages matched to them in mT5 token length (Figure 7). PC1: top-PC share. $\mathrm{PC1}_{\max}$ ($\ell$): peak PC1 over the 12 layers, at layer $\ell$.

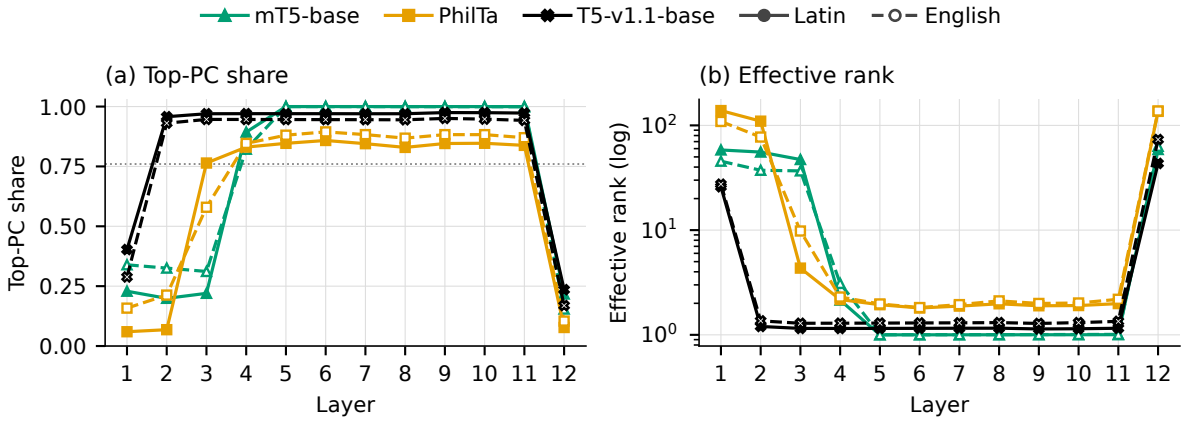



Figure 7: Label-free geometry of three T5 encoders on the Latin corpus and on English court opinions matched in mT5 token length: (a) top-PC share, dotted line at 0.76; (b) effective rank, log scale. On both texts every model is low-rank from layer 2 to 4 through layer 11 and recovers at layer 12.

| | AUROC | | Top-PC share | | Eff. rank | |
|---|---|---|---|---|---|---|
| Layer | Before | After | Before | After | Before | After |
| 1 | 0.934 | 0.936 | 0.050 | 0.050 | 134.33 | 134.44 |
| 2 | 0.563 | 0.569 | 0.771 | 0.759 | 4.61 | 4.92 |
| 3 | 0.513 | 0.512 | 0.935 | 0.929 | 1.59 | 1.64 |
| 4 | 0.500 | 0.499 | 0.952 | 0.945 | 1.38 | 1.43 |
| 5 | 0.496 | 0.500 | 0.943 | 0.931 | 1.41 | 1.48 |
| 6 | 0.496 | 0.500 | 0.934 | 0.921 | 1.44 | 1.51 |
| 7 | 0.498 | 0.501 | 0.937 | 0.923 | 1.43 | 1.50 |
| 8 | 0.502 | 0.505 | 0.950 | 0.940 | 1.36 | 1.42 |
| 9 | 0.503 | 0.507 | 0.943 | 0.931 | 1.40 | 1.48 |
| 10 | 0.505 | 0.510 | 0.942 | 0.930 | 1.43 | 1.50 |
| 11 | 0.508 | 0.512 | 0.939 | 0.927 | 1.47 | 1.55 |
| 12 | 0.938 | 0.984 | 0.081 | 0.069 | 132.22 | 137.68 |

Table 11: LaTa before and after contrastive fine-tuning on training pairs, with no post-hoc correction. AUROC: test ranking AUROC. Top-PC share and effective rank are read on the training passages, which include those of the fine-tuning pairs. Fine-tuning lifts the last layer and leaves the middle layers collapsed.

## E Localizing the Collapse: Details

This appendix gives the per-layer tables and the further controls behind Section 6. Collapsed layers are the 26 layers of LaTa, PhilTa and mT5-base with baseline AUROC below 0.70. Every coordinate ranking, component, token ranking and threshold is fit on training passages only, and every AUROC is test ranking AUROC. A layer counts as restored when it reaches 0.90.

**Coordinate ablation.** Table 12 reads each model at its lowest-AUROC baseline layer. The two coordinate rankings pick nearly the same coordinates at the collapsed layers: on average 9.1 of the top ten are shared. Massive coordinates alone do not mark a collapse. At LaTa layer 6, the lowest-AUROC layer of the panel, the three pooled coordinates of largest mean magnitude reach 720, 684 and 676, against a median of 8.5. At mT5-base layer 1, a healthy layer (AUROC 0.822), they reach 135, 131 and 63, against a median of 1.9. Per-coordinate standardization with training statistics (Timkey and van Schijndel, 2021) is a partial repair. It recovers a median 79% of the gain from removing ten components, but it restores only 9 of the 26 layers (1 of 10 in LaTa, 5 of 9 in PhilTa, 3 of 7 in mT5-base). The embedding-trained models barely react to zeroing: at the train-selected layers of LaBSE, Qwen3-0.6B and KaLM-mini, zeroing up to ten

coordinates changes AUROC by at most 0.007.

**Zeroing more coordinates.** Figure 8 extends the zeroing to 400 coordinates. Under the variance ranking, zeroing restores none of the 26 layers up to 30 coordinates, 7 at 100, and 19 at 300 and 400. The 19 are every collapsed layer of LaTa and PhilTa; no mT5-base layer is restored. At the lowest-AUROC layers plotted in Figure 8, LaTa and PhilTa pass 0.90 only after 200 and 150 coordinates are zeroed (0.924 and 0.937 at $k = 400$), and mT5-base reaches only 0.779 at $k = 400$, where its top-PC share is still 0.949. Under the mean-absolute-value ranking of Figure 3a, the first layers are restored at 50 coordinates and all 19 at 400. The random-coordinate curves in Figures 3a and 8 are means over five random draws of the $k$ coordinates. Zeroing 400 random coordinates removes a median 55% of the training variance, yet the median AUROC stays at 0.538, against 0.541 uncorrected. At the layers plotted in Figure 8, random zeroing moves the AUROC of LaTa, PhilTa and mT5-base by at most 0.010, and zeroing by variance moves the embedding-trained models by at most 0.038. Top-PC share falls before AUROC recovers: at 100 zeroed coordinates it is below 0.2 at 19 layers, while only 7 are restored.

**What survives zeroing.** A follow-up zeroes the ten highest-variance coordinates and asks which direction then dominates. These ten coordinates carry a median 92% of the first component's squared loading. The new dominant direction is the remainder of the first component on the other coordinates (median cosine 0.999 with that remainder). It holds a median 72% of the remaining variance. Removing the original second and third components while keeping the first restores none of the 26 layers. Zeroing followed by ABTT with one, two or three components restores 9, 21 and 26 layers, against 1, 13 and 26 without zeroing.

**Number of directions.** Figure 9 plots the sweep over $D$ at each model's lowest-AUROC layer, and Table 13 gives medians over groups of layers. LaTa reaches the full gain of removing ten components with two (median 98%), and PhilTa with three (95%). mT5-base reaches a median 85% with three and keeps rising up to ten. With three components every collapsed layer recovers at least 80% of that gain. The failure of a single component is not an estimation artifact. At every collapsed layer the first component fit on training passages and the one

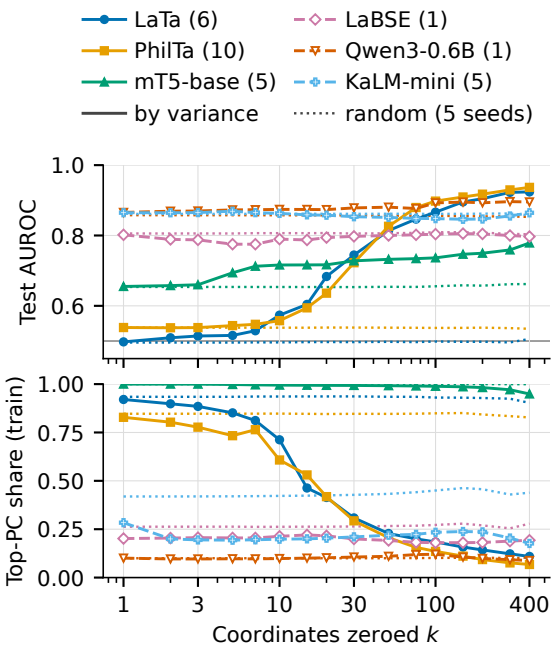



Figure 8: Test ranking AUROC (top) and training top-PC share (bottom) against zeroed coordinates $k$ (log axis), each model at its lowest-AUROC baseline layer (in parentheses). Filled, solid (raw T5-family) and hollow, dashed (embedding-trained): zeroing by variance; dotted: random coordinates (five seeds); gray line: chance. At these layers zeroing by variance passes 0.90 for LaTa and PhilTa only past 100 coordinates, and not for mT5-base.

fit on test passages agree to $|\cos| \geq 0.994$. The selection grid stops at $D = 10$; extending it to $D = 50$ changes test AUROC by at most 0.023.

**Removed and retained subspaces.** Table 14 ranks passages within the removed and within the retained subspace. At the collapsed layers the removed subspace holds a median 99% of the centered variance of vectors that already rank at chance, so its chance-level ranking there is close to a tautology. The next ten components alone rank at a median 0.919 at those layers, a dimension-matched control (Table 14). The healthy layers are more informative. At the 74 healthy panel layers the removed subspace still ranks at 0.736 to 0.937, so ABTT discards some signal. Yet at all 36 layers of the three pretrained T5 encoders the removed directions rank below the next $D$ components. At LaTa's last layer, contrastive fine-tuning moves the top ten directions from below the next ten (0.868 against 0.907) to above them (0.953 against 0.900), consistent with Rajaee and Pilehvar (2021).

| | | | Zeroed, ranked by mean $\lvert x\rvert$ | | | | Zeroed, ranked by variance | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Model | L | Base | 1 | 3 | 5 | 10 | 1 | 3 | 5 | 10 | Std. | $D$=0 | ABTT$_{D=1}$ | ABTT$_{D=3}$ |
| *Test ranking AUROC* | | | | | | | | | | | | | | |
| LaTa | 6 | 0.496 | 0.498 | 0.514 | 0.512 | 0.574 | 0.498 | 0.514 | 0.516 | 0.574 | 0.822 | 0.482 | 0.671 | 0.973 |
| PhilTa | 10 | 0.538 | 0.537 | 0.540 | 0.540 | 0.557 | 0.538 | 0.538 | 0.544 | 0.558 | 0.907 | 0.501 | 0.602 | 0.964 |
| mT5-base | 5 | 0.654 | 0.655 | 0.660 | 0.694 | 0.716 | 0.655 | 0.660 | 0.694 | 0.716 | 0.806 | 0.670 | 0.835 | 0.914 |
| LaBSE | 1 | 0.806 | 0.804 | 0.793 | 0.794 | 0.788 | 0.802 | 0.788 | 0.775 | 0.790 | 0.907 | 0.867 | 0.886 | 0.929 |
| Qwen3-0.6B | 1 | 0.858 | 0.866 | 0.874 | 0.877 | 0.880 | 0.866 | 0.870 | 0.873 | 0.874 | 0.963 | 0.954 | 0.959 | 0.970 |
| KaLM-mini | 5 | 0.861 | 0.862 | 0.857 | 0.857 | 0.862 | 0.865 | 0.865 | 0.869 | 0.863 | 0.933 | 0.917 | 0.926 | 0.969 |
| *Top-PC share of the training vectors* | | | | | | | | | | | | | | |
| LaTa | 6 | 0.934 | 0.921 | 0.885 | 0.859 | 0.712 | 0.921 | 0.885 | 0.851 | 0.712 | 0.211 | 0.934 | 0.712 | 0.071 |
| PhilTa | 10 | 0.847 | 0.827 | 0.796 | 0.760 | 0.580 | 0.828 | 0.777 | 0.733 | 0.608 | 0.143 | 0.847 | 0.763 | 0.088 |
| mT5-base | 5 | 1.000 | 1.000 | 1.000 | 0.998 | 0.995 | 1.000 | 1.000 | 0.998 | 0.995 | 0.910 | 1.000 | 0.458 | 0.138 |
| LaBSE | 1 | 0.263 | 0.263 | 0.267 | 0.258 | 0.199 | 0.202 | 0.206 | 0.205 | 0.213 | 0.189 | 0.263 | 0.138 | 0.077 |
| Qwen3-0.6B | 1 | 0.100 | 0.101 | 0.096 | 0.095 | 0.095 | 0.101 | 0.096 | 0.097 | 0.098 | 0.102 | 0.100 | 0.077 | 0.049 |
| KaLM-mini | 5 | 0.419 | 0.298 | 0.301 | 0.306 | 0.192 | 0.284 | 0.195 | 0.193 | 0.198 | 0.196 | 0.419 | 0.210 | 0.067 |

Table 12: Coordinate ablation at each model's lowest-AUROC baseline layer (L), fit on training embeddings. Std.: per-coordinate standardization; $D$=0: centering alone, which leaves top-PC share unchanged by definition. Figure 8 extends the zeroing to 400 coordinates. In the three T5-family rows, zeroing ten coordinates leaves AUROC at or below 0.716, while standardization reaches 0.806 to 0.907 and ABTT$_{D=3}$ 0.914 to 0.973.

| | | | **Removed components $D$** | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Layers** | $n$ | raw | 0 | 1 | 2 | 3 | 5 | 7 | 10 | 15 | 20 | 30 | 50 |
| *Lowest-AUROC baseline layer of each model* | | | | | | | | | | | | | |
| LaTa (6) | 1 | 0.496 | 0.482 | 0.671 | 0.940 | 0.973 | 0.973 | 0.977 | 0.964 | 0.973 | 0.966 | 0.981 | 0.986 |
| PhilTa (10) | 1 | 0.538 | 0.501 | 0.602 | 0.888 | 0.964 | 0.977 | 0.982 | 0.982 | 0.975 | 0.977 | 0.973 | 0.969 |
| mT5-base (5) | 1 | 0.654 | 0.670 | 0.835 | 0.892 | 0.914 | 0.951 | 0.971 | 0.978 | 0.972 | 0.965 | 0.965 | 0.967 |
| LaBSE (1) | 1 | 0.806 | 0.867 | 0.886 | 0.915 | 0.929 | 0.962 | 0.967 | 0.977 | 0.977 | 0.977 | 0.977 | 0.979 |
| Qwen3-0.6B (1) | 1 | 0.858 | 0.954 | 0.959 | 0.962 | 0.970 | 0.972 | 0.979 | 0.978 | 0.980 | 0.981 | 0.976 | 0.984 |
| KaLM-mini (5) | 1 | 0.861 | 0.917 | 0.926 | 0.969 | 0.969 | 0.983 | 0.982 | 0.983 | 0.978 | 0.974 | 0.974 | 0.976 |
| *Median over model-layers* | | | | | | | | | | | | | |
| Collapsed T5 | 26 | 0.541 | 0.503 | 0.713 | 0.902 | 0.964 | 0.973 | 0.977 | 0.977 | 0.974 | 0.975 | 0.974 | 0.971 |
| LaTa | 10 | 0.502 | 0.488 | 0.713 | 0.956 | 0.973 | 0.972 | 0.976 | 0.970 | 0.971 | 0.969 | 0.980 | 0.983 |
| PhilTa | 9 | 0.542 | 0.502 | 0.602 | 0.872 | 0.963 | 0.980 | 0.981 | 0.981 | 0.975 | 0.978 | 0.973 | 0.970 |
| mT5-base | 7 | 0.657 | 0.678 | 0.865 | 0.894 | 0.934 | 0.955 | 0.974 | 0.978 | 0.975 | 0.974 | 0.965 | 0.962 |
| Other T5 | 10 | 0.874 | 0.936 | 0.952 | 0.967 | 0.973 | 0.979 | 0.976 | 0.973 | 0.972 | 0.972 | 0.973 | 0.971 |
| Embedding-trained | 64 | 0.903 | 0.944 | 0.966 | 0.971 | 0.976 | 0.979 | 0.980 | 0.978 | 0.977 | 0.975 | 0.974 | 0.973 |
| All | 100 | 0.885 | 0.936 | 0.958 | 0.968 | 0.973 | 0.978 | 0.979 | 0.977 | 0.975 | 0.974 | 0.974 | 0.972 |

Table 13: Test ranking AUROC against the number $D$ of components ABTT removes, fit on training embeddings ($D = 0$: centering alone). Top: each model at its lowest-AUROC baseline layer (in parentheses). Bottom: medians over each group's $n$ model-layers. Three components bring every collapsed layer to 0.91 or above; mT5-base keeps improving to ten.

**Token audit.** Table 15 gives the token audit at each model's lowest-AUROC layer. Its pooling columns compare mean pooling with four variants: without special tokens, SIF weights with special tokens kept, SIF with training-only token frequencies (which also drops special tokens), and without the 100 most frequent training tokens. The ablation column pools without the $m$ token types that contribute most to the top principal components, for $m$ from 1 to 100 under either of two token rankings; its control is the mean over five random sets of other token types, sampled in proportion to training count, that hold at least as many training tokens as the dropped types. The random directions of the token-mix columns are 20 directions orthogonal to the top ten principal components. The group shares split the variance of the first component's score by covariance among special tokens, the 100 most frequent tokens and all other tokens. $\lvert\rho\rvert$ is the absolute Spearman correlation between that score and the log token count of test passages, since the sign of a principal component is a convention. In LaTa, a token mix with one mean contribution per token type, fit on training tokens, explains a median 0.985 of the variance of the first component's score on test passages. For random directions the same statistic is 0.18. At layer 6 the comma alone holds 0.962 of the first component's score variance over training passages. Ranking token types by the first component alone restores at most 2 of LaTa's 10

| | | | | Full vector | | Removed subspace | | | Retained subspace | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Model** | $\ell$ | $D$ | Var. | Raw | Cent. | PC1 | PCs 2–$D$ | PCs 1–$D$ | Next $D$ | Rand. $D$ | All |
| *Pretrained, train-selected layer and D* | | | | | | | | | | | |
| LaTa | 12 | 10 | 0.35 | 0.938 | 0.972 | 0.670 | 0.825 | 0.868 | 0.907 | 0.861 | 0.971 |
| PhilTa | 9 | 10 | 0.99 | 0.539 | 0.501 | 0.462 | 0.545 | 0.487 | 0.922 | 0.873 | 0.983 |
| mT5-base | 2 | 10 | 0.57 | 0.822 | 0.910 | 0.610 | 0.735 | 0.751 | 0.896 | 0.866 | 0.975 |
| LaBSE | 11 | 10 | 0.48 | 0.960 | 0.958 | 0.647 | 0.932 | 0.909 | 0.958 | 0.902 | 0.987 |
| Qwen3-0.6B | 2 | 10 | 0.44 | 0.863 | 0.940 | 0.607 | 0.792 | 0.798 | 0.906 | 0.882 | 0.973 |
| KaLM-mini | 1 | 10 | 0.41 | 0.887 | 0.958 | 0.518 | 0.846 | 0.819 | 0.910 | 0.880 | 0.981 |
| Collapsed T5, median (26) | – | – | 0.99 | 0.541 | 0.503 | 0.463 | 0.601 | 0.489 | 0.919 | 0.874 | 0.977 |
| *Fine-tuning contrast, same layer, $D = 10$* | | | | | | | | | | | |
| LaTa, pretrained | 12 | 10 | 0.35 | 0.938 | 0.972 | 0.670 | 0.825 | 0.868 | 0.907 | 0.861 | 0.971 |
| LaTa, fine-tuned | 12 | 10 | 0.33 | 0.984 | 0.985 | 0.697 | 0.935 | 0.953 | 0.900 | 0.874 | 0.970 |
| Qwen3-0.6B, pretrained | 27 | 10 | 0.41 | 0.967 | 0.969 | 0.643 | 0.913 | 0.917 | 0.909 | 0.884 | 0.982 |
| Qwen3-0.6B, fine-tuned | 27 | 10 | 0.31 | 0.993 | 0.996 | 0.684 | 0.980 | 0.983 | 0.969 | 0.927 | 0.992 |
| KaLM-mini, pretrained | 24 | 10 | 0.32 | 0.956 | 0.979 | 0.663 | 0.909 | 0.924 | 0.909 | 0.855 | 0.981 |
| KaLM-mini, fine-tuned | 24 | 10 | 0.23 | 0.997 | 0.996 | 0.699 | 0.977 | 0.978 | 0.967 | 0.918 | 0.994 |

Table 14: Test ranking AUROC of cosine within parts of the pooled vector, with components fit on training embeddings. Var.: share of centered training variance removed; PC1: first component alone (cosine is a sign match); Next $D$: PCs $D{+}1$ to $2D$; Rand. $D$: random $D$-dimensional projections of the retained vectors (mean of five seeds). At collapsed layers the signal lies outside the removed directions.

| | | AUROC by pooling | | | | | Token ablation | | Token-mix EV | | Share by token group | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Model | L | Mean | No spec. | SIF+spec. | SIF | No freq. | Drop | Rand. | PC1 | Rand. | Spec. | Freq. | Other | $\lvert\rho\rvert$ |
| LaTa | 6 | 0.496 | 0.493 | 0.860 | 0.878 | 0.904 | 0.922 | 0.483 | 0.98 | 0.19 | 0.01 | 0.97 | 0.02 | 0.38 |
| PhilTa | 10 | 0.538 | 0.496 | 0.677 | 0.879 | 0.692 | 0.917 | 0.540 | 0.44 | 0.14 | 0.23 | 0.75 | 0.02 | 0.11 |
| mT5-base | 5 | 0.654 | 0.653 | 0.675 | 0.673 | 0.726 | 0.724 | 0.656 | 0.07 | 0.24 | 0.00 | 0.53 | 0.47 | 0.20 |
| LaBSE | 1 | 0.806 | 0.809 | 0.812 | 0.818 | 0.816 | 0.827 | 0.800 | 0.36 | 0.73 | 0.09 | 0.37 | 0.53 | 0.95 |
| Qwen3-0.6B | 1 | 0.858 | 0.855 | 0.856 | 0.852 | 0.869 | 0.885 | 0.836 | 0.99 | 0.71 | 0.74 | 0.08 | 0.18 | 0.58 |
| KaLM-mini | 5 | 0.861 | 0.859 | 0.867 | 0.863 | 0.872 | 0.879 | 0.854 | 0.98 | 0.47 | 0.01 | 0.02 | 0.97 | 0.51 |

Table 15: Token audit (test ranking AUROC) at each model's lowest-AUROC baseline layer (L). No spec.: no special tokens; SIF+spec.: SIF keeping special tokens; No freq.: no 100 most frequent tokens; Drop: no top $m$ token types, $m$ and ranking chosen on training AUROC; Rand.: mass-matched random types. Token-mix EV: PC1 score variance explained by token mix; Rand.: random directions. Group shares split it and sum to one, but can fall outside [0, 1]. $|\rho|$: Spearman correlation with log token count.

collapsed layers, so the second and third components matter too. Two pooling variants separate the two changes that SIF makes. In LaTa, dropping the 100 most frequent training tokens recovers at least 80% of SIF's gain at all 10 layers (median AUROC 0.901). Dropping special tokens alone recovers it at none (median 0.498). In PhilTa the token mix explains only a median 0.42 of the first component's score. Its ablation needs more types: on test passages, dropping the top 3 restores 1 of 9 layers, the top 10 restore 8, and the top 30 restore all 9 (median 0.925); on training passages the counts are 1, 6 and 7. With the ranking and $m$ chosen per layer on training AUROC, every collapsed layer of both models is restored: test medians 0.926 (LaTa) and 0.928 (PhilTa), minima 0.921 and 0.908, against mass-matched random controls of 0.491 and 0.542. At $m = 30$, dropping random types of the same training mass stays at a median of 0.543. In mT5-base the token mix explains a median 0.066 of the first component's score, against a median 0.160 for random directions. Dropping its top 100 types lifts AUROC to 0.724 to 0.748, short of 0.90. In none of the three models does the first token carry the dominant direction (share at most 0.005). The same three delimiter roles carry outlier activations in BERT (Bondarenko et al., 2023), and delimiter tokens carry massive activations in large language models (Sun et al., 2024). Removing punctuation and special tokens before pooling also helps BERT (Fuster Baggetto and Fresno, 2022); here it separates chance from recovery.

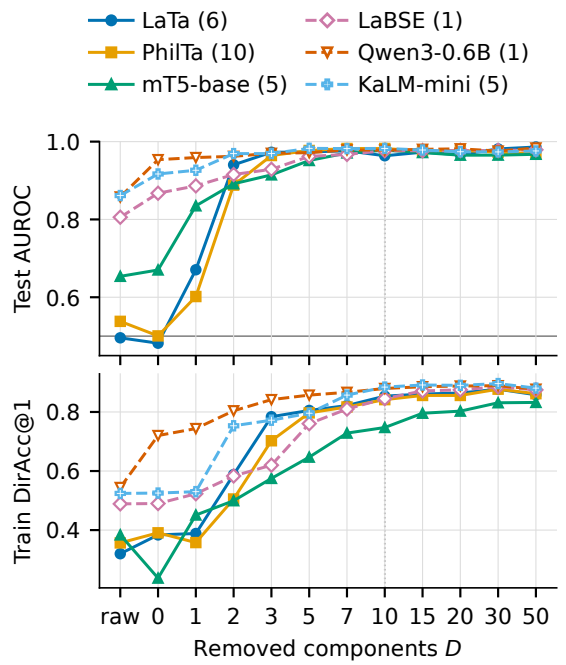



Figure 9: Test ranking AUROC (top) and training DirAcc@1 as a fraction (bottom) against removed components $D$, layers and markers and line styles as in Figure 8. Raw: uncorrected; $D = 0$: centering only; ticks unevenly spaced; dotted: $D = 10$, top of the selection grid. ABTT fit on training embeddings. LaTa and PhilTa recover nearly fully with two or three components; mT5-base keeps improving to ten.

## F Bootstrap Intervals and Routing Checks

This appendix gives the intervals behind Table 2 and the routing checks that Section 7 quotes.

**Intervals.** A bootstrap replicate draws the 514 test directories with replacement ($B = 10{,}000$, seed 233). A drawn directory keeps every file and pair: a directory drawn $c$ times weights its files and within-directory pairs by $c$ and its pairs with a directory drawn $c'$ times by $cc'$, and no pair is formed between two copies of one directory. The layer, threshold, ABTT components, $D$, and SIF weights stay fit on the training split, and every score is computed once on the full test set, so a replicate only reweights test pairs and files. Intervals are 95% percentiles; a paired difference recomputes both cells on each replicate. Before resampling, every cell was recomputed with the paper's evaluation code and matched its printed value. Table 16 gives every cell and Table 17 the contrasts quoted in Section 7.

**Routing checks.** Table 18 reports, at the headline routing cells, the AUROC of each test witness's maximum cosine for existing against new witnesses (no threshold), the gap between the best test threshold and the training threshold, the skewness of the 10-occurrence distribution (Radovanović et al., 2010), centering alone ($D = 0$), and assignment accuracy with $\tau$ refit as the exact best-F1 cut over all training pair scores.

**Threshold grid.** We keep the grid of 200 evenly spaced thresholds in $[0, 1]$ used for every printed number. Every finer grid optimizes the same training objective, the best F1 over same- against different-directory training pairs, and differs only in resolution. With the layer and $D$ fixed, a grid of 10,001 values moves 45 of the 66 printed routing numbers, by a median of 3 files (0.35 points); ABTT, SIF+ABTT, and fine-tuned cells move by at most 0.8 points, and Base and SIF cells by up to 2.6 (mT5-base's baseline). The exact best-F1 cut moves 48 of them. Re-selecting the layer and $D$ under the new grid as well moves 42 of the 60 encoder routing numbers with the fine grid and 44 with the exact cut; the lexical references have no layer or $D$ to re-select. Refitting would trade one set of differences of one to a few files for another; it changes no conclusion of Section 7.

| | | Ranking | | Routing | |
|---|---|---|---|---|---|
| **Model** | **Setting** | AUROC | Cosine gap | Assignment acc. | DirAcc@1 |
| LaTa | Base | 0.938 [0.910, 0.963] | 0.237 [0.222, 0.253] | 73.8 [70.2, 77.3] | 72.1 [68.4, 75.8] |
| | SIF | 0.948 [0.915, 0.974] | 0.366 [0.342, 0.390] | 83.2 [80.1, 86.3] | 81.8 [78.5, 85.1] |
| | ABTT | 0.971 [0.948, 0.991] | 0.525 [0.491, 0.559] | 88.5 [85.8, 90.9] | 86.1 [83.2, 88.9] |
| | SIF+ABTT | 0.966 [0.938, 0.989] | 0.580 [0.541, 0.617] | 90.2 [87.6, 92.6] | 88.2 [85.4, 90.9] |
| PhilTa | Base | 0.939 [0.918, 0.958] | 0.231 [0.217, 0.247] | 70.9 [66.9, 74.9] | 69.3 [65.2, 73.5] |
| | SIF | 0.955 [0.935, 0.972] | 0.319 [0.300, 0.339] | 79.0 [75.6, 82.4] | 77.7 [74.2, 81.3] |
| | ABTT | 0.983 [0.969, 0.993] | 0.540 [0.508, 0.571] | 91.1 [88.9, 93.2] | 88.3 [85.6, 90.9] |
| | SIF+ABTT | 0.982 [0.969, 0.993] | 0.573 [0.538, 0.608] | 91.4 [89.1, 93.6] | 88.9 [86.2, 91.5] |
| mT5-base | Base | 0.838 [0.810, 0.866] | 0.081 [0.072, 0.092] | 47.2 [43.1, 51.6] | 46.6 [42.5, 51.0] |
| | SIF | 0.845 [0.816, 0.876] | 0.036 [0.031, 0.042] | 54.5 [50.3, 58.8] | 54.1 [49.8, 58.4] |
| | ABTT | 0.975 [0.954, 0.992] | 0.536 [0.498, 0.574] | 90.3 [87.9, 92.7] | 88.5 [85.7, 91.2] |
| | SIF+ABTT | 0.974 [0.950, 0.993] | 0.519 [0.480, 0.559] | 90.3 [88.0, 92.6] | 87.5 [84.8, 90.2] |
| LaBSE | Base | 0.956 [0.932, 0.977] | 0.183 [0.171, 0.195] | 83.3 [80.2, 86.4] | 81.5 [78.1, 84.7] |
| | SIF | 0.956 [0.931, 0.977] | 0.184 [0.172, 0.197] | 83.4 [80.2, 86.5] | 81.9 [78.6, 85.2] |
| | ABTT | 0.987 [0.977, 0.995] | 0.611 [0.575, 0.646] | 90.8 [88.5, 92.9] | 88.5 [85.8, 90.9] |
| | SIF+ABTT | 0.976 [0.960, 0.989] | 0.580 [0.543, 0.617] | 89.6 [87.2, 91.9] | 86.9 [84.2, 89.5] |
| Qwen3-0.6B | Base | 0.966 [0.950, 0.981] | 0.024 [0.023, 0.026] | 82.3 [79.1, 85.4] | 80.3 [76.9, 83.6] |
| | SIF | 0.942 [0.919, 0.963] | 0.053 [0.048, 0.060] | 76.0 [72.5, 79.4] | 74.2 [70.6, 77.8] |
| | ABTT | 0.973 [0.948, 0.990] | 0.553 [0.513, 0.590] | 91.5 [89.1, 93.7] | 89.4 [86.7, 92.0] |
| | SIF+ABTT | 0.972 [0.947, 0.991] | 0.540 [0.499, 0.580] | 90.8 [88.3, 93.1] | 88.6 [85.8, 91.2] |
| KaLM-mini | Base | 0.972 [0.955, 0.986] | 0.056 [0.053, 0.060] | 87.5 [84.8, 90.1] | 85.9 [82.9, 88.7] |
| | SIF | 0.964 [0.944, 0.982] | 0.056 [0.051, 0.062] | 85.7 [82.8, 88.5] | 84.0 [80.9, 87.1] |
| | ABTT | 0.981 [0.963, 0.993] | 0.568 [0.529, 0.605] | 91.7 [89.5, 93.8] | 89.4 [86.7, 91.9] |
| | SIF+ABTT | 0.981 [0.962, 0.995] | 0.535 [0.496, 0.573] | 90.8 [88.4, 93.0] | 88.0 [85.2, 90.7] |
| LaTa (fine-tuned) | Base | 0.984 [0.973, 0.992] | 0.387 [0.368, 0.406] | 83.4 [80.2, 86.5] | 81.6 [78.1, 84.9] |
| | ABTT | 0.970 [0.947, 0.990] | 0.548 [0.511, 0.583] | 87.8 [85.2, 90.2] | 85.2 [82.3, 87.9] |
| Qwen3-0.6B (fine-tuned) | Base | 0.996 [0.993, 0.999] | 0.684 [0.653, 0.714] | 91.4 [88.7, 93.8] | 90.8 [88.1, 93.3] |
| | ABTT | 0.994 [0.989, 0.998] | 0.716 [0.680, 0.751] | 92.1 [89.6, 94.3] | 91.4 [88.8, 93.7] |
| KaLM-mini (fine-tuned) | Base | 0.997 [0.993, 0.999] | 0.636 [0.605, 0.666] | 92.4 [90.1, 94.6] | 91.7 [89.3, 94.0] |
| | ABTT | 0.994 [0.988, 0.998] | 0.651 [0.612, 0.687] | 93.4 [91.2, 95.4] | 92.5 [90.3, 94.7] |
| BM25 (word) | – | 0.974 [0.956, 0.987] | 0.243 [0.222, 0.266] | 62.5 [58.2, 66.6] | 56.3 [52.0, 60.5] |
| Char. n-grams | – | 0.987 [0.976, 0.996] | 0.555 [0.519, 0.591] | 91.8 [89.7, 93.8] | 89.9 [87.3, 92.3] |
| Levenshtein | – | 0.952 [0.927, 0.975] | 0.362 [0.332, 0.393] | 87.8 [85.0, 90.4] | 86.2 [83.2, 89.2] |

Table 16: Every headline cell (Table 2, plus SIF, SIF+ABTT, BM25 and Levenshtein) at its train-selected layer, with 95% intervals from a bootstrap over the 514 test directories in brackets. Routing in percent. The intervals cover test sampling, not the choice of split.

| Contrast | Model | Δ AUROC | Δ Assignment acc. | Δ DirAcc@1 |
|---|---|---|---|---|
| ABTT − Base | LaTa | +0.034 [0.017, 0.054] | +14.7 [11.1, 18.3] | +14.0 [10.4, 17.6] |
| | PhilTa | +0.043 [0.029, 0.060] | +20.3 [16.2, 24.5] | +19.0 [15.0, 23.1] |
| | mT5-base | +0.137 [0.109, 0.166] | +43.1 [38.6, 47.6] | +41.8 [37.2, 46.4] |
| | LaBSE | +0.030 [0.014, 0.049] | +7.5 [4.6, 10.4] | +7.0 [4.1, 10.0] |
| | Qwen3-0.6B | +0.006 [−0.010, 0.019] | +9.2 [6.3, 12.2] | +9.1 [6.1, 12.2] |
| | KaLM-mini | +0.009 [−0.002, 0.021] | +4.2 [1.7, 6.7] | +3.5 [1.0, 6.0] |
| Fine-tuned: ABTT − Base | LaTa | −0.014 [−0.032, 0.001] | +4.3 [1.7, 7.1] | +3.6 [1.0, 6.3] |
| | Qwen3-0.6B | −0.002 [−0.005, 0.001] | +0.7 [−0.7, 2.1] | +0.6 [−0.8, 2.0] |
| | KaLM-mini | −0.003 [−0.007, 0.001] | +0.9 [−0.3, 2.2] | +0.8 [−0.5, 2.1] |
| Fine-tuned ABTT − zero-shot ABTT | LaTa | −0.002 [−0.010, 0.005] | −0.7 [−2.6, 1.2] | −0.9 [−2.8, 1.0] |
| | Qwen3-0.6B | +0.022 [0.006, 0.044] | +0.6 [−1.6, 2.9] | +2.0 [0.0, 4.0] |
| | KaLM-mini | +0.013 [0.002, 0.029] | +1.6 [−0.3, 3.6] | +3.1 [1.2, 5.2] |
| Char. n-grams − ABTT | LaTa | +0.016 [0.003, 0.030] | +3.4 [1.1, 5.7] | +3.7 [1.4, 6.2] |
| | PhilTa | +0.004 [−0.002, 0.011] | +0.7 [−1.0, 2.4] | +1.5 [−0.2, 3.3] |
| | mT5-base | +0.012 [0.000, 0.027] | +1.5 [−0.6, 3.7] | +1.4 [−0.7, 3.6] |
| | LaBSE | 0.000 [−0.004, 0.005] | +1.0 [−1.0, 3.1] | +1.4 [−0.8, 3.6] |
| | Qwen3-0.6B | +0.014 [0.002, 0.031] | +0.3 [−1.6, 2.3] | +0.5 [−1.4, 2.5] |
| | KaLM-mini | +0.006 [−0.002, 0.016] | +0.1 [−1.6, 1.9] | +0.5 [−1.3, 2.3] |
| Centering − Base (ABTT layer) | LaTa | +0.034 [0.021, 0.049] | −2.2 [−5.2, 0.9] | +6.8 [4.6, 9.1] |
| | PhilTa | −0.038 [−0.060, −0.016] | +12.9 [9.0, 16.9] | +12.4 [8.5, 16.3] |
| | mT5-base | +0.088 [0.066, 0.109] | +20.5 [16.0, 24.8] | +15.6 [11.3, 19.9] |
| | LaBSE | −0.002 [−0.009, 0.004] | −4.7 [−7.4, −2.0] | −4.7 [−7.3, −2.0] |
| | Qwen3-0.6B | +0.076 [0.061, 0.092] | +19.2 [15.4, 23.2] | +16.7 [12.7, 20.7] |
| | KaLM-mini | +0.070 [0.052, 0.089] | +19.3 [15.5, 23.2] | +16.9 [13.1, 20.7] |
| ABTT − centering (ABTT layer) | LaTa | −0.001 [−0.016, 0.012] | +40.4 [35.8, 44.8] | +49.2 [44.1, 53.9] |
| | PhilTa | +0.482 [0.460, 0.504] | +7.3 [4.9, 10.0] | +6.6 [4.1, 9.3] |
| | mT5-base | +0.065 [0.050, 0.081] | +24.5 [20.7, 28.3] | +27.7 [23.7, 31.7] |
| | LaBSE | +0.029 [0.013, 0.048] | +12.1 [8.8, 15.5] | +11.7 [8.4, 15.0] |
| | Qwen3-0.6B | +0.033 [0.018, 0.050] | +25.9 [22.0, 29.9] | +27.7 [23.5, 32.1] |
| | KaLM-mini | +0.023 [0.008, 0.040] | +12.0 [9.1, 15.1] | +13.8 [10.7, 17.0] |
| Spread over six models | Base | 0.134 [0.110, 0.158] | 40.3 [35.7, 44.7] | 39.3 [34.6, 43.7] |
| | ABTT | 0.015 [0.006, 0.034] | 3.3 [1.8, 5.8] | 3.3 [1.8, 6.0] |

Table 17: Paired differences between headline cells at train-selected layers, with 95% intervals from the bootstrap of Table 16. Centering rows use each column's ABTT layer. Spread: largest minus smallest of the six zero-shot models. Routing in points. Differences use unrounded cells, so they can differ by one last digit from printed-cell differences.

| Model | Setting | L | Cos. SD | Exist./new AUROC | Assign. | Oracle gap | $\tau$ | Exact | Skew |
|---|---|---|---|---|---|---|---|---|---|
| LaTa | Base | 1 | 0.112 | 0.869 [0.840, 0.896] | 73.8 | +5.6 [2.8, 9.4] | 0.844 | 74.6 | 2.81 |
| | Base @ ABTT L | 8 | 0.632 | 0.621 [0.583, 0.658] | 50.2 | +16.4 [11.9, 21.9] | 0.995 | 44.1 | 1.59 |
| | Centering @ ABTT L | 8 | 0.901 | 0.604 [0.564, 0.644] | 48.0 | +14.3 [9.4, 21.8] | 0.995 | 42.8 | 1.97 |
| | ABTT | 8 | 0.100 | 0.943 [0.925, 0.960] | 88.5 | +0.6 [0.2, 1.7] | 0.472 | 88.5 | 1.03 |
| PhilTa | Base | 1 | 0.111 | 0.843 [0.813, 0.873] | 70.9 | +7.0 [3.7, 11.5] | 0.894 | 71.0 | 3.15 |
| | Centering @ ABTT L | 1 | 0.126 | 0.906 [0.882, 0.928] | 83.8 | +1.2 [0.6, 2.5] | 0.548 | 84.4 | 1.78 |
| | ABTT | 1 | 0.086 | 0.960 [0.944, 0.974] | 91.1 | +0.7 [0.2, 1.8] | 0.417 | 91.1 | 0.72 |
| mT5-base | Base | 12 | 0.104 | 0.685 [0.642, 0.726] | 47.2 | +19.6 [14.3, 26.8] | 0.970 | 49.5 | 2.59 |
| | Base @ ABTT L | 1 | 0.065 | 0.696 [0.654, 0.737] | 45.3 | +23.1 [16.5, 30.2] | 0.995 | 52.2 | 3.83 |
| | Centering @ ABTT L | 1 | 0.242 | 0.765 [0.729, 0.801] | 65.9 | +5.6 [2.0, 10.5] | 0.688 | 66.0 | 3.29 |
| | ABTT | 1 | 0.081 | 0.952 [0.935, 0.967] | 90.3 | +0.9 [0.2, 2.4] | 0.437 | 91.1 | 1.13 |
| LaBSE | Base | 11 | 0.039 | 0.922 [0.901, 0.942] | 83.3 | +1.0 [0.1, 3.5] | 0.975 | 82.5 | 2.00 |
| | Centering @ ABTT L | 11 | 0.209 | 0.891 [0.866, 0.914] | 78.7 | +2.8 [0.7, 6.3] | 0.704 | 79.0 | 1.84 |
| | ABTT | 11 | 0.102 | 0.960 [0.944, 0.973] | 90.8 | +0.6 [0.1, 1.6] | 0.452 | 90.8 | 0.58 |
| Qwen3-0.6B | Base | 28 | 0.043 | 0.900 [0.874, 0.923] | 82.3 | +1.0 [0.2, 2.6] | 0.945 | 82.6 | 2.51 |
| | Base @ ABTT L | 5 | 0.034 | 0.708 [0.666, 0.748] | 46.4 | +23.4 [17.7, 30.4] | 0.990 | 53.0 | 2.49 |
| | Centering @ ABTT L | 5 | 0.171 | 0.802 [0.767, 0.835] | 65.6 | +9.6 [5.9, 14.5] | 0.653 | 66.6 | 1.77 |
| | ABTT | 5 | 0.086 | 0.954 [0.935, 0.971] | 91.5 | +0.7 [0.2, 1.7] | 0.417 | 91.5 | 1.07 |
| KaLM-mini | Base | 23 | 0.021 | 0.931 [0.910, 0.950] | 87.5 | 0.0 [0.0, 1.1] | 0.965 | 86.2 | 1.97 |
| | Base @ ABTT L | 3 | 0.054 | 0.756 [0.719, 0.792] | 60.4 | +11.3 [6.7, 17.0] | 0.975 | 58.3 | 3.58 |
| | Centering @ ABTT L | 3 | 0.135 | 0.881 [0.853, 0.906] | 79.7 | +2.7 [1.2, 4.8] | 0.558 | 79.5 | 1.75 |
| | ABTT | 3 | 0.085 | 0.958 [0.941, 0.973] | 91.7 | +0.2 [0.0, 1.6] | 0.397 | 91.7 | 0.87 |

Table 18: Routing checks at the headline routing cells: Base and ABTT at their train-selected layers (L), baseline and centering ($D = 0$) at the ABTT layer. Cos. SD: SD of test pairwise cosines; $\tau$: threshold fit on training pairs. Exist./new AUROC: threshold-free, from each witness's maximum cosine. Assign.: assignment accuracy (%) at the training $\tau$. Oracle gap: accuracy at the best test threshold minus Assign. Exact: Assign. with $\tau$ at the exact training best-F1 cut. Skew: skewness of the 10-occurrence distribution (Radovanović et al., 2010). Brackets: 95% bootstrap intervals.

## G Routing Gain Without a Threshold

This appendix gives the paired contrasts behind the second finding of Section 7, for the three embedding-trained encoders. Each contrast compares the ABTT cell with the baseline cell, each at its own train-selected routing layer, and its 95% interval comes from the directory-level bootstrap of Appendix F. Per-cell values are in Table 18.

**No threshold.** The existing-versus-new AUROC of each test witness's best-match cosine rises by 0.037 (0.021 to 0.055) for LaBSE, 0.055 (0.034 to 0.077) for Qwen3-0.6B, and 0.027 (0.012 to 0.043) for KaLM-mini.

**Best test threshold.** At the baseline cells the train-fit threshold comes within 1.0 point of the best test threshold. When both cells use their best test threshold, ABTT still raises assignment accuracy by 7.0 points (4.2 to 9.3) for LaBSE, 8.9 (6.1 to 11.7) for Qwen3-0.6B, and 4.4 (2.2 to 6.8) for KaLM-mini.

**The training mean.** Centering alone ($D = 0$) at the baseline cell's layer changes assignment accuracy by $-4.7$ points ($-7.4$ to $-2.0$) for LaBSE, $+2.6$ ($-0.4$ to $5.5$) for Qwen3-0.6B, and $-1.9$ ($-3.7$ to $-0.1$) for KaLM-mini. At the ABTT cell's layer, the removed components add 12.1, 25.9 and 12.0 points on top of centering (Table 17).

**Hubness.** From the baseline cell to the ABTT cell, the skewness of the 10-occurrence distribution (Radovanović et al., 2010) falls from 2.0–2.5 to 0.6–1.1, while centering alone at the ABTT layer leaves it near 1.8. Across representations, low hubness does not track good routing. The character n-gram reference has a skewness of 10.6 and routes as well as the best projected encoder, and SIF raises skewness in LaTa and PhilTa while it raises their routing. We therefore report the drop in hubness as an observation and do not test it as a cause.

## H Reference Systems

Table 2 carries four reference rows below the six encoders: one supervised fine-tuning ceiling for each of the three fine-tuned models, and the character 3–5-gram TF-IDF reference. This appendix describes them. Table 19 gives the three fine-tuning ceilings with the zero-shot rows they are compared against. Some routing numbers are means over *five reseedings*: each reseeding keeps the training/test split and redraws, within each directory's test witnesses, which serve as queries and which as references; queries are routed against the reference directories. Every model we fine-tuned is reported, on whichever side of its own zero-shot row it lands.

Three encoders are fine-tuned under one recipe, LaTa, Qwen3-0.6B and KaLM-mini. Fine-tuning only the Latin-pretrained model would leave the ceiling confounded with Latin pretraining, so the second model is a multilingual decoder-only encoder with no Latin-specific pretraining, Qwen3-0.6B. The third, KaLM-mini, has the best ABTT cell in the paper (91.7 assignment accuracy; 89.4 DirAcc@1, tied with Qwen3-0.6B), so it tests whether supervision can clear that cell.

All three are trained with symmetric InfoNCE over positive pairs and in-batch negatives (no two pairs from one directory in a batch), temperature 0.05, AdamW at learning rate $2 \times 10^{-5}$ with weight decay 0.01, linear warmup over the first 10 percent of steps then linear decay, gradient clipping at 1.0, 16 pairs per batch, up to 8 epochs with early stopping after 3 epochs without dev improvement, and

seed 42. All three are mean-pooled with the same token filter as their zero-shot rows, so the fine-tuned vectors live in the space those rows are measured in; Qwen3-0.6B and KaLM-mini add gradient checkpointing, which recomputes activations rather than storing them and leaves the gradients unchanged.

The dev slice holds out 28 of the 190 train directories with at least two files (71 files), the same directories for every model, so each trains on 499 of the 565 positive train pairs, and the checkpoint is chosen by dev DirAcc@1 with dev AUROC as the tie-break. Accuracy differences at or below one dev file are treated as ties, because one file out of 71 is the pool's whole resolution; without that, a model whose zero-shot encoder already routes all 71 dev files can never select a trained checkpoint. A 71-file pool is the weak link in this protocol: one borderline file flipped between two otherwise identical Qwen3-0.6B runs at a training loss of $6\times10^{-4}$, which is GPU non-determinism rather than a difference in the model, and it is worth 1.4 points of the selection criterion. LaTa selects epoch 7, the last epoch run before early stopping fired, and its ABTT sweep selects $D = 10$, the top of the grid, at every layer, so the tuned and fixed ABTT variants coincide there. Qwen3-0.6B selects epoch 3 of 6 run, and its sweep is a genuine one, selecting the top of the grid at 12 of 28 layers and $D = 2$ at the layer its routing row reports. KaLM-mini selects epoch 1 of 4 run, the only model of the three whose zero-shot encoder leaves the dev pool room to improve, and its sweep selects the top of the grid at 19 of 24 layers including the layer its routing row reports. All three rows are ceilings at this training budget rather than asymptotes: training loss reaches $4 \times 10^{-5}$ on Qwen3-0.6B by epoch 4 and $3 \times 10^{-5}$ on KaLM-mini, so the 499 pairs are memorized and more epochs buy nothing.

**Compute.** Most hidden-state extraction and all fine-tuning ran on a single NVIDIA A100 GPU; the control-model extraction and the token audit ran on a single NVIDIA A40, and T5-efficient-base, with a re-extraction of the four controls for Table 9, on CPU. The whole project, exploratory runs included, used about 160 GPU-hours and under 200 CPU core-hours. Every post-hoc analysis (ABTT, whitening, zeroing, bootstrap intervals) ran on CPU from cached vectors.

**Fine-tuning ties the projection on LaTa routing and adds about two points on Qwen3-0.6B and KaLM-mini.** Fine-tuning LaTa on the positive training pairs gives an uncorrected ranking AUROC of 0.984, above every zero-shot baseline cell (0.838 to 0.972), and 83.4 assignment accuracy. With ABTT on top it reaches 87.8 assignment accuracy and 85.2 DirAcc@1. In point estimate that is below every zero-shot ABTT cell (88.5 to 91.7 and 86.1 to 89.4). Against LaTa's own zero-shot ABTT cell, however, the DirAcc@1 difference is $-0.9$ points ($-2.8$ to 1.0; Table 17), and over five reseedings both reach 87.7. ABTT lowers the fine-tuned row's AUROC from 0.984 to 0.970 in point estimate ($-0.014$, interval $-0.032$ to $0.001$) while it widens the cosine gap from 0.387 to 0.548. On the other two encoders the same recipe goes further. With ABTT, fine-tuned Qwen3-0.6B reaches 92.1 assignment accuracy and 91.4 DirAcc@1, and fine-tuned KaLM-mini 93.4 and 92.5, both above every zero-shot ABTT cell in point estimate. On the single split only KaLM-mini's DirAcc@1 lead over its zero-shot ABTT cell excludes zero ($+3.1$, 1.2 to 5.2). Over five reseedings the fine-tuned encoder with ABTT beats its zero-shot ABTT row by 1.8 points on Qwen3-0.6B (92.3 against 90.5) and by 2.1 points on KaLM-mini (93.0 against 90.9). The tie with the projection is therefore a statement about LaTa, not about post-processing in general. All three ceilings are also flattered: 206 of the 535 test witnesses whose directory already exists (38.5 percent) sit in a directory that supplied a training pair, and witnesses inside a directory are near-duplicates, so each margin is an upper bound.

**Lexical reference systems.** Table 20 gives the three lexical reference systems (BM25, character 3–5-gram TF-IDF, and Levenshtein similarity) on both tasks; the character n-gram row is the reference quoted in Section 7. BM25 uses $k_1 = 1.5$ and $b = 0.75$, with document frequencies, vocabulary and average document length taken from training witnesses only. The character TF-IDF vectorizer is also fit on training witnesses only, and normalized Levenshtein similarity has nothing to fit. Every score matrix is min-max rescaled using its training block. The map is monotone, so AUROC is unchanged, but $\tau$ is fit on a fixed grid after it. That matters for BM25, whose $\tau$ would pin at the grid ceiling without rescaling, and is the identity for character TF-IDF, whose training scores already span $[0, 1]$. BM25 routes poorly because a raw BM25 score grows with query length, so a witness's maximum score tracks document length as much as relatedness, and the $\tau$ fit on the

| System | Ranking | | Routing | | |
|---|---|---|---|---|---|
| | AUROC | Cosine gap | Assign. acc. | DirAcc@1 | DirAcc@1 5 reseedings |
| LaTa (zero-shot) | 0.938 | 0.237 | 73.8 | 72.1 | 73.1 ± 1.0 |
| LaTa (zero-shot) + ABTT | 0.971 | 0.525 | 88.5 | 86.1 | 87.7 ± 0.4 |
| LaTa (fine-tuned) | 0.984 | 0.387 | 83.4 | 81.6 | 83.4 ± 0.9 |
| LaTa (fine-tuned) + ABTT | 0.970 | 0.548 | 87.8 | 85.2 | 87.7 ± 0.8 |
| Qwen3-0.6B (zero-shot) | 0.966 | 0.024 | 82.3 | 80.3 | 82.4 ± 1.0 |
| Qwen3-0.6B (zero-shot) + ABTT | 0.973 | 0.553 | 91.5 | 89.4 | 90.5 ± 0.4 |
| Qwen3-0.6B (fine-tuned) | 0.996 | 0.684 | 91.4 | 90.8 | 91.7 ± 0.6 |
| Qwen3-0.6B (fine-tuned) + ABTT | 0.994 | 0.716 | 92.1 | 91.4 | 92.3 ± 0.2 |
| KaLM-mini (zero-shot) | 0.972 | 0.056 | 87.5 | 85.9 | 87.3 ± 0.7 |
| KaLM-mini (zero-shot) + ABTT | 0.981 | 0.568 | 91.7 | 89.4 | 90.9 ± 0.6 |
| KaLM-mini (fine-tuned) | 0.997 | 0.636 | 92.4 | 91.7 | 92.6 ± 0.8 |
| KaLM-mini (fine-tuned) + ABTT | 0.994 | 0.651 | 93.4 | 92.5 | 93.0 ± 0.5 |

Table 19: Fine-tuning ceiling for LaTa, Qwen3-0.6B and KaLM-mini on 499 training pairs; layers as in Table 21. Test scores; routing in percent; last column: mean ± s.d. over five reseedings. Fine-tuning raises ranking AUROC and ABTT lowers it in point estimate; on reseeded routing, fine-tuned ABTT ties zero-shot ABTT for LaTa and exceeds it for Qwen3-0.6B and KaLM-mini.

| System | Ranking | | Routing | |
|---|---|---|---|---|
| | AUROC | Gap | Assign. | DirAcc@1 |
| BM25 (word) | 0.974 | 0.243 | 62.5 | 56.3 |
| TF-IDF char 3–5 | 0.987 | 0.555 | 91.8 | 89.9 |
| Levenshtein | 0.952 | 0.362 | 87.8 | 86.2 |

Table 20: Test scores of lexical reference systems on the same training/test split (847/858 witnesses) and evaluation code as the encoders of Table 2. Gap: mean same-source minus mean different-source score. Routing in percent; Assign.: existing-versus-new accuracy, with $\tau$ fit on the training split. TF-IDF char 3–5 routes best; BM25 ranks pairs well but routes poorly.

training split transfers badly. Over five reseedings, DirAcc@1 is 68.3 ± 7.1 for BM25, 91.1 ± 0.5 for character TF-IDF and 87.2 ± 0.3 for Levenshtein similarity.

# I Per-Layer Results

This appendix gives the per-layer numbers behind Figure 1 and the routing results at every layer. Table 21 lists the train-selected layer behind every headline cell. Tables 22, 23 and 24 give ranking AUROC, routing assignment accuracy and routing DirAcc@1 at every layer of LaTa, PhilTa and mT5-base. Tables 25, 26 and 27 give the same for LaBSE, Qwen3-0.6B and KaLM-mini. Each compares baseline mean pooling with ABTT, its $D$ tuned per layer on the training split. Fixing the number of removed components at ten gives the same range (AUROC 0.962 to 0.987 at all 100 panel layers), so the result does not rest on how that number is chosen.

| | Ranking layer | | | | Routing layer | | | |
|---|---|---|---|---|---|---|---|---|
| **Model** | Base | SIF | ABTT | SIF+ABTT | Base | SIF | ABTT | SIF+ABTT |
| LaTa | 12 | 1 | 12 | 7 | 1 | 1 | 8 | 12 |
| PhilTa | 1 | 1 | 9 | 8 | 1 | 1 | 1 | 1 |
| mT5-base | 12 | 1 | 2 | 1 | 12 | 1 | 1 | 2 |
| LaBSE | 12 | 12 | 11 | 10 | 11 | 12 | 11 | 11 |
| Qwen3-0.6B | 26 | 22 | 2 | 2 | 28 | 27 | 5 | 5 |
| KaLM-mini | 23 | 23 | 1 | 2 | 23 | 22 | 3 | 1 |
| LaTa (fine-tuned) | 12 | – | 12 | – | 12 | – | 12 | – |
| Qwen3-0.6B (fine-tuned) | 28 | – | 27 | – | 28 | – | 27 | – |
| KaLM-mini (fine-tuned) | 24 | – | 24 | – | 24 | – | 24 | – |

Table 21: Train-selected layer behind each encoder cell of Tables 2, 16 and 19: the layer of highest training AUROC (ranking) or training DirAcc@1 (routing). Layer 1 is the first transformer block. Fine-tuned rows have no SIF setting.

| Model | Layer | Ranking (T5 encoders) | | | |
|---|---|---|---|---|---|
| | | AUROC (Base) | AUROC (ABTT) | Cosine gap (Base) | Cosine gap (ABTT) |
| LaTa | 1 | 0.934 | 0.970 | 0.286 | 0.566 |
| | 2 | 0.563 | 0.969 | 0.019 | 0.554 |
| | 3 | 0.513 | 0.973 | -0.048 | 0.555 |
| | 4 | 0.500 | 0.971 | -0.067 | 0.565 |
| | 5 | 0.496 | 0.972 | -0.069 | 0.572 |
| | 6 | 0.496 | 0.964 | -0.069 | 0.569 |
| | 7 | 0.498 | 0.962 | -0.072 | 0.564 |
| | 8 | 0.502 | 0.965 | -0.074 | 0.558 |
| | 9 | 0.503 | 0.965 | -0.073 | 0.557 |
| | 10 | 0.505 | 0.971 | -0.072 | 0.551 |
| | 11 | 0.508 | 0.973 | -0.067 | 0.550 |
| | **12** | 0.938 | **0.971** | 0.237 | **0.525** |
| PhilTa | 1 | 0.939 | 0.977 | 0.231 | 0.574 |
| | 2 | 0.911 | 0.973 | 0.161 | 0.553 |
| | 3 | 0.601 | 0.973 | 0.085 | 0.549 |
| | 4 | 0.551 | 0.978 | 0.063 | 0.551 |
| | 5 | 0.542 | 0.979 | 0.053 | 0.551 |
| | 6 | 0.541 | 0.981 | 0.047 | 0.551 |
| | 7 | 0.542 | 0.979 | 0.046 | 0.554 |
| | 8 | 0.542 | 0.984 | 0.046 | 0.555 |
| | **9** | 0.539 | **0.983** | 0.041 | **0.540** |
| | 10 | 0.538 | 0.982 | 0.040 | 0.531 |
| | 11 | 0.539 | 0.982 | 0.041 | 0.511 |
| | 12 | 0.911 | 0.982 | 0.181 | 0.487 |
| mT5-base | 1 | 0.822 | 0.979 | 0.027 | 0.550 |
| | **2** | 0.822 | **0.975** | 0.026 | **0.536** |
| | 3 | 0.804 | 0.968 | 0.025 | 0.517 |
| | 4 | 0.799 | 0.974 | 0.040 | 0.502 |
| | 5 | 0.654 | 0.978 | 0.219 | 0.482 |
| | 6 | 0.657 | 0.978 | 0.238 | 0.483 |
| | 7 | 0.656 | 0.980 | 0.244 | 0.482 |
| | 8 | 0.656 | 0.982 | 0.249 | 0.471 |
| | 9 | 0.660 | 0.982 | 0.235 | 0.475 |
| | 10 | 0.659 | 0.977 | 0.226 | 0.459 |
| | 11 | 0.659 | 0.974 | 0.213 | 0.449 |
| | 12 | 0.838 | 0.972 | 0.081 | 0.442 |

Table 22: Per-layer test ranking AUROC and cosine gap for LaTa, PhilTa and mT5-base. Base: mean pooling; ABTT: fit on the training split, $D$ tuned per layer. Bold: the layer behind the ABTT cell of Table 16 (best training AUROC under ABTT) and its ABTT values, not always the best test layer.

| Model | Layer | Routing (T5 encoders, witness-level, $\tau$-thresholded) | | | | | |
|---|---|---|---|---|---|---|---|
| | | Existing (Base) | Existing (ABTT) | New (Base) | New (ABTT) | Overall (Base) | Overall (ABTT) |
| LaTa | 1 | 0.630 | 0.910 | 0.916 | 0.907 | 0.738 | 0.909 |
| | 2 | 0.153 | 0.905 | 0.913 | 0.892 | 0.439 | 0.900 |
| | 3 | 0.256 | 0.882 | 0.842 | 0.950 | 0.477 | 0.908 |
| | 4 | 0.389 | 0.893 | 0.718 | 0.920 | 0.513 | 0.903 |
| | 5 | 0.374 | 0.892 | 0.715 | 0.913 | 0.502 | 0.900 |
| | 6 | 0.378 | 0.884 | 0.712 | 0.913 | 0.503 | 0.895 |
| | 7 | 0.372 | 0.875 | 0.715 | 0.916 | 0.501 | 0.890 |
| | **8** | 0.379 | **0.884** | 0.706 | **0.885** | 0.502 | **0.885** |
| | 9 | 0.366 | 0.886 | 0.724 | 0.873 | 0.501 | 0.881 |
| | 10 | 0.344 | 0.873 | 0.752 | 0.882 | 0.498 | 0.876 |
| | 11 | 0.310 | 0.867 | 0.786 | 0.901 | 0.490 | 0.880 |
| | 12 | 0.576 | 0.890 | 0.932 | 0.833 | 0.710 | 0.868 |
| PhilTa | **1** | 0.591 | **0.918** | 0.904 | **0.901** | 0.709 | **0.911** |
| | 2 | 0.479 | 0.895 | 0.867 | 0.885 | 0.625 | 0.892 |
| | 3 | 0.193 | 0.897 | 0.889 | 0.879 | 0.455 | 0.890 |
| | 4 | 0.148 | 0.886 | 0.923 | 0.873 | 0.439 | 0.881 |
| | 5 | 0.252 | 0.879 | 0.796 | 0.873 | 0.457 | 0.876 |
| | 6 | 0.338 | 0.905 | 0.700 | 0.817 | 0.474 | 0.872 |
| | 7 | 0.301 | 0.901 | 0.746 | 0.830 | 0.469 | 0.874 |
| | 8 | 0.262 | 0.875 | 0.808 | 0.885 | 0.467 | 0.879 |
| | 9 | 0.303 | 0.890 | 0.762 | 0.870 | 0.476 | 0.882 |
| | 10 | 0.299 | 0.908 | 0.780 | 0.793 | 0.480 | 0.865 |
| | 11 | 0.256 | 0.869 | 0.824 | 0.836 | 0.470 | 0.857 |
| | 12 | 0.338 | 0.880 | 0.898 | 0.876 | 0.549 | 0.879 |
| mT5-base | **1** | 0.135 | **0.877** | 0.981 | **0.947** | 0.453 | **0.903** |
| | 2 | 0.110 | 0.890 | 0.975 | 0.910 | 0.436 | 0.897 |
| | 3 | 0.161 | 0.794 | 0.926 | 0.941 | 0.449 | 0.850 |
| | 4 | 0.269 | 0.787 | 0.858 | 0.882 | 0.491 | 0.823 |
| | 5 | 0.422 | 0.746 | 0.659 | 0.864 | 0.512 | 0.790 |
| | 6 | 0.636 | 0.766 | 0.489 | 0.876 | 0.580 | 0.808 |
| | 7 | 0.594 | 0.793 | 0.517 | 0.851 | 0.565 | 0.815 |
| | 8 | 0.557 | 0.809 | 0.560 | 0.842 | 0.558 | 0.822 |
| | 9 | 0.364 | 0.841 | 0.746 | 0.814 | 0.508 | 0.831 |
| | 10 | 0.314 | 0.794 | 0.783 | 0.876 | 0.491 | 0.825 |
| | 11 | 0.297 | 0.763 | 0.780 | 0.854 | 0.479 | 0.797 |
| | 12 | 0.198 | 0.834 | 0.926 | 0.799 | 0.472 | 0.821 |

Table 23: Per-layer test routing assignment accuracy for LaTa, PhilTa and mT5-base, overall and among existing and new witnesses, as fractions. Base: mean pooling; ABTT: fit on the training split, $D$ tuned per layer. Bold: the layer behind the ABTT cell of Table 16 (best training DirAcc@1 under ABTT) and its ABTT values, not always the best test layer.

| | | Routing: DirAcc@1 (T5 encoders, single seed) | | | | | |
|---|---|---|---|---|---|---|---|
| **Model** | **Layer** | DirAcc@1 (Base) | DirAcc@1 (ABTT) | Existing (Base) | Existing (ABTT) | New (Base) | New (ABTT) |
| LaTa | 1 | 0.721 | 0.880 | 0.630 | 0.910 | 0.916 | 0.907 |
| | 2 | 0.383 | 0.869 | 0.153 | 0.905 | 0.913 | 0.892 |
| | 3 | 0.350 | 0.882 | 0.256 | 0.882 | 0.842 | 0.950 |
| | 4 | 0.303 | 0.881 | 0.389 | 0.893 | 0.718 | 0.920 |
| | 5 | 0.297 | 0.875 | 0.374 | 0.892 | 0.715 | 0.913 |
| | 6 | 0.297 | 0.874 | 0.378 | 0.884 | 0.712 | 0.913 |
| | 7 | 0.302 | 0.868 | 0.372 | 0.875 | 0.715 | 0.916 |
| | **8** | 0.302 | **0.861** | 0.379 | **0.884** | 0.706 | **0.885** |
| | 9 | 0.309 | 0.858 | 0.366 | 0.886 | 0.724 | 0.873 |
| | 10 | 0.317 | 0.852 | 0.344 | 0.873 | 0.752 | 0.882 |
| | 11 | 0.331 | 0.857 | 0.310 | 0.867 | 0.786 | 0.901 |
| | 12 | 0.689 | 0.841 | 0.576 | 0.890 | 0.932 | 0.833 |
| PhilTa | **1** | 0.693 | **0.883** | 0.591 | **0.918** | 0.904 | **0.901** |
| | 2 | 0.603 | 0.869 | 0.479 | 0.895 | 0.867 | 0.885 |
| | 3 | 0.389 | 0.865 | 0.193 | 0.897 | 0.889 | 0.879 |
| | 4 | 0.369 | 0.859 | 0.148 | 0.886 | 0.923 | 0.873 |
| | 5 | 0.334 | 0.854 | 0.252 | 0.879 | 0.796 | 0.873 |
| | 6 | 0.310 | 0.845 | 0.338 | 0.905 | 0.700 | 0.817 |
| | 7 | 0.325 | 0.847 | 0.301 | 0.901 | 0.746 | 0.830 |
| | 8 | 0.350 | 0.857 | 0.262 | 0.875 | 0.808 | 0.885 |
| | 9 | 0.337 | 0.857 | 0.303 | 0.890 | 0.762 | 0.870 |
| | 10 | 0.344 | 0.833 | 0.299 | 0.908 | 0.780 | 0.793 |
| | 11 | 0.355 | 0.829 | 0.256 | 0.869 | 0.824 | 0.836 |
| | 12 | 0.528 | 0.854 | 0.338 | 0.880 | 0.898 | 0.876 |
| mT5-base | **1** | 0.451 | **0.885** | 0.135 | **0.877** | 0.981 | **0.947** |
| | 2 | 0.435 | 0.874 | 0.110 | 0.890 | 0.975 | 0.910 |
| | 3 | 0.445 | 0.830 | 0.161 | 0.794 | 0.926 | 0.941 |
| | 4 | 0.467 | 0.798 | 0.269 | 0.787 | 0.858 | 0.882 |
| | 5 | 0.352 | 0.756 | 0.422 | 0.746 | 0.659 | 0.864 |
| | 6 | 0.346 | 0.775 | 0.636 | 0.766 | 0.489 | 0.876 |
| | 7 | 0.352 | 0.775 | 0.594 | 0.793 | 0.517 | 0.851 |
| | 8 | 0.366 | 0.784 | 0.557 | 0.809 | 0.560 | 0.842 |
| | 9 | 0.394 | 0.787 | 0.364 | 0.841 | 0.746 | 0.814 |
| | 10 | 0.387 | 0.788 | 0.314 | 0.794 | 0.783 | 0.876 |
| | 11 | 0.378 | 0.765 | 0.297 | 0.763 | 0.780 | 0.854 |
| | 12 | 0.466 | 0.779 | 0.198 | 0.834 | 0.926 | 0.799 |

Table 24: Per-layer test routing DirAcc@1 for LaTa, PhilTa and mT5-base, as fractions, computed once, without reseedings. Existing, New: assignment accuracy on existing or new witnesses. Base: mean pooling; ABTT: fit on the training split, $D$ tuned per layer. Bold: the layer behind the ABTT cell of Table 16 (best training DirAcc@1 under ABTT) and its ABTT values, not always the best test layer.

| Model | Layer | Ranking (embedding-trained models) | | | |
|---|---|---|---|---|---|
| | | AUROC (Base) | AUROC (ABTT) | Cosine gap (Base) | Cosine gap (ABTT) |
| LaBSE | 1 | 0.806 | 0.977 | 0.060 | 0.520 |
| | 2 | 0.833 | 0.979 | 0.037 | 0.531 |
| | 3 | 0.853 | 0.978 | 0.062 | 0.543 |
| | 4 | 0.862 | 0.977 | 0.073 | 0.525 |
| | 5 | 0.864 | 0.968 | 0.066 | 0.501 |
| | 6 | 0.894 | 0.976 | 0.078 | 0.532 |
| | 7 | 0.887 | 0.975 | 0.088 | 0.524 |
| | 8 | 0.879 | 0.975 | 0.080 | 0.522 |
| | 9 | 0.916 | 0.979 | 0.090 | 0.565 |
| | 10 | 0.934 | 0.982 | 0.103 | 0.586 |
| | **11** | 0.960 | **0.987** | 0.061 | **0.611** |
| | 12 | 0.956 | 0.981 | 0.183 | 0.592 |
| Qwen3-0.6B | 1 | 0.858 | 0.978 | 0.033 | 0.564 |
| | **2** | 0.863 | **0.973** | 0.024 | **0.553** |
| | 3 | 0.862 | 0.980 | 0.037 | 0.556 |
| | 4 | 0.860 | 0.986 | 0.036 | 0.547 |
| | 5 | 0.861 | 0.981 | 0.030 | 0.545 |
| | 6 | 0.867 | 0.983 | 0.035 | 0.545 |
| | 7 | 0.876 | 0.980 | 0.033 | 0.544 |
| | 8 | 0.882 | 0.972 | 0.037 | 0.532 |
| | 9 | 0.881 | 0.977 | 0.034 | 0.532 |
| | 10 | 0.887 | 0.976 | 0.031 | 0.529 |
| | 11 | 0.895 | 0.977 | 0.030 | 0.530 |
| | 12 | 0.901 | 0.973 | 0.027 | 0.530 |
| | 13 | 0.907 | 0.974 | 0.026 | 0.523 |
| | 14 | 0.905 | 0.974 | 0.024 | 0.519 |
| | 15 | 0.901 | 0.971 | 0.022 | 0.514 |
| | 16 | 0.910 | 0.975 | 0.026 | 0.509 |
| | 17 | 0.916 | 0.979 | 0.032 | 0.521 |
| | 18 | 0.933 | 0.983 | 0.032 | 0.537 |
| | 19 | 0.936 | 0.980 | 0.035 | 0.541 |
| | 20 | 0.943 | 0.982 | 0.041 | 0.559 |
| | 21 | 0.953 | 0.982 | 0.047 | 0.562 |
| | 22 | 0.963 | 0.983 | 0.044 | 0.583 |
| | 23 | 0.961 | 0.984 | 0.038 | 0.578 |
| | 24 | 0.963 | 0.984 | 0.032 | 0.576 |
| | 25 | 0.961 | 0.983 | 0.027 | 0.578 |
| | 26 | 0.966 | 0.985 | 0.024 | 0.568 |
| | 27 | 0.967 | 0.982 | 0.017 | 0.559 |
| | 28 | 0.958 | 0.975 | 0.098 | 0.556 |
| KaLM-mini | **1** | 0.887 | **0.981** | 0.063 | **0.568** |
| | 2 | 0.906 | 0.983 | 0.044 | 0.566 |
| | 3 | 0.881 | 0.981 | 0.062 | 0.551 |
| | 4 | 0.871 | 0.983 | 0.060 | 0.553 |
| | 5 | 0.861 | 0.983 | 0.062 | 0.541 |
| | 6 | 0.872 | 0.983 | 0.063 | 0.529 |
| | 7 | 0.888 | 0.979 | 0.072 | 0.524 |
| | 8 | 0.890 | 0.976 | 0.072 | 0.524 |
| | 9 | 0.892 | 0.974 | 0.066 | 0.507 |
| | 10 | 0.893 | 0.971 | 0.067 | 0.494 |
| | 11 | 0.894 | 0.970 | 0.061 | 0.475 |
| | 12 | 0.905 | 0.970 | 0.071 | 0.475 |
| | 13 | 0.900 | 0.968 | 0.066 | 0.467 |
| | 14 | 0.904 | 0.971 | 0.063 | 0.471 |
| | 15 | 0.907 | 0.973 | 0.067 | 0.468 |
| | 16 | 0.917 | 0.972 | 0.071 | 0.489 |
| | 17 | 0.926 | 0.975 | 0.075 | 0.501 |
| | 18 | 0.945 | 0.977 | 0.060 | 0.519 |
| | 19 | 0.956 | 0.979 | 0.061 | 0.531 |
| | 20 | 0.960 | 0.980 | 0.054 | 0.555 |
| | 21 | 0.962 | 0.981 | 0.043 | 0.555 |
| | 22 | 0.971 | 0.985 | 0.050 | 0.555 |
| | 23 | 0.972 | 0.986 | 0.056 | 0.552 |

*(continued on next page)*

| Model | Layer | Ranking (embedding-trained models) | | | |
|---|---|---|---|---|---|
| | | AUROC (Base) | AUROC (ABTT) | Cosine gap (Base) | Cosine gap (ABTT) |
| | 24 | 0.956 | 0.981 | 0.243 | 0.520 |

Table 25: Per-layer test ranking AUROC and cosine gap for the embedding-trained models LaBSE, Qwen3-0.6B and KaLM-mini. Layout and markers as in Table 22.

| Model | Layer | Routing (embedding-trained models) | | | | | |
|---|---|---|---|---|---|---|---|
| | | Existing (Base) | Existing (ABTT) | New (Base) | New (ABTT) | Overall (Base) | Overall (ABTT) |
| LaBSE | 1 | 0.222 | 0.879 | 0.898 | 0.873 | 0.477 | 0.876 |
| | 2 | 0.381 | 0.875 | 0.811 | 0.827 | 0.543 | 0.857 |
| | 3 | 0.277 | 0.886 | 0.907 | 0.867 | 0.514 | 0.879 |
| | 4 | 0.372 | 0.897 | 0.885 | 0.836 | 0.565 | 0.874 |
| | 5 | 0.327 | 0.845 | 0.901 | 0.851 | 0.543 | 0.847 |
| | 6 | 0.548 | 0.860 | 0.882 | 0.851 | 0.674 | 0.857 |
| | 7 | 0.398 | 0.849 | 0.892 | 0.830 | 0.584 | 0.841 |
| | 8 | 0.385 | 0.869 | 0.870 | 0.743 | 0.568 | 0.822 |
| | 9 | 0.421 | 0.871 | 0.929 | 0.817 | 0.612 | 0.851 |
| | 10 | 0.606 | 0.892 | 0.892 | 0.805 | 0.713 | 0.859 |
| | **11** | 0.785 | **0.923** | 0.913 | **0.882** | 0.833 | **0.908** |
| | 12 | 0.729 | 0.914 | 0.938 | 0.864 | 0.808 | 0.895 |
| Qwen3-0.6B | 1 | 0.264 | 0.921 | 0.960 | 0.885 | 0.526 | 0.908 |
| | 2 | 0.437 | 0.869 | 0.814 | 0.916 | 0.579 | 0.887 |
| | 3 | 0.250 | 0.914 | 0.947 | 0.907 | 0.513 | 0.911 |
| | 4 | 0.411 | 0.936 | 0.848 | 0.842 | 0.576 | 0.901 |
| | **5** | 0.181 | **0.910** | 0.932 | **0.923** | 0.464 | **0.915** |
| | 6 | 0.422 | 0.935 | 0.830 | 0.796 | 0.576 | 0.882 |
| | 7 | 0.250 | 0.890 | 0.947 | 0.876 | 0.513 | 0.885 |
| | 8 | 0.409 | 0.931 | 0.895 | 0.799 | 0.592 | 0.881 |
| | 9 | 0.460 | 0.901 | 0.851 | 0.851 | 0.607 | 0.882 |
| | 10 | 0.533 | 0.892 | 0.836 | 0.864 | 0.647 | 0.881 |
| | 11 | 0.293 | 0.903 | 0.978 | 0.808 | 0.551 | 0.867 |
| | 12 | 0.407 | 0.912 | 0.938 | 0.817 | 0.607 | 0.876 |
| | 13 | 0.432 | 0.908 | 0.944 | 0.836 | 0.625 | 0.881 |
| | 14 | 0.456 | 0.907 | 0.926 | 0.811 | 0.633 | 0.871 |
| | 15 | 0.469 | 0.901 | 0.920 | 0.845 | 0.639 | 0.880 |
| | 16 | 0.639 | 0.897 | 0.830 | 0.808 | 0.711 | 0.864 |
| | 17 | 0.480 | 0.886 | 0.935 | 0.851 | 0.652 | 0.873 |
| | 18 | 0.553 | 0.882 | 0.944 | 0.839 | 0.700 | 0.866 |
| | 19 | 0.514 | 0.871 | 0.966 | 0.858 | 0.684 | 0.866 |
| | 20 | 0.654 | 0.865 | 0.898 | 0.892 | 0.746 | 0.875 |
| | 21 | 0.647 | 0.912 | 0.950 | 0.898 | 0.761 | 0.907 |
| | 22 | 0.731 | 0.888 | 0.938 | 0.916 | 0.809 | 0.899 |
| | 23 | 0.622 | 0.914 | 0.966 | 0.895 | 0.752 | 0.907 |
| | 24 | 0.748 | 0.901 | 0.935 | 0.935 | 0.818 | 0.914 |
| | 25 | 0.574 | 0.893 | 0.975 | 0.923 | 0.725 | 0.904 |
| | 26 | 0.613 | 0.899 | 0.975 | 0.895 | 0.749 | 0.897 |
| | 27 | 0.793 | 0.877 | 0.833 | 0.923 | 0.808 | 0.894 |
| | 28 | 0.796 | 0.877 | 0.867 | 0.913 | 0.823 | 0.890 |
| KaLM-mini | 1 | 0.391 | 0.921 | 0.907 | 0.920 | 0.585 | 0.921 |
| | 2 | 0.493 | 0.916 | 0.848 | 0.879 | 0.627 | 0.902 |
| | **3** | 0.443 | **0.923** | 0.870 | **0.907** | 0.604 | **0.917** |
| | 4 | 0.308 | 0.931 | 0.935 | 0.895 | 0.544 | 0.917 |
| | 5 | 0.338 | 0.905 | 0.882 | 0.923 | 0.543 | 0.911 |
| | 6 | 0.256 | 0.895 | 0.910 | 0.920 | 0.502 | 0.904 |
| | 7 | 0.475 | 0.864 | 0.854 | 0.944 | 0.618 | 0.894 |
| | 8 | 0.467 | 0.862 | 0.870 | 0.913 | 0.619 | 0.881 |
| | 9 | 0.469 | 0.873 | 0.885 | 0.870 | 0.626 | 0.872 |
| | 10 | 0.449 | 0.875 | 0.885 | 0.870 | 0.613 | 0.873 |
| | 11 | 0.458 | 0.862 | 0.885 | 0.864 | 0.619 | 0.862 |
| | 12 | 0.507 | 0.862 | 0.895 | 0.904 | 0.653 | 0.878 |

*(continued on next page)*

| Model | Layer | Routing (embedding-trained models) | | | | | |
|---|---|---|---|---|---|---|---|
| | | Existing (Base) | Existing (ABTT) | New (Base) | New (ABTT) | Overall (Base) | Overall (ABTT) |
| | 13 | 0.540 | 0.860 | 0.895 | 0.870 | 0.674 | 0.864 |
| | 14 | 0.593 | 0.836 | 0.845 | 0.885 | 0.688 | 0.854 |
| | 15 | 0.566 | 0.864 | 0.898 | 0.827 | 0.691 | 0.850 |
| | 16 | 0.550 | 0.849 | 0.935 | 0.830 | 0.695 | 0.841 |
| | 17 | 0.675 | 0.864 | 0.920 | 0.889 | 0.767 | 0.873 |
| | 18 | 0.697 | 0.875 | 0.960 | 0.898 | 0.796 | 0.883 |
| | 19 | 0.721 | 0.873 | 0.966 | 0.892 | 0.814 | 0.880 |
| | 20 | 0.705 | 0.791 | 0.966 | 0.913 | 0.803 | 0.837 |
| | 21 | 0.727 | 0.811 | 0.966 | 0.926 | 0.817 | 0.854 |
| | 22 | 0.850 | 0.879 | 0.916 | 0.923 | 0.875 | 0.895 |
| | 23 | 0.849 | 0.884 | 0.920 | 0.898 | 0.875 | 0.889 |
| | 24 | 0.787 | 0.892 | 0.944 | 0.867 | 0.846 | 0.882 |

Table 26: Per-layer test routing assignment accuracy, as fractions, for the embedding-trained models LaBSE, Qwen3-0.6B and KaLM-mini. Layout and markers as in Table 23.

| Model | Layer | Routing: DirAcc@1 (embedding-trained models, single seed) | | | | | |
|---|---|---|---|---|---|---|---|
| | | DirAcc@1 (Base) | DirAcc@1 (ABTT) | Existing (Base) | Existing (ABTT) | New (Base) | New (ABTT) |
| LaBSE | 1 | 0.465 | 0.851 | 0.222 | 0.879 | 0.898 | 0.873 |
| | 2 | 0.494 | 0.830 | 0.381 | 0.875 | 0.811 | 0.827 |
| | 3 | 0.500 | 0.848 | 0.277 | 0.886 | 0.907 | 0.867 |
| | 4 | 0.542 | 0.841 | 0.372 | 0.897 | 0.885 | 0.836 |
| | 5 | 0.521 | 0.811 | 0.327 | 0.845 | 0.901 | 0.851 |
| | 6 | 0.640 | 0.831 | 0.548 | 0.860 | 0.882 | 0.851 |
| | 7 | 0.561 | 0.810 | 0.398 | 0.849 | 0.892 | 0.830 |
| | 8 | 0.544 | 0.779 | 0.385 | 0.869 | 0.870 | 0.743 |
| | 9 | 0.608 | 0.823 | 0.421 | 0.871 | 0.929 | 0.817 |
| | 10 | 0.693 | 0.832 | 0.606 | 0.892 | 0.892 | 0.805 |
| | **11** | 0.815 | **0.885** | 0.785 | **0.923** | 0.913 | **0.882** |
| | 12 | 0.791 | 0.868 | 0.729 | 0.914 | 0.938 | 0.864 |
| Qwen3-0.6B | 1 | 0.521 | 0.875 | 0.264 | 0.921 | 0.960 | 0.885 |
| | 2 | 0.558 | 0.871 | 0.437 | 0.869 | 0.814 | 0.916 |
| | 3 | 0.508 | 0.890 | 0.250 | 0.914 | 0.947 | 0.907 |
| | 4 | 0.559 | 0.874 | 0.411 | 0.936 | 0.848 | 0.842 |
| | **5** | 0.450 | **0.894** | 0.181 | **0.910** | 0.932 | **0.923** |
| | 6 | 0.551 | 0.858 | 0.422 | 0.935 | 0.830 | 0.796 |
| | 7 | 0.507 | 0.859 | 0.250 | 0.890 | 0.947 | 0.876 |
| | 8 | 0.580 | 0.852 | 0.409 | 0.931 | 0.895 | 0.799 |
| | 9 | 0.587 | 0.852 | 0.460 | 0.901 | 0.851 | 0.851 |
| | 10 | 0.622 | 0.848 | 0.533 | 0.892 | 0.836 | 0.864 |
| | 11 | 0.548 | 0.830 | 0.293 | 0.903 | 0.978 | 0.808 |
| | 12 | 0.603 | 0.848 | 0.407 | 0.912 | 0.938 | 0.817 |
| | 13 | 0.618 | 0.851 | 0.432 | 0.908 | 0.944 | 0.836 |
| | 14 | 0.621 | 0.840 | 0.456 | 0.907 | 0.926 | 0.811 |
| | 15 | 0.627 | 0.848 | 0.469 | 0.901 | 0.920 | 0.845 |
| | 16 | 0.682 | 0.830 | 0.639 | 0.897 | 0.830 | 0.808 |
| | 17 | 0.641 | 0.840 | 0.480 | 0.886 | 0.935 | 0.851 |
| | 18 | 0.691 | 0.837 | 0.553 | 0.882 | 0.944 | 0.839 |
| | 19 | 0.676 | 0.837 | 0.514 | 0.871 | 0.966 | 0.858 |
| | 20 | 0.733 | 0.852 | 0.654 | 0.865 | 0.898 | 0.892 |
| | 21 | 0.752 | 0.881 | 0.647 | 0.912 | 0.950 | 0.898 |
| | 22 | 0.795 | 0.876 | 0.731 | 0.888 | 0.938 | 0.916 |
| | 23 | 0.742 | 0.886 | 0.622 | 0.914 | 0.966 | 0.895 |
| | 24 | 0.808 | 0.896 | 0.748 | 0.901 | 0.935 | 0.935 |
| | 25 | 0.718 | 0.886 | 0.574 | 0.893 | 0.975 | 0.923 |
| | 26 | 0.740 | 0.881 | 0.613 | 0.899 | 0.975 | 0.895 |
| | 27 | 0.787 | 0.876 | 0.793 | 0.877 | 0.833 | 0.923 |
| | 28 | 0.803 | 0.872 | 0.796 | 0.877 | 0.867 | 0.913 |
| KaLM-mini | 1 | 0.577 | 0.902 | 0.391 | 0.921 | 0.907 | 0.920 |

*(continued on next page)*

| Model | Layer | Routing: DirAcc@1 (embedding-trained models, single seed) | | | | | |
|---|---|---|---|---|---|---|---|
| | | DirAcc@1 (Base) | DirAcc@1 (ABTT) | Existing (Base) | Existing (ABTT) | New (Base) | New (ABTT) |
| | 2 | 0.607 | 0.879 | 0.493 | 0.916 | 0.848 | 0.879 |
| | **3** | 0.587 | **0.894** | 0.443 | **0.923** | 0.870 | **0.907** |
| | 4 | 0.538 | 0.887 | 0.308 | 0.931 | 0.935 | 0.895 |
| | 5 | 0.524 | 0.882 | 0.338 | 0.905 | 0.882 | 0.923 |
| | 6 | 0.490 | 0.876 | 0.256 | 0.895 | 0.910 | 0.920 |
| | 7 | 0.592 | 0.873 | 0.475 | 0.864 | 0.854 | 0.944 |
| | 8 | 0.597 | 0.860 | 0.467 | 0.862 | 0.870 | 0.913 |
| | 9 | 0.601 | 0.843 | 0.469 | 0.873 | 0.885 | 0.870 |
| | 10 | 0.584 | 0.844 | 0.449 | 0.875 | 0.885 | 0.870 |
| | 11 | 0.586 | 0.836 | 0.458 | 0.862 | 0.885 | 0.864 |
| | 12 | 0.622 | 0.850 | 0.507 | 0.862 | 0.895 | 0.904 |
| | 13 | 0.633 | 0.840 | 0.540 | 0.860 | 0.895 | 0.870 |
| | 14 | 0.636 | 0.829 | 0.593 | 0.836 | 0.845 | 0.885 |
| | 15 | 0.650 | 0.817 | 0.566 | 0.864 | 0.898 | 0.827 |
| | 16 | 0.676 | 0.819 | 0.550 | 0.849 | 0.935 | 0.830 |
| | 17 | 0.747 | 0.852 | 0.675 | 0.864 | 0.920 | 0.889 |
| | 18 | 0.777 | 0.865 | 0.697 | 0.875 | 0.960 | 0.898 |
| | 19 | 0.801 | 0.861 | 0.721 | 0.873 | 0.966 | 0.892 |
| | 20 | 0.789 | 0.817 | 0.705 | 0.791 | 0.966 | 0.913 |
| | 21 | 0.802 | 0.831 | 0.727 | 0.811 | 0.966 | 0.926 |
| | 22 | 0.857 | 0.875 | 0.850 | 0.879 | 0.916 | 0.923 |
| | 23 | 0.859 | 0.872 | 0.849 | 0.884 | 0.920 | 0.898 |
| | 24 | 0.833 | 0.858 | 0.787 | 0.892 | 0.944 | 0.867 |

Table 27: Per-layer test routing DirAcc@1, as fractions, for the embedding-trained models LaBSE, Qwen3-0.6B and KaLM-mini, computed once, without reseedings. Layout and markers as in Table 24.

## J Last-Token versus Mean Pooling

Decoder-only embedding models often use last-token pooling, so this appendix checks that mean pooling, which every main-text result uses, does not handicap them. We compare the two poolings under the same ABTT pipeline, training and test split, and ABTT choices. Table 28 reports, for each model and pooling, the layer chosen on the training split by DirAcc@1, the rule behind the routing cells of Table 2, with test assignment accuracy, DirAcc@1 and cosine gap at that layer. Mean pooling routes better for each of the five models in this run, the two decoders included; mT5-base was not part of it.

| Model | Pooling | Layer | $D$ | Assignment acc. | DirAcc@1 | Cosine gap |
|---|---|---|---|---|---|---|
| LaTa | Mean | 8 | 10 | 0.885 | 0.861 | 0.558 |
| | Last token | 7 | 10 | 0.664 | 0.466 | 0.308 |
| PhilTa | Mean | 1 | 10 | 0.911 | 0.883 | 0.573 |
| | Last token | 11 | 10 | 0.589 | 0.406 | 0.205 |
| LaBSE | Mean | 11 | 10 | 0.908 | 0.885 | 0.610 |
| | Last token | 12 | 10 | 0.895 | 0.871 | 0.586 |
| Qwen3-0.6B | Mean | 5 | 10 | 0.915 | 0.894 | 0.545 |
| | Last token | 26 | 5 | 0.881 | 0.864 | 0.580 |
| KaLM-mini | Mean | 3 | 10 | 0.917 | 0.894 | 0.552 |
| | Last token | 24 | 10 | 0.875 | 0.847 | 0.483 |

Table 28: Mean versus last-token pooling under ABTT for five models; test scores, accuracies as fractions. $D$ (components removed) and the layer (highest training DirAcc@1) are chosen on the training split. Mean pooling routes better for all five.

## K SIF and Whitening Baselines

Section 3 compares SIF frequency weighting and reduced-rank whitening with ABTT; this appendix gives both. Tables 31 and 32 report SIF alone, SIF followed by ABTT with $D = 10$, and SIF followed by ABTT with $D$ tuned per layer, at every layer of the six panel models, with one primary metric per task. Table 33 gives DirAcc@1 under the baseline and SIF+ABTT as means over five reseedings. Five-reseeding values for ABTT alone, at the train-selected layers of LaTa, Qwen3-0.6B and KaLM-mini, are in Table 19.

**SIF.** SIF acts before pooling. It can down-weight the frequent tokens that feed the dominant direction (Appendix C), but it removes no direction from the pooled vector. At every collapsed layer it stays below ABTT.

**Shortlist accuracy.** In the shortlist form of routing, each directory receives its best witness match score, a pseudo-option `NEW` receives score $\tau$, and the simulated expert succeeds when the labeled directory appears in the shortlist. Shortlist accuracy, DirAcc@$K$, is the fraction of test queries whose labeled directory is among the top $K$ options. Table 29 gives this shortlist view under SIF+ABTT, over five reseedings (Appendix H). DirAcc@2 ranges from 95.1 to 96.1 percent across the six models, and little is gained after DirAcc@3.

**Whitening.** Full-rank PCA whitening (Su et al., 2021) is ill-conditioned here, since it estimates 768 to 1,024 dimensions from 847 training vectors (condition numbers $3 \times 10^5$ to $10^{16}$). It ranks below ABTT at every layer: its test AUROC is 0.820–0.876 for the T5 encoders and LaBSE, and 0.552–0.643 for the two decoders. It also breaks the routing threshold. Full-rank whitening leaves the different-directory cosine mean at or below 0.015, so across all 100 whitening cells every file is routed to an existing directory, existing accuracy is 1.000, and new accuracy never exceeds one file in 323. Assignment accuracy then sits at the 62.4 percent class prior. That prior beats the baseline in 60 of 100 cells while DirAcc@1 never exceeds 41.5 percent, so the apparent gain is a thresholding artifact. Whitening to a reduced dimension, as Su et al. (2021) also do, drops the near-null directions and is the fairer comparison (Table 30). Fit on training embeddings with $r = 128$ or 256 components, it ranks on par with ABTT. At each model's train-selected layer, $r = 128$ reaches test AUROC 0.973–0.984, against 0.971–0.987 for ABTT. Over all 100 panel layers, collapsed layers included, both $r = 128$ and $r = 256$ stay within 0.021 of ABTT; $r = 64$ falls slightly below. On ranking, the repair therefore does not depend on ABTT in particular. On routing it does. At each model's train-selected routing layer, $r = 128$ reaches DirAcc@1 1.9 to 4.5 points below ABTT, and $r = 256$ up to 15 points below. We therefore keep ABTT as the reference correction.

| Model | DirAcc@1 | DirAcc@2 | DirAcc@3 | DirAcc@4 | DirAcc@5 |
|---|---|---|---|---|---|
| LaTa | $88.9 \pm 0.5$ | $96.1 \pm 0.5$ | $97.4 \pm 0.6$ | $97.9 \pm 0.3$ | $98.2 \pm 0.2$ |
| PhilTa | $89.8 \pm 0.8$ | $95.8 \pm 0.6$ | $97.4 \pm 0.3$ | $97.7 \pm 0.3$ | $97.8 \pm 0.4$ |
| mT5-base | $89.3 \pm 0.5$ | $95.1 \pm 0.2$ | $96.7 \pm 0.7$ | $97.1 \pm 0.4$ | $97.2 \pm 0.4$ |
| LaBSE | $90.6 \pm 1.0$ | $95.5 \pm 0.5$ | $96.4 \pm 0.5$ | $97.2 \pm 0.6$ | $97.8 \pm 0.5$ |
| Qwen3-0.6B | $89.3 \pm 0.7$ | $95.7 \pm 0.4$ | $97.1 \pm 0.3$ | $97.5 \pm 0.4$ | $97.6 \pm 0.4$ |
| KaLM-mini | $89.3 \pm 0.5$ | $95.1 \pm 0.4$ | $96.6 \pm 0.7$ | $97.2 \pm 0.7$ | $97.6 \pm 0.5$ |

Table 29: Test routing shortlist accuracy DirAcc@$K$ (%) under SIF+ABTT, mean $\pm$ std over five reseedings. Each model's layer is its best by training DirAcc@1 under SIF+ABTT in the single run (bold in Table 32), not always the best over the five reseedings. DirAcc@2 is 95.1 to 96.1 for all six models, with little gain after DirAcc@3.

| | Ranking AUROC | | | | | | Routing DirAcc@1 | | | | | Routing assignment | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | whitening, $r$ | | | | | whitening, $r$ | | | | | whitening, $r$ | | | |
| **Model** | Base | ABTT | 64 | 128 | 256 | full | ABTT | 64 | 128 | 256 | full | ABTT | 64 | 128 | 256 | full |
| LaTa | 0.938 | 0.971 | 0.969 | 0.973 | 0.983 | 0.872 | 86.1 | 81.2 | 83.1 | 77.0 | 41.1 | 88.5 | 84.7 | 86.2 | 80.7 | 62.5 |
| PhilTa | 0.939 | 0.983 | 0.974 | 0.977 | 0.979 | 0.854 | 88.3 | 83.0 | 86.4 | 83.7 | 41.0 | 91.1 | 86.5 | 88.8 | 86.8 | 62.5 |
| mT5-base | 0.838 | 0.975 | 0.971 | 0.979 | 0.974 | 0.845 | 88.5 | 78.9 | 85.8 | 83.8 | 41.0 | 90.3 | 82.6 | 88.6 | 87.2 | 62.5 |
| LaBSE | 0.956 | 0.987 | 0.979 | 0.984 | 0.976 | 0.866 | 88.5 | 86.1 | 84.0 | 73.5 | 40.3 | 90.8 | 88.8 | 86.7 | 78.0 | 62.5 |
| Qwen3-0.6B | 0.966 | 0.973 | 0.973 | 0.978 | 0.983 | 0.594 | 89.4 | 84.8 | 85.0 | 75.9 | 20.5 | 91.5 | 87.2 | 88.0 | 79.3 | 62.4 |
| KaLM-mini | 0.972 | 0.981 | 0.976 | 0.981 | 0.981 | 0.562 | 89.4 | 82.6 | 86.7 | 84.5 | 16.8 | 91.7 | 85.4 | 89.2 | 87.8 | 62.4 |

Table 30: PCA whitening to $r$ components against the baseline and ABTT, fit on training embeddings, at each setting's train-selected layer. Whitening keeps the top $r$ principal components, rescaled to unit variance; full keeps all, at most 847 (the number of training passages). Routing in percent. Whitening ranks on par with ABTT, but routes below it.

| Model | Layer | Ranking: SIF-suite AUROC (six models) | | | |
|---|---|---|---|---|---|
| | | AUROC (Base) | AUROC (SIF) | AUROC (SIF+ABTT$_{10}$) | AUROC (SIF+ABTT) |
| LaTa | 1 | 0.934 | 0.948 | 0.972 | 0.976 |
| | 2 | 0.563 | 0.938 | 0.964 | 0.971 |
| | 3 | 0.513 | 0.921 | 0.967 | 0.966 |
| | 4 | 0.500 | 0.890 | 0.970 | 0.968 |
| | 5 | 0.496 | 0.878 | 0.970 | 0.970 |
| | 6 | 0.496 | 0.865 | 0.965 | 0.965 |
| | **7** | 0.498 | 0.858 | 0.966 | **0.966** |
| | 8 | 0.502 | 0.842 | 0.964 | 0.964 |
| | 9 | 0.503 | 0.854 | 0.964 | 0.964 |
| | 10 | 0.505 | 0.859 | 0.966 | 0.966 |
| | 11 | 0.508 | 0.867 | 0.965 | 0.965 |
| | 12 | 0.938 | 0.948 | 0.963 | 0.963 |
| PhilTa | 1 | 0.939 | 0.955 | 0.975 | 0.975 |
| | 2 | 0.911 | 0.941 | 0.973 | 0.973 |
| | 3 | 0.601 | 0.928 | 0.968 | 0.968 |
| | 4 | 0.551 | 0.916 | 0.972 | 0.972 |
| | 5 | 0.542 | 0.904 | 0.972 | 0.972 |
| | 6 | 0.541 | 0.884 | 0.981 | 0.981 |
| | 7 | 0.542 | 0.891 | 0.981 | 0.981 |
| | **8** | 0.542 | 0.895 | 0.982 | **0.982** |
| | 9 | 0.539 | 0.881 | 0.977 | 0.977 |
| | 10 | 0.538 | 0.874 | 0.977 | 0.977 |
| | 11 | 0.539 | 0.872 | 0.979 | 0.979 |
| | 12 | 0.911 | 0.929 | 0.969 | 0.969 |
| mT5-base | **1** | 0.822 | 0.845 | 0.974 | **0.974** |
| | 2 | 0.822 | 0.835 | 0.966 | 0.966 |
| | 3 | 0.804 | 0.808 | 0.958 | 0.958 |
| | 4 | 0.799 | 0.809 | 0.966 | 0.966 |
| | 5 | 0.654 | 0.672 | 0.974 | 0.974 |
| | 6 | 0.657 | 0.678 | 0.971 | 0.971 |
| | 7 | 0.656 | 0.680 | 0.967 | 0.967 |
| | 8 | 0.656 | 0.681 | 0.968 | 0.968 |

*(continued on next page)*

| Model | Layer | Ranking: SIF-suite AUROC (six models) | | | |
|---|---|---|---|---|---|
| | | AUROC (Base) | AUROC (SIF) | AUROC (SIF+ABTT$_{10}$) | AUROC (SIF+ABTT) |
| | 9 | 0.660 | 0.685 | 0.967 | 0.967 |
| | 10 | 0.659 | 0.684 | 0.966 | 0.966 |
| | 11 | 0.659 | 0.689 | 0.966 | 0.966 |
| | 12 | 0.838 | 0.833 | 0.970 | 0.970 |
| LaBSE | 1 | 0.806 | 0.818 | 0.974 | 0.974 |
| | 2 | 0.833 | 0.840 | 0.975 | 0.975 |
| | 3 | 0.853 | 0.862 | 0.968 | 0.968 |
| | 4 | 0.862 | 0.869 | 0.966 | 0.966 |
| | 5 | 0.864 | 0.869 | 0.964 | 0.964 |
| | 6 | 0.894 | 0.895 | 0.965 | 0.965 |
| | 7 | 0.887 | 0.892 | 0.964 | 0.964 |
| | 8 | 0.879 | 0.883 | 0.968 | 0.968 |
| | 9 | 0.916 | 0.915 | 0.973 | 0.973 |
| | **10** | 0.934 | 0.934 | 0.976 | **0.976** |
| | 11 | 0.960 | 0.959 | 0.983 | 0.983 |
| | 12 | 0.956 | 0.956 | 0.980 | 0.984 |
| Qwen3-0.6B | 1 | 0.858 | 0.852 | 0.982 | 0.981 |
| | **2** | 0.863 | 0.853 | 0.972 | **0.972** |
| | 3 | 0.862 | 0.858 | 0.977 | 0.975 |
| | 4 | 0.860 | 0.860 | 0.976 | 0.977 |
| | 5 | 0.861 | 0.858 | 0.971 | 0.971 |
| | 6 | 0.867 | 0.861 | 0.974 | 0.974 |
| | 7 | 0.876 | 0.866 | 0.971 | 0.971 |
| | 8 | 0.882 | 0.868 | 0.969 | 0.969 |
| | 9 | 0.881 | 0.866 | 0.968 | 0.968 |
| | 10 | 0.887 | 0.870 | 0.966 | 0.966 |
| | 11 | 0.895 | 0.875 | 0.964 | 0.964 |
| | 12 | 0.901 | 0.881 | 0.966 | 0.966 |
| | 13 | 0.907 | 0.887 | 0.967 | 0.967 |
| | 14 | 0.905 | 0.885 | 0.966 | 0.966 |
| | 15 | 0.901 | 0.883 | 0.968 | 0.968 |
| | 16 | 0.910 | 0.894 | 0.965 | 0.965 |
| | 17 | 0.916 | 0.899 | 0.967 | 0.974 |
| | 18 | 0.933 | 0.914 | 0.966 | 0.973 |
| | 19 | 0.936 | 0.915 | 0.966 | 0.976 |
| | 20 | 0.943 | 0.918 | 0.971 | 0.977 |
| | 21 | 0.953 | 0.931 | 0.971 | 0.971 |
| | 22 | 0.963 | 0.942 | 0.973 | 0.973 |
| | 23 | 0.961 | 0.941 | 0.973 | 0.973 |
| | 24 | 0.963 | 0.943 | 0.976 | 0.974 |
| | 25 | 0.961 | 0.938 | 0.978 | 0.975 |
| | 26 | 0.966 | 0.942 | 0.980 | 0.980 |
| | 27 | 0.967 | 0.936 | 0.978 | 0.978 |
| | 28 | 0.958 | 0.925 | 0.968 | 0.968 |
| KaLM-mini | 1 | 0.887 | 0.880 | 0.982 | 0.980 |
| | **2** | 0.906 | 0.899 | 0.981 | **0.981** |
| | 3 | 0.881 | 0.882 | 0.977 | 0.977 |
| | 4 | 0.871 | 0.873 | 0.974 | 0.974 |
| | 5 | 0.861 | 0.862 | 0.971 | 0.971 |
| | 6 | 0.872 | 0.867 | 0.967 | 0.967 |
| | 7 | 0.888 | 0.879 | 0.972 | 0.972 |
| | 8 | 0.890 | 0.880 | 0.973 | 0.975 |
| | 9 | 0.892 | 0.882 | 0.968 | 0.968 |
| | 10 | 0.893 | 0.882 | 0.964 | 0.964 |
| | 11 | 0.894 | 0.885 | 0.969 | 0.969 |
| | 12 | 0.905 | 0.896 | 0.966 | 0.966 |
| | 13 | 0.900 | 0.894 | 0.964 | 0.964 |
| | 14 | 0.904 | 0.898 | 0.970 | 0.970 |
| | 15 | 0.907 | 0.901 | 0.970 | 0.970 |
| | 16 | 0.917 | 0.913 | 0.967 | 0.967 |
| | 17 | 0.926 | 0.923 | 0.968 | 0.968 |
| | 18 | 0.945 | 0.939 | 0.977 | 0.977 |
| | 19 | 0.956 | 0.950 | 0.978 | 0.978 |

*(continued on next page)*

| Model | Layer | Ranking: SIF-suite AUROC (six models) | | | |
|---|---|---|---|---|---|
| | | AUROC (Base) | AUROC (SIF) | AUROC (SIF+ABTT$_{10}$) | AUROC (SIF+ABTT) |
| | 20 | 0.960 | 0.956 | 0.978 | 0.978 |
| | 21 | 0.962 | 0.958 | 0.980 | 0.983 |
| | 22 | 0.971 | 0.967 | 0.983 | 0.984 |
| | 23 | 0.972 | 0.964 | 0.986 | 0.985 |
| | 24 | 0.956 | 0.956 | 0.976 | 0.976 |

Table 31: Per-layer test ranking AUROC under the SIF variants for the six panel models. SIF: frequency-weighted pooling; SIF+ABTT$_{10}$: SIF, then ABTT with $D$=10; SIF+ABTT: $D$ tuned per layer on the training split. Base and markers as in Table 22, with bold layers selected under SIF+ABTT.

| Model | Layer | Routing: SIF-suite assignment accuracy (six models) | | | |
|---|---|---|---|---|---|
| | | Overall (Base) | Overall (SIF) | Overall (SIF+ABTT$_{10}$) | Overall (SIF+ABTT) |
| LaTa | 1 | 0.738 | 0.832 | 0.920 | 0.899 |
| | 2 | 0.439 | 0.791 | 0.904 | 0.900 |
| | 3 | 0.477 | 0.720 | 0.910 | 0.907 |
| | 4 | 0.513 | 0.668 | 0.900 | 0.902 |
| | 5 | 0.502 | 0.648 | 0.906 | 0.906 |
| | 6 | 0.503 | 0.668 | 0.895 | 0.895 |
| | 7 | 0.501 | 0.642 | 0.903 | 0.903 |
| | 8 | 0.502 | 0.614 | 0.892 | 0.892 |
| | 9 | 0.501 | 0.618 | 0.889 | 0.889 |
| | 10 | 0.498 | 0.655 | 0.890 | 0.890 |
| | 11 | 0.490 | 0.621 | 0.880 | 0.880 |
| | **12** | 0.710 | 0.804 | 0.902 | **0.902** |
| PhilTa | **1** | 0.709 | 0.790 | 0.914 | **0.914** |
| | 2 | 0.625 | 0.738 | 0.892 | 0.892 |
| | 3 | 0.455 | 0.713 | 0.892 | 0.892 |
| | 4 | 0.439 | 0.663 | 0.886 | 0.886 |
| | 5 | 0.457 | 0.626 | 0.899 | 0.899 |
| | 6 | 0.474 | 0.612 | 0.906 | 0.906 |
| | 7 | 0.469 | 0.625 | 0.906 | 0.906 |
| | 8 | 0.467 | 0.614 | 0.886 | 0.886 |
| | 9 | 0.476 | 0.590 | 0.895 | 0.895 |
| | 10 | 0.480 | 0.580 | 0.888 | 0.888 |
| | 11 | 0.470 | 0.565 | 0.886 | 0.886 |
| | 12 | 0.549 | 0.667 | 0.899 | 0.899 |
| mT5-base | 1 | 0.453 | 0.545 | 0.907 | 0.907 |
| | **2** | 0.436 | 0.537 | 0.903 | **0.903** |
| | 3 | 0.449 | 0.568 | 0.885 | 0.885 |
| | 4 | 0.491 | 0.488 | 0.857 | 0.857 |
| | 5 | 0.512 | 0.483 | 0.826 | 0.826 |
| | 6 | 0.580 | 0.558 | 0.817 | 0.817 |
| | 7 | 0.565 | 0.554 | 0.832 | 0.832 |
| | 8 | 0.558 | 0.548 | 0.834 | 0.834 |
| | 9 | 0.508 | 0.495 | 0.845 | 0.845 |
| | 10 | 0.491 | 0.495 | 0.832 | 0.832 |
| | 11 | 0.479 | 0.484 | 0.816 | 0.816 |
| | 12 | 0.472 | 0.541 | 0.858 | 0.858 |
| LaBSE | 1 | 0.477 | 0.521 | 0.886 | 0.886 |
| | 2 | 0.543 | 0.508 | 0.875 | 0.875 |
| | 3 | 0.514 | 0.577 | 0.860 | 0.860 |
| | 4 | 0.565 | 0.587 | 0.869 | 0.869 |
| | 5 | 0.543 | 0.598 | 0.857 | 0.857 |
| | 6 | 0.674 | 0.648 | 0.872 | 0.872 |
| | 7 | 0.584 | 0.634 | 0.858 | 0.858 |
| | 8 | 0.568 | 0.571 | 0.854 | 0.854 |
| | 9 | 0.612 | 0.613 | 0.859 | 0.859 |
| | 10 | 0.713 | 0.698 | 0.864 | 0.864 |

*(continued on next page)*

| Model | Layer | Routing: SIF-suite assignment accuracy (six models) | | | |
|---|---|---|---|---|---|
| | | Overall (Base) | Overall (SIF) | Overall (SIF+ABTT$_{10}$) | Overall (SIF+ABTT) |
| | **11** | 0.833 | 0.807 | 0.896 | **0.896** |
| | 12 | 0.808 | 0.834 | 0.890 | 0.880 |
| Qwen3-0.6B | 1 | 0.526 | 0.566 | 0.914 | 0.909 |
| | 2 | 0.579 | 0.524 | 0.894 | 0.894 |
| | 3 | 0.513 | 0.537 | 0.904 | 0.901 |
| | 4 | 0.576 | 0.531 | 0.899 | 0.904 |
| | **5** | 0.464 | 0.537 | 0.908 | **0.908** |
| | 6 | 0.576 | 0.526 | 0.906 | 0.906 |
| | 7 | 0.513 | 0.573 | 0.911 | 0.911 |
| | 8 | 0.592 | 0.627 | 0.911 | 0.911 |
| | 9 | 0.607 | 0.576 | 0.910 | 0.910 |
| | 10 | 0.647 | 0.599 | 0.900 | 0.900 |
| | 11 | 0.551 | 0.606 | 0.893 | 0.893 |
| | 12 | 0.607 | 0.663 | 0.900 | 0.900 |
| | 13 | 0.625 | 0.675 | 0.897 | 0.897 |
| | 14 | 0.633 | 0.571 | 0.893 | 0.893 |
| | 15 | 0.639 | 0.600 | 0.893 | 0.893 |
| | 16 | 0.711 | 0.679 | 0.880 | 0.880 |
| | 17 | 0.652 | 0.686 | 0.900 | 0.876 |
| | 18 | 0.700 | 0.731 | 0.899 | 0.882 |
| | 19 | 0.684 | 0.700 | 0.886 | 0.850 |
| | 20 | 0.746 | 0.676 | 0.882 | 0.866 |
| | 21 | 0.761 | 0.739 | 0.892 | 0.892 |
| | 22 | 0.809 | 0.719 | 0.893 | 0.893 |
| | 23 | 0.752 | 0.759 | 0.893 | 0.893 |
| | 24 | 0.818 | 0.706 | 0.900 | 0.899 |
| | 25 | 0.725 | 0.748 | 0.897 | 0.893 |
| | 26 | 0.749 | 0.772 | 0.903 | 0.903 |
| | 27 | 0.808 | 0.760 | 0.899 | 0.899 |
| | 28 | 0.823 | 0.774 | 0.890 | 0.890 |
| KaLM-mini | **1** | 0.585 | 0.604 | 0.906 | **0.908** |
| | 2 | 0.627 | 0.594 | 0.908 | 0.908 |
| | 3 | 0.604 | 0.576 | 0.907 | 0.907 |
| | 4 | 0.544 | 0.577 | 0.901 | 0.901 |
| | 5 | 0.543 | 0.533 | 0.897 | 0.897 |
| | 6 | 0.502 | 0.558 | 0.900 | 0.900 |
| | 7 | 0.618 | 0.593 | 0.901 | 0.901 |
| | 8 | 0.619 | 0.607 | 0.892 | 0.878 |
| | 9 | 0.626 | 0.663 | 0.876 | 0.876 |
| | 10 | 0.613 | 0.603 | 0.866 | 0.866 |
| | 11 | 0.619 | 0.627 | 0.874 | 0.874 |
| | 12 | 0.653 | 0.642 | 0.874 | 0.874 |
| | 13 | 0.674 | 0.666 | 0.864 | 0.864 |
| | 14 | 0.688 | 0.648 | 0.861 | 0.861 |
| | 15 | 0.691 | 0.698 | 0.866 | 0.866 |
| | 16 | 0.695 | 0.738 | 0.859 | 0.859 |
| | 17 | 0.767 | 0.763 | 0.882 | 0.882 |
| | 18 | 0.796 | 0.796 | 0.879 | 0.879 |
| | 19 | 0.814 | 0.804 | 0.878 | 0.878 |
| | 20 | 0.803 | 0.821 | 0.876 | 0.876 |
| | 21 | 0.817 | 0.832 | 0.874 | 0.876 |
| | 22 | 0.875 | 0.857 | 0.876 | 0.876 |
| | 23 | 0.875 | 0.862 | 0.880 | 0.881 |
| | 24 | 0.846 | 0.847 | 0.869 | 0.869 |

Table 32: Per-layer test routing assignment accuracy, as fractions, for the six panel models. Variants and markers as in Table 31, with bold layers selected by training DirAcc@1.

| Model | Layer | Routing: DirAcc@1 (five reseedings, mean ± std, SIF+ABTT) | | | | | |
|---|---|---|---|---|---|---|---|
| | | DirAcc@1 (Base) | DirAcc@1 (SIF+ABTT) | Existing (Base) | Existing (SIF+ABTT) | New (Base) | New (SIF+ABTT) |
| LaTa | 1 | 0.731 ± 0.010 | 0.889 ± 0.010 | 0.563 ± 0.007 | 0.844 ± 0.011 | 0.949 ± 0.011 | 0.948 ± 0.010 |
| | 2 | 0.429 ± 0.011 | 0.888 ± 0.007 | 0.049 ± 0.006 | 0.841 ± 0.007 | 0.920 ± 0.008 | 0.948 ± 0.007 |
| | 3 | 0.393 ± 0.011 | 0.891 ± 0.010 | 0.047 ± 0.006 | 0.847 ± 0.010 | 0.840 ± 0.020 | 0.947 ± 0.012 |
| | 4 | 0.331 ± 0.019 | 0.889 ± 0.009 | 0.048 ± 0.009 | 0.860 ± 0.009 | 0.696 ± 0.030 | 0.925 ± 0.013 |
| | 5 | 0.326 ± 0.021 | 0.891 ± 0.009 | 0.038 ± 0.005 | 0.858 ± 0.008 | 0.698 ± 0.028 | 0.934 ± 0.014 |
| | 6 | 0.322 ± 0.018 | 0.891 ± 0.009 | 0.038 ± 0.007 | 0.863 ± 0.010 | 0.688 ± 0.026 | 0.927 ± 0.013 |
| | 7 | 0.328 ± 0.020 | 0.896 ± 0.008 | 0.045 ± 0.007 | 0.848 ± 0.009 | 0.693 ± 0.028 | 0.957 ± 0.010 |
| | 8 | 0.325 ± 0.022 | 0.884 ± 0.008 | 0.046 ± 0.010 | 0.834 ± 0.008 | 0.685 ± 0.032 | 0.949 ± 0.011 |
| | 9 | 0.332 ± 0.019 | 0.886 ± 0.011 | 0.045 ± 0.010 | 0.848 ± 0.009 | 0.701 ± 0.024 | 0.935 ± 0.015 |
| | 10 | 0.346 ± 0.017 | 0.889 ± 0.004 | 0.045 ± 0.010 | 0.846 ± 0.008 | 0.735 ± 0.021 | 0.944 ± 0.012 |
| | 11 | 0.361 ± 0.018 | 0.882 ± 0.009 | 0.045 ± 0.009 | 0.862 ± 0.010 | 0.767 ± 0.029 | 0.907 ± 0.014 |
| | **12** | 0.707 ± 0.014 | **0.889 ± 0.005** | 0.511 ± 0.016 | **0.834 ± 0.005** | 0.961 ± 0.008 | **0.961 ± 0.010** |
| PhilTa | **1** | 0.705 ± 0.015 | **0.898 ± 0.008** | 0.531 ± 0.016 | **0.858 ± 0.010** | 0.931 ± 0.012 | **0.951 ± 0.011** |
| | 2 | 0.626 ± 0.019 | 0.888 ± 0.010 | 0.410 ± 0.017 | 0.866 ± 0.010 | 0.906 ± 0.018 | 0.917 ± 0.017 |
| | 3 | 0.440 ± 0.013 | 0.893 ± 0.009 | 0.071 ± 0.014 | 0.858 ± 0.010 | 0.917 ± 0.013 | 0.939 ± 0.011 |
| | 4 | 0.421 ± 0.016 | 0.891 ± 0.009 | 0.027 ± 0.007 | 0.858 ± 0.008 | 0.928 ± 0.015 | 0.935 ± 0.013 |
| | 5 | 0.375 ± 0.016 | 0.895 ± 0.010 | 0.042 ± 0.013 | 0.852 ± 0.010 | 0.804 ± 0.018 | 0.950 ± 0.014 |
| | 6 | 0.337 ± 0.015 | 0.897 ± 0.009 | 0.060 ± 0.012 | 0.861 ± 0.010 | 0.696 ± 0.026 | 0.944 ± 0.013 |
| | 7 | 0.361 ± 0.017 | 0.900 ± 0.009 | 0.056 ± 0.012 | 0.860 ± 0.008 | 0.755 ± 0.020 | 0.952 ± 0.013 |
| | 8 | 0.389 ± 0.016 | 0.891 ± 0.010 | 0.058 ± 0.014 | 0.859 ± 0.009 | 0.818 ± 0.018 | 0.932 ± 0.015 |
| | 9 | 0.371 ± 0.016 | 0.893 ± 0.008 | 0.066 ± 0.014 | 0.861 ± 0.009 | 0.766 ± 0.027 | 0.934 ± 0.016 |
| | 10 | 0.377 ± 0.014 | 0.892 ± 0.009 | 0.066 ± 0.014 | 0.855 ± 0.005 | 0.778 ± 0.023 | 0.940 ± 0.017 |
| | 11 | 0.394 ± 0.014 | 0.888 ± 0.008 | 0.059 ± 0.010 | 0.851 ± 0.010 | 0.827 ± 0.021 | 0.935 ± 0.013 |
| | 12 | 0.566 ± 0.013 | 0.897 ± 0.010 | 0.282 ± 0.014 | 0.863 ± 0.010 | 0.934 ± 0.007 | 0.941 ± 0.018 |
| mT5-base | 1 | 0.487 ± 0.012 | 0.899 ± 0.005 | 0.100 ± 0.009 | 0.856 ± 0.007 | 0.987 ± 0.007 | 0.954 ± 0.011 |
| | **2** | 0.476 ± 0.012 | **0.893 ± 0.005** | 0.083 ± 0.008 | **0.851 ± 0.012** | 0.984 ± 0.008 | **0.947 ± 0.013** |
| | 3 | 0.483 ± 0.014 | 0.883 ± 0.011 | 0.124 ± 0.009 | 0.817 ± 0.014 | 0.947 ± 0.008 | 0.970 ± 0.007 |
| | 4 | 0.500 ± 0.015 | 0.852 ± 0.007 | 0.206 ± 0.006 | 0.772 ± 0.010 | 0.879 ± 0.020 | 0.956 ± 0.010 |
| | 5 | 0.371 ± 0.014 | 0.825 ± 0.008 | 0.155 ± 0.006 | 0.724 ± 0.007 | 0.651 ± 0.023 | 0.956 ± 0.006 |
| | 6 | 0.361 ± 0.012 | 0.810 ± 0.005 | 0.263 ± 0.021 | 0.765 ± 0.012 | 0.488 ± 0.018 | 0.868 ± 0.016 |
| | 7 | 0.370 ± 0.010 | 0.827 ± 0.008 | 0.255 ± 0.014 | 0.760 ± 0.011 | 0.519 ± 0.017 | 0.914 ± 0.013 |
| | 8 | 0.382 ± 0.008 | 0.824 ± 0.012 | 0.244 ± 0.007 | 0.773 ± 0.012 | 0.561 ± 0.017 | 0.891 ± 0.016 |
| | 9 | 0.428 ± 0.014 | 0.838 ± 0.010 | 0.171 ± 0.008 | 0.783 ± 0.013 | 0.759 ± 0.024 | 0.909 ± 0.006 |
| | 10 | 0.421 ± 0.010 | 0.821 ± 0.017 | 0.140 ± 0.011 | 0.767 ± 0.015 | 0.783 ± 0.021 | 0.892 ± 0.019 |
| | 11 | 0.420 ± 0.015 | 0.803 ± 0.012 | 0.131 ± 0.012 | 0.721 ± 0.013 | 0.792 ± 0.016 | 0.910 ± 0.015 |
| | 12 | 0.507 ± 0.009 | 0.841 ± 0.009 | 0.169 ± 0.007 | 0.772 ± 0.016 | 0.942 ± 0.011 | 0.931 ± 0.011 |
| LaBSE | 1 | 0.502 ± 0.012 | 0.858 ± 0.007 | 0.170 ± 0.003 | 0.838 ± 0.007 | 0.929 ± 0.016 | 0.885 ± 0.009 |
| | 2 | 0.523 ± 0.011 | 0.860 ± 0.004 | 0.274 ± 0.005 | 0.817 ± 0.003 | 0.844 ± 0.022 | 0.915 ± 0.008 |
| | 3 | 0.535 ± 0.010 | 0.847 ± 0.006 | 0.218 ± 0.009 | 0.831 ± 0.009 | 0.945 ± 0.008 | 0.867 ± 0.007 |
| | 4 | 0.569 ± 0.008 | 0.850 ± 0.006 | 0.299 ± 0.004 | 0.837 ± 0.006 | 0.917 ± 0.014 | 0.866 ± 0.011 |
| | 5 | 0.551 ± 0.012 | 0.845 ± 0.005 | 0.255 ± 0.012 | 0.794 ± 0.005 | 0.933 ± 0.013 | 0.911 ± 0.010 |
| | 6 | 0.661 ± 0.011 | 0.859 ± 0.004 | 0.461 ± 0.019 | 0.817 ± 0.008 | 0.919 ± 0.018 | 0.914 ± 0.015 |
| | 7 | 0.596 ± 0.010 | 0.849 ± 0.004 | 0.333 ± 0.016 | 0.808 ± 0.006 | 0.935 ± 0.016 | 0.902 ± 0.010 |
| | 8 | 0.572 ± 0.012 | 0.847 ± 0.006 | 0.314 ± 0.019 | 0.802 ± 0.009 | 0.904 ± 0.016 | 0.905 ± 0.008 |
| | 9 | 0.631 ± 0.008 | 0.854 ± 0.007 | 0.381 ± 0.015 | 0.828 ± 0.002 | 0.954 ± 0.008 | 0.887 ± 0.012 |
| | 10 | 0.712 ± 0.016 | 0.858 ± 0.005 | 0.551 ± 0.019 | 0.836 ± 0.005 | 0.920 ± 0.015 | 0.887 ± 0.010 |
| | **11** | 0.820 ± 0.010 | **0.906 ± 0.010** | 0.731 ± 0.012 | **0.875 ± 0.013** | 0.935 ± 0.012 | **0.944 ± 0.017** |
| | 12 | 0.793 ± 0.015 | 0.884 ± 0.006 | 0.670 ± 0.019 | 0.850 ± 0.012 | 0.952 ± 0.013 | 0.927 ± 0.015 |
| Qwen3-0.6B | 1 | 0.550 ± 0.008 | 0.894 ± 0.005 | 0.223 ± 0.006 | 0.840 ± 0.010 | 0.972 ± 0.003 | 0.965 ± 0.011 |
| | 2 | 0.572 ± 0.015 | 0.882 ± 0.004 | 0.363 ± 0.014 | 0.828 ± 0.005 | 0.841 ± 0.032 | 0.952 ± 0.007 |
| | 3 | 0.541 ± 0.007 | 0.889 ± 0.006 | 0.216 ± 0.007 | 0.827 ± 0.008 | 0.962 ± 0.007 | 0.968 ± 0.011 |
| | 4 | 0.580 ± 0.012 | 0.891 ± 0.004 | 0.353 ± 0.008 | 0.845 ± 0.007 | 0.874 ± 0.028 | 0.950 ± 0.006 |
| | **5** | 0.487 ± 0.017 | **0.893 ± 0.007** | 0.133 ± 0.012 | **0.839 ± 0.011** | 0.944 ± 0.014 | **0.964 ± 0.009** |
| | 6 | 0.568 ± 0.012 | 0.897 ± 0.005 | 0.350 ± 0.010 | 0.857 ± 0.008 | 0.850 ± 0.023 | 0.948 ± 0.012 |
| | 7 | 0.541 ± 0.007 | 0.902 ± 0.006 | 0.211 ± 0.009 | 0.862 ± 0.008 | 0.967 ± 0.012 | 0.953 ± 0.012 |
| | 8 | 0.609 ± 0.009 | 0.903 ± 0.006 | 0.362 ± 0.013 | 0.861 ± 0.010 | 0.929 ± 0.017 | 0.956 ± 0.005 |
| | 9 | 0.612 ± 0.013 | 0.900 ± 0.004 | 0.393 ± 0.016 | 0.868 ± 0.008 | 0.894 ± 0.021 | 0.943 ± 0.006 |
| | 10 | 0.638 ± 0.013 | 0.890 ± 0.009 | 0.452 ± 0.009 | 0.847 ± 0.014 | 0.878 ± 0.022 | 0.944 ± 0.010 |
| | 11 | 0.578 ± 0.006 | 0.885 ± 0.005 | 0.257 ± 0.010 | 0.838 ± 0.014 | 0.991 ± 0.005 | 0.947 ± 0.012 |
| | 12 | 0.627 ± 0.009 | 0.892 ± 0.005 | 0.366 ± 0.016 | 0.844 ± 0.010 | 0.964 ± 0.013 | 0.955 ± 0.007 |
| | 13 | 0.639 ± 0.010 | 0.887 ± 0.006 | 0.383 ± 0.014 | 0.849 ± 0.011 | 0.970 ± 0.015 | 0.937 ± 0.011 |
| | 14 | 0.646 ± 0.009 | 0.885 ± 0.006 | 0.402 ± 0.017 | 0.852 ± 0.009 | 0.960 ± 0.016 | 0.927 ± 0.007 |

*(continued on next page)*

| Model | Layer | Routing: DirAcc@1 (five reseedings, mean ± std, SIF+ABTT) | | | | | |
|---|---|---|---|---|---|---|---|
| | | DirAcc@1 (Base) | DirAcc@1 (SIF+ABTT) | Existing (Base) | Existing (SIF+ABTT) | New (Base) | New (SIF+ABTT) |
| | 15 | 0.648 ± 0.012 | 0.885 ± 0.007 | 0.410 ± 0.017 | 0.852 ± 0.010 | 0.954 ± 0.017 | 0.929 ± 0.018 |
| | 16 | 0.691 ± 0.016 | 0.873 ± 0.005 | 0.546 ± 0.017 | 0.837 ± 0.009 | 0.878 ± 0.017 | 0.920 ± 0.016 |
| | 17 | 0.658 ± 0.010 | 0.862 ± 0.006 | 0.420 ± 0.018 | 0.807 ± 0.012 | 0.966 ± 0.013 | 0.933 ± 0.013 |
| | 18 | 0.709 ± 0.012 | 0.868 ± 0.002 | 0.503 ± 0.018 | 0.833 ± 0.013 | 0.975 ± 0.006 | 0.914 ± 0.017 |
| | 19 | 0.691 ± 0.010 | 0.874 ± 0.002 | 0.462 ± 0.018 | 0.860 ± 0.007 | 0.986 ± 0.006 | 0.893 ± 0.012 |
| | 20 | 0.743 ± 0.020 | 0.860 ± 0.004 | 0.588 ± 0.023 | 0.797 ± 0.011 | 0.943 ± 0.018 | 0.941 ± 0.010 |
| | 21 | 0.758 ± 0.017 | 0.879 ± 0.006 | 0.585 ± 0.021 | 0.836 ± 0.008 | 0.981 ± 0.010 | 0.935 ± 0.010 |
| | 22 | 0.799 ± 0.014 | 0.882 ± 0.006 | 0.671 ± 0.019 | 0.855 ± 0.002 | 0.965 ± 0.006 | 0.918 ± 0.015 |
| | 23 | 0.757 ± 0.012 | 0.887 ± 0.006 | 0.577 ± 0.014 | 0.857 ± 0.007 | 0.989 ± 0.004 | 0.927 ± 0.011 |
| | 24 | 0.810 ± 0.014 | 0.893 ± 0.005 | 0.695 ± 0.021 | 0.861 ± 0.004 | 0.959 ± 0.003 | 0.935 ± 0.011 |
| | 25 | 0.732 ± 0.011 | 0.890 ± 0.009 | 0.527 ± 0.016 | 0.866 ± 0.006 | 0.997 ± 0.003 | 0.920 ± 0.014 |
| | 26 | 0.749 ± 0.013 | 0.895 ± 0.008 | 0.562 ± 0.017 | 0.858 ± 0.004 | 0.991 ± 0.003 | 0.943 ± 0.015 |
| | 27 | 0.812 ± 0.004 | 0.881 ± 0.005 | 0.741 ± 0.008 | 0.824 ± 0.006 | 0.903 ± 0.011 | 0.954 ± 0.011 |
| | 28 | 0.824 ± 0.010 | 0.871 ± 0.006 | 0.736 ± 0.015 | 0.802 ± 0.010 | 0.939 ± 0.005 | 0.961 ± 0.007 |
| KaLM-mini | **1** | 0.596 ± 0.012 | **0.893 ± 0.005** | 0.344 ± 0.003 | **0.846 ± 0.005** | 0.922 ± 0.022 | **0.953 ± 0.006** |
| | 2 | 0.620 ± 0.018 | 0.893 ± 0.003 | 0.422 ± 0.014 | 0.845 ± 0.005 | 0.876 ± 0.023 | 0.955 ± 0.008 |
| | 3 | 0.605 ± 0.011 | 0.893 ± 0.004 | 0.379 ± 0.008 | 0.843 ± 0.000 | 0.896 ± 0.018 | 0.959 ± 0.010 |
| | 4 | 0.562 ± 0.015 | 0.889 ± 0.005 | 0.265 ± 0.009 | 0.842 ± 0.008 | 0.945 ± 0.016 | 0.949 ± 0.008 |
| | 5 | 0.550 ± 0.013 | 0.890 ± 0.004 | 0.270 ± 0.005 | 0.841 ± 0.004 | 0.910 ± 0.019 | 0.952 ± 0.009 |
| | 6 | 0.522 ± 0.016 | 0.893 ± 0.004 | 0.203 ± 0.012 | 0.850 ± 0.005 | 0.934 ± 0.014 | 0.949 ± 0.014 |
| | 7 | 0.604 ± 0.018 | 0.893 ± 0.003 | 0.393 ± 0.018 | 0.840 ± 0.008 | 0.876 ± 0.023 | 0.962 ± 0.008 |
| | 8 | 0.607 ± 0.014 | 0.866 ± 0.003 | 0.388 ± 0.011 | 0.798 ± 0.008 | 0.891 ± 0.017 | 0.954 ± 0.007 |
| | 9 | 0.614 ± 0.011 | 0.874 ± 0.005 | 0.387 ± 0.008 | 0.821 ± 0.008 | 0.908 ± 0.013 | 0.943 ± 0.008 |
| | 10 | 0.602 ± 0.013 | 0.859 ± 0.008 | 0.360 ± 0.005 | 0.774 ± 0.010 | 0.916 ± 0.017 | 0.969 ± 0.010 |
| | 11 | 0.601 ± 0.012 | 0.864 ± 0.008 | 0.361 ± 0.007 | 0.792 ± 0.010 | 0.911 ± 0.020 | 0.956 ± 0.008 |
| | 12 | 0.640 ± 0.010 | 0.868 ± 0.006 | 0.416 ± 0.006 | 0.801 ± 0.007 | 0.929 ± 0.019 | 0.954 ± 0.006 |
| | 13 | 0.645 ± 0.011 | 0.862 ± 0.005 | 0.436 ± 0.005 | 0.793 ± 0.004 | 0.915 ± 0.018 | 0.951 ± 0.012 |
| | 14 | 0.654 ± 0.010 | 0.859 ± 0.006 | 0.479 ± 0.004 | 0.795 ± 0.008 | 0.880 ± 0.017 | 0.941 ± 0.011 |
| | 15 | 0.661 ± 0.009 | 0.870 ± 0.006 | 0.460 ± 0.009 | 0.814 ± 0.009 | 0.921 ± 0.014 | 0.941 ± 0.006 |
| | 16 | 0.685 ± 0.008 | 0.870 ± 0.007 | 0.477 ± 0.009 | 0.824 ± 0.008 | 0.953 ± 0.015 | 0.930 ± 0.008 |
| | 17 | 0.757 ± 0.006 | 0.886 ± 0.010 | 0.612 ± 0.016 | 0.841 ± 0.005 | 0.944 ± 0.010 | 0.943 ± 0.016 |
| | 18 | 0.781 ± 0.008 | 0.880 ± 0.008 | 0.636 ± 0.013 | 0.840 ± 0.002 | 0.967 ± 0.007 | 0.931 ± 0.016 |
| | 19 | 0.805 ± 0.012 | 0.877 ± 0.008 | 0.673 ± 0.020 | 0.831 ± 0.007 | 0.976 ± 0.003 | 0.935 ± 0.011 |
| | 20 | 0.794 ± 0.010 | 0.878 ± 0.006 | 0.650 ± 0.018 | 0.831 ± 0.003 | 0.979 ± 0.004 | 0.937 ± 0.011 |
| | 21 | 0.805 ± 0.010 | 0.871 ± 0.007 | 0.673 ± 0.016 | 0.831 ± 0.004 | 0.975 ± 0.006 | 0.924 ± 0.012 |
| | 22 | 0.871 ± 0.006 | 0.873 ± 0.006 | 0.811 ± 0.005 | 0.813 ± 0.007 | 0.949 ± 0.006 | 0.951 ± 0.007 |
| | 23 | 0.873 ± 0.007 | 0.885 ± 0.005 | 0.811 ± 0.006 | 0.835 ± 0.007 | 0.953 ± 0.007 | 0.950 ± 0.006 |
| | 24 | 0.848 ± 0.009 | 0.856 ± 0.004 | 0.749 ± 0.017 | 0.807 ± 0.006 | 0.975 ± 0.007 | 0.918 ± 0.009 |

Table 33: Per-layer test routing DirAcc@1 for the six panel models, overall and on existing or new witnesses, as fractions (mean ± std over five reseedings). Bold: the reported layer and its SIF+ABTT values. Each model's layer is its best by training DirAcc@1 under SIF+ABTT in the single run (bold in Table 32), not always the best over the five reseedings.

## L Cluster Geometry

This appendix shows the effect of ABTT on the whole corpus at the routing layers of Table 2. Figures 10 and 11 give the t-SNE (van der Maaten and Hinton, 2008) and UMAP (McInnes et al., 2018) panels for LaTa, PhilTa, and mT5-base, and Figures 12 and 13 extend them to LaBSE, Qwen3-0.6B, and KaLM-mini. All four draw all 1,705 witnesses, color the same six largest directories, and compare baseline mean pooling with ABTT-only at $D = 10$. Each model sits at the layer chosen on the training split by DirAcc@1 under ABTT, the routing ABTT layer of Table 21. In both maps, ABTT draws the witnesses of each colored directory into tighter groups than baseline mean pooling does. The maps are two-dimensional, so we read them as illustrations and compute no statistic on them.

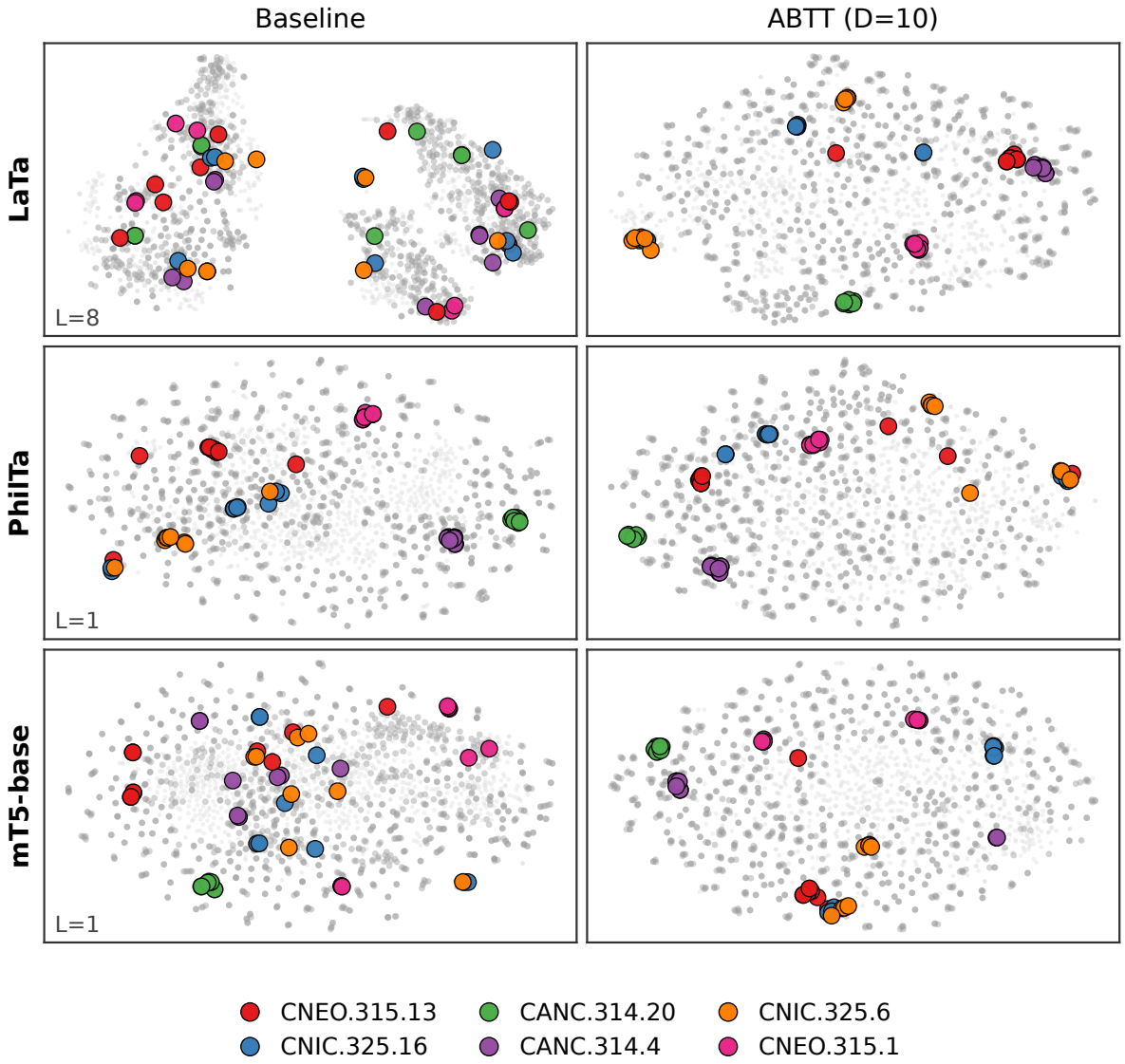



Figure 10: t-SNE maps of all 1,705 witnesses for LaTa, PhilTa and mT5-base at the train-selected routing ABTT layer (L; Table 21), under baseline mean pooling and ABTT alone ($D = 10$, fit on training embeddings). Mid gray: other non-singleton directories; light gray: singletons. ABTT draws each colored directory into a tighter group.

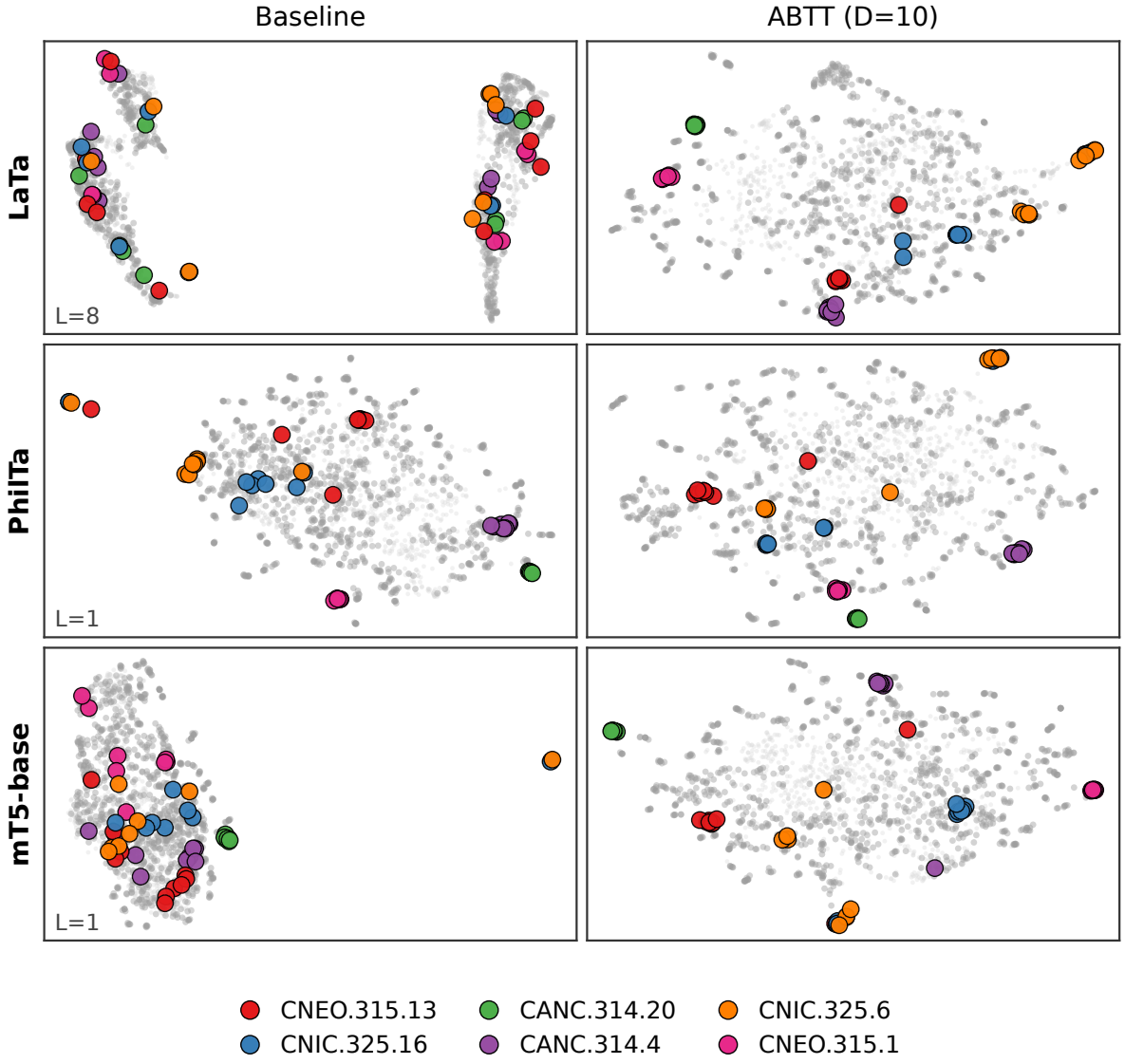



Figure 11: UMAP maps configured as in Figure 10. Consistent with it, ABTT draws each colored directory into a tighter group.

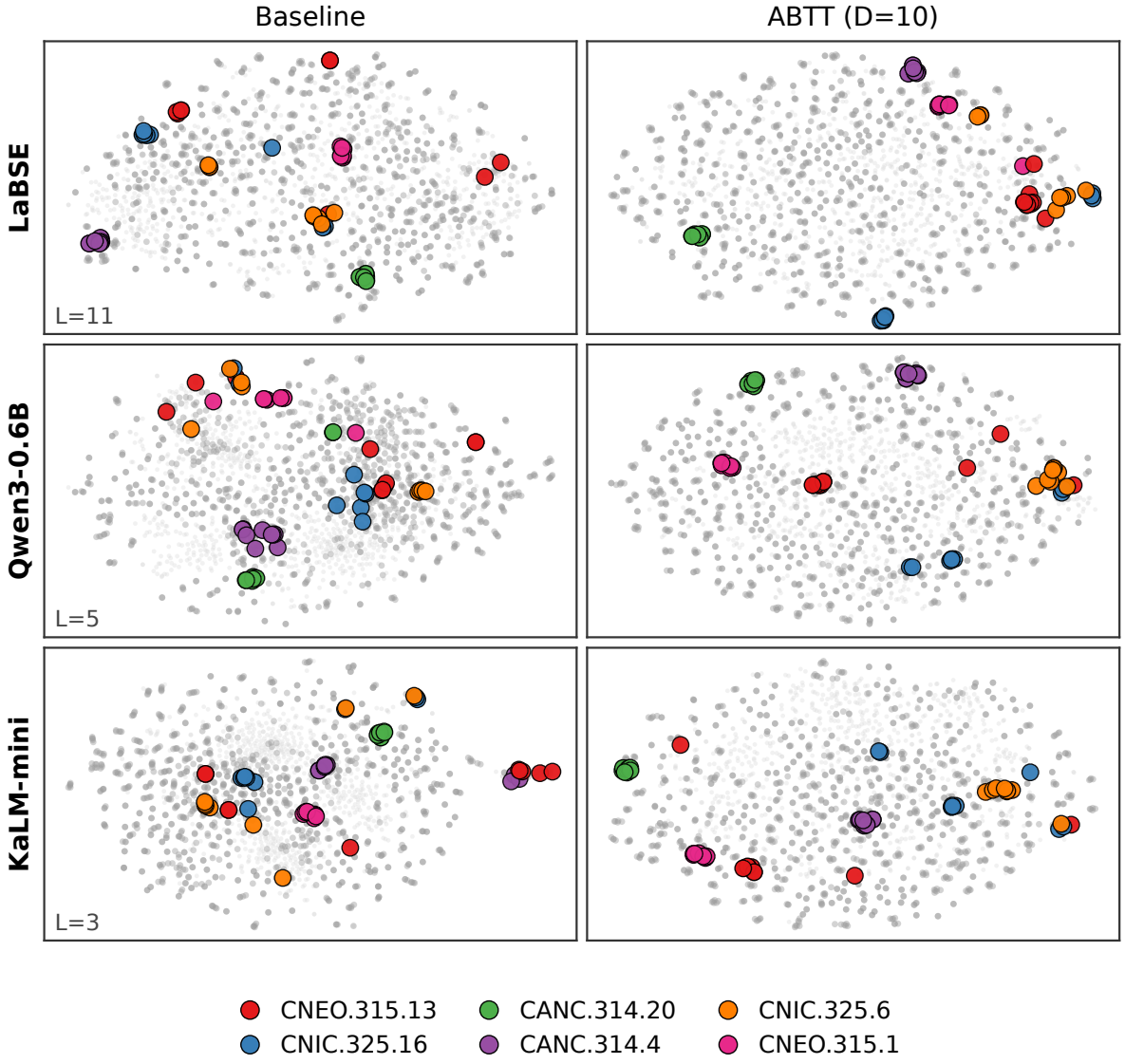



Figure 12: t-SNE maps of all 1,705 witnesses for LaBSE, Qwen3-0.6B and KaLM-mini, with layers, columns and colors as in Figure 10.

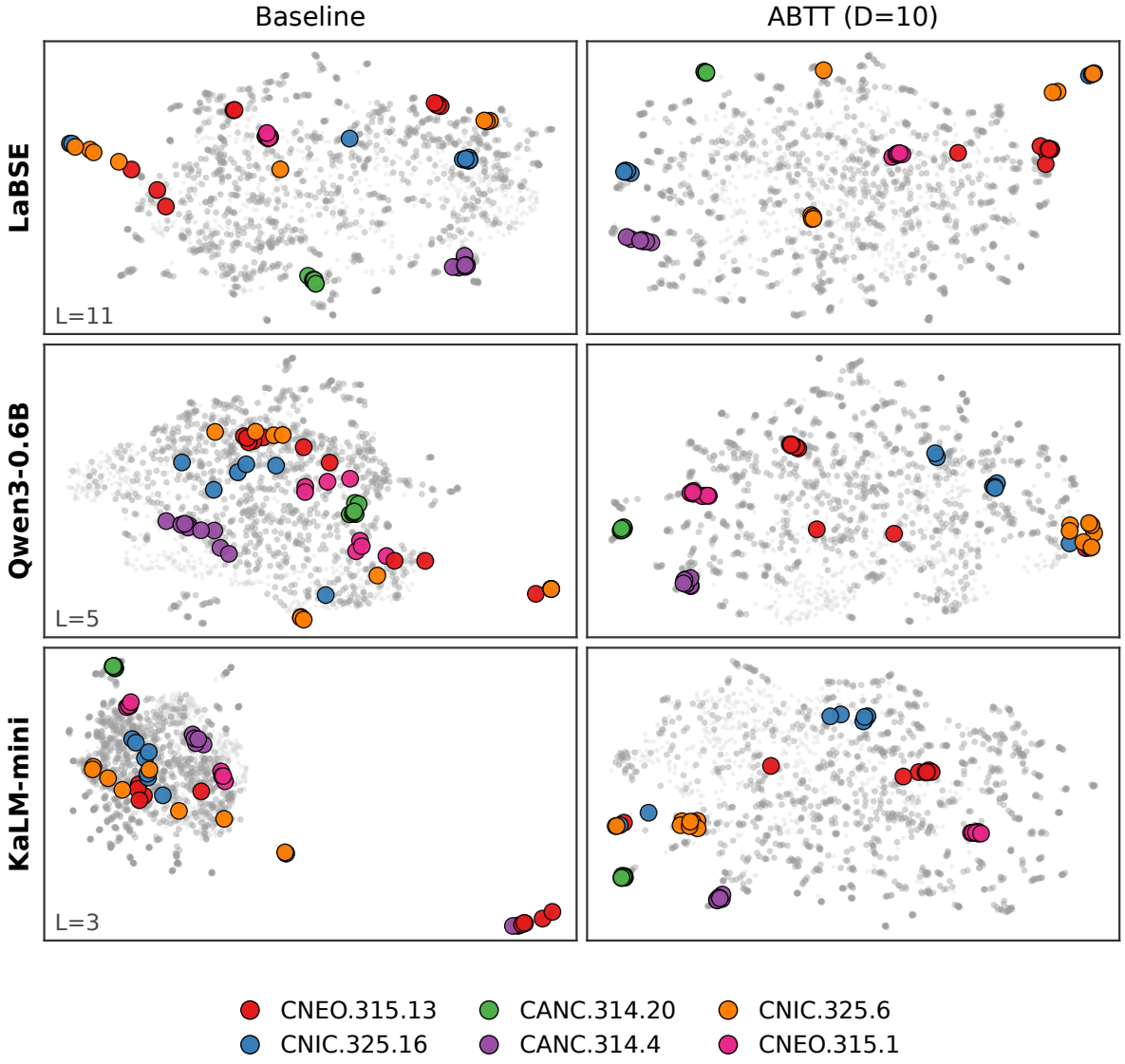



Figure 13: UMAP maps matching Figure 12. The cluster-tightening pattern appears under a second mapping method.